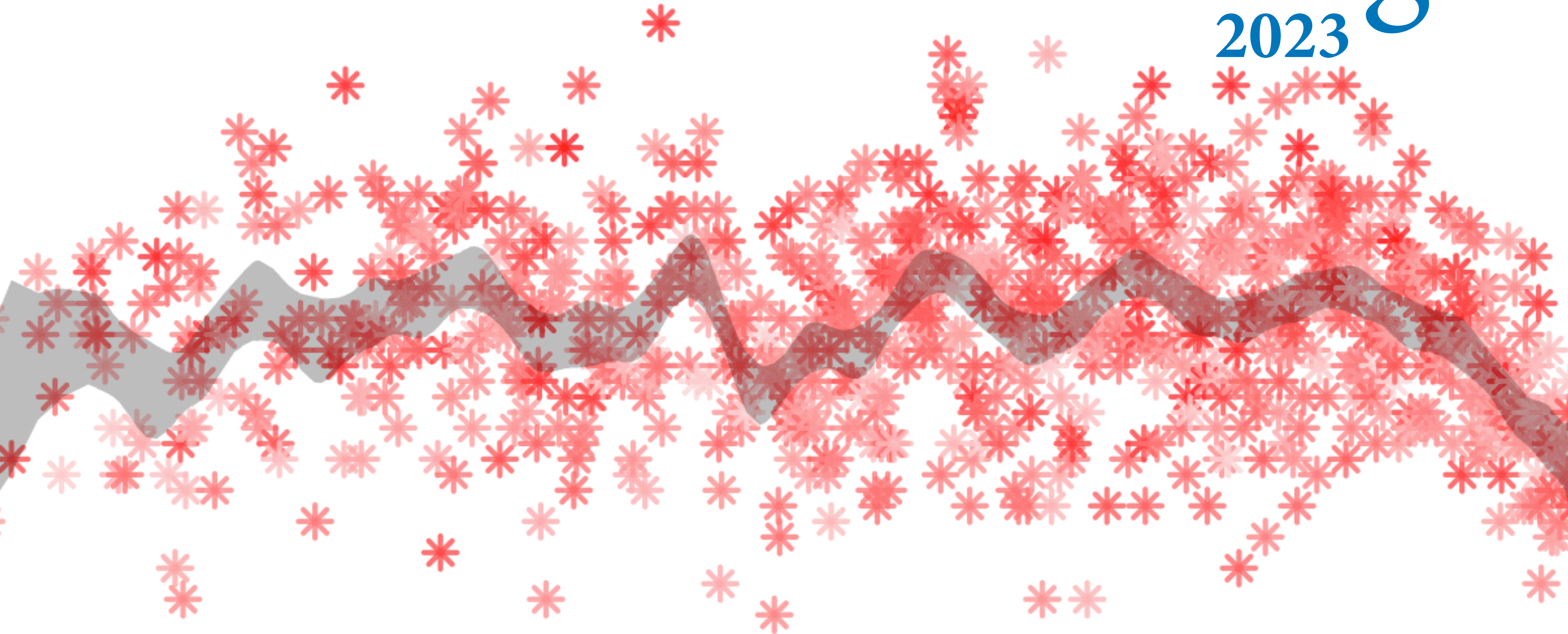


Statistical Rethinking

2023



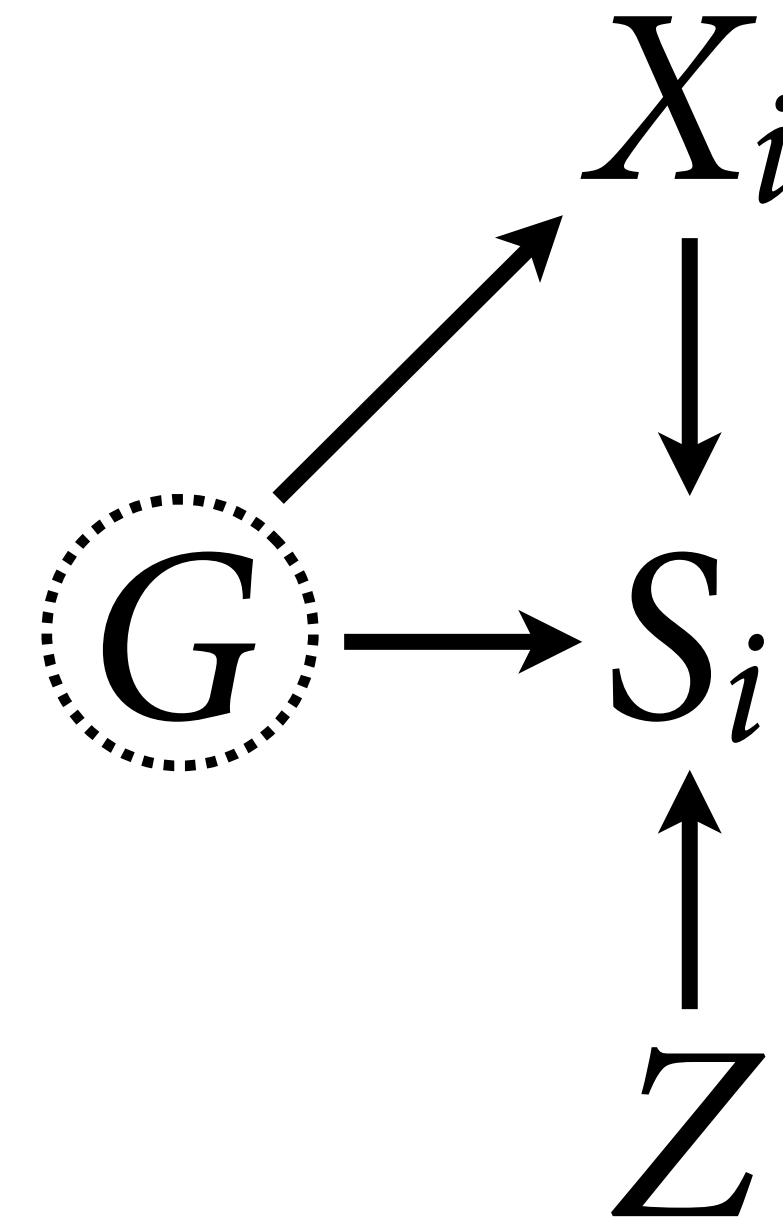
14. Correlated Features

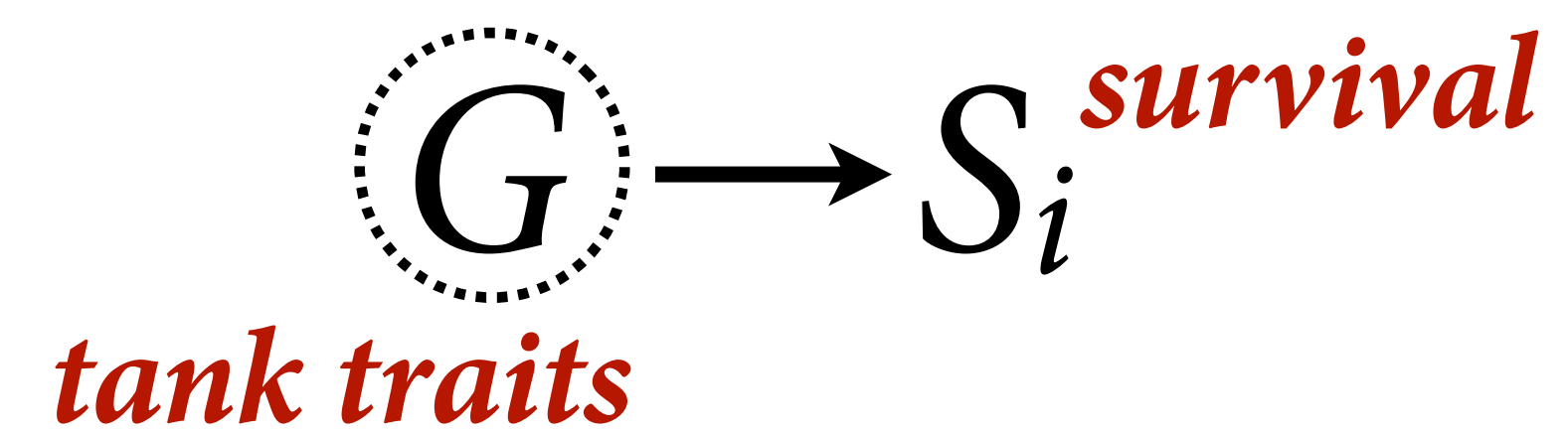
Random confounds

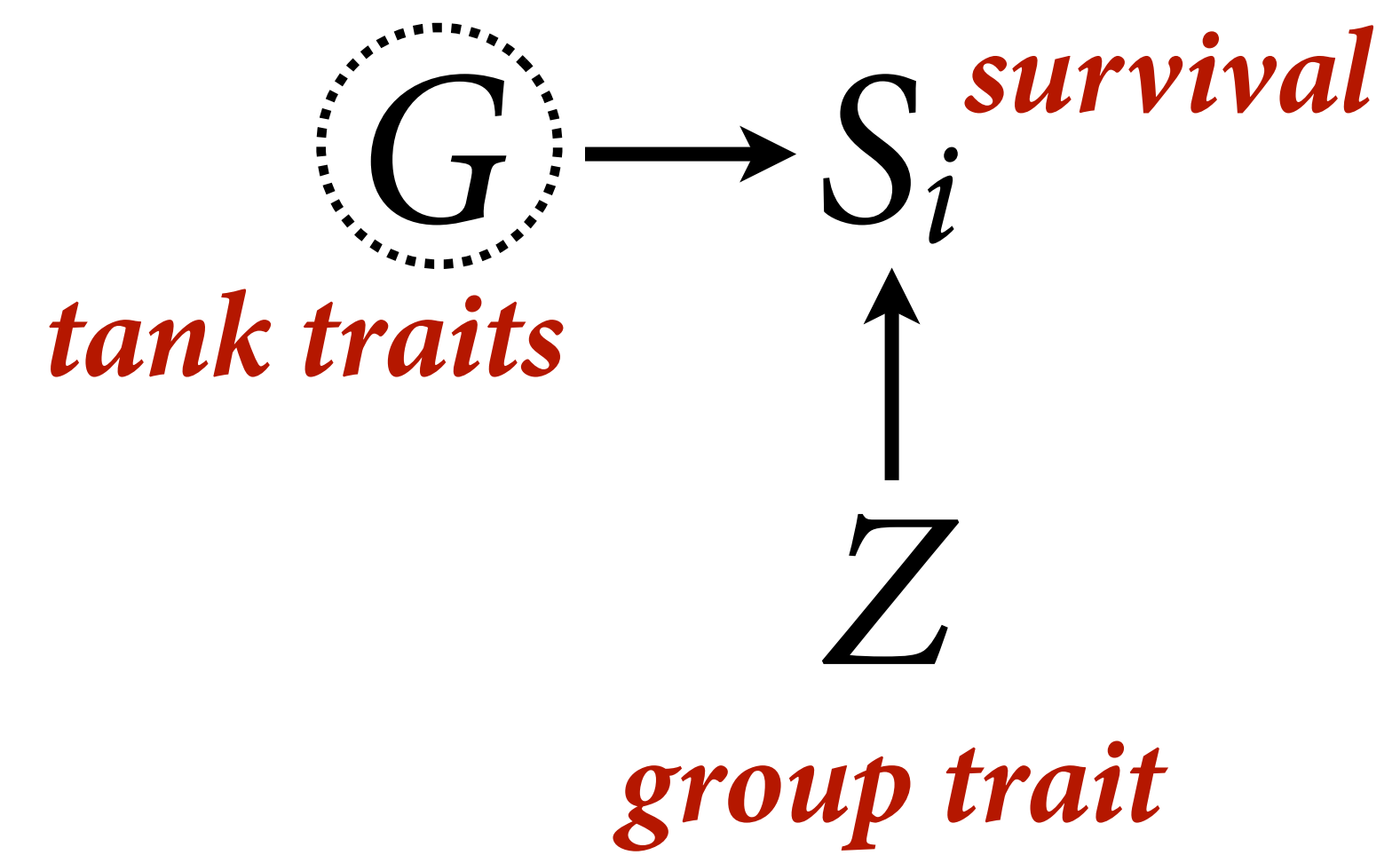
When unobserved group features influence individually-varying causes

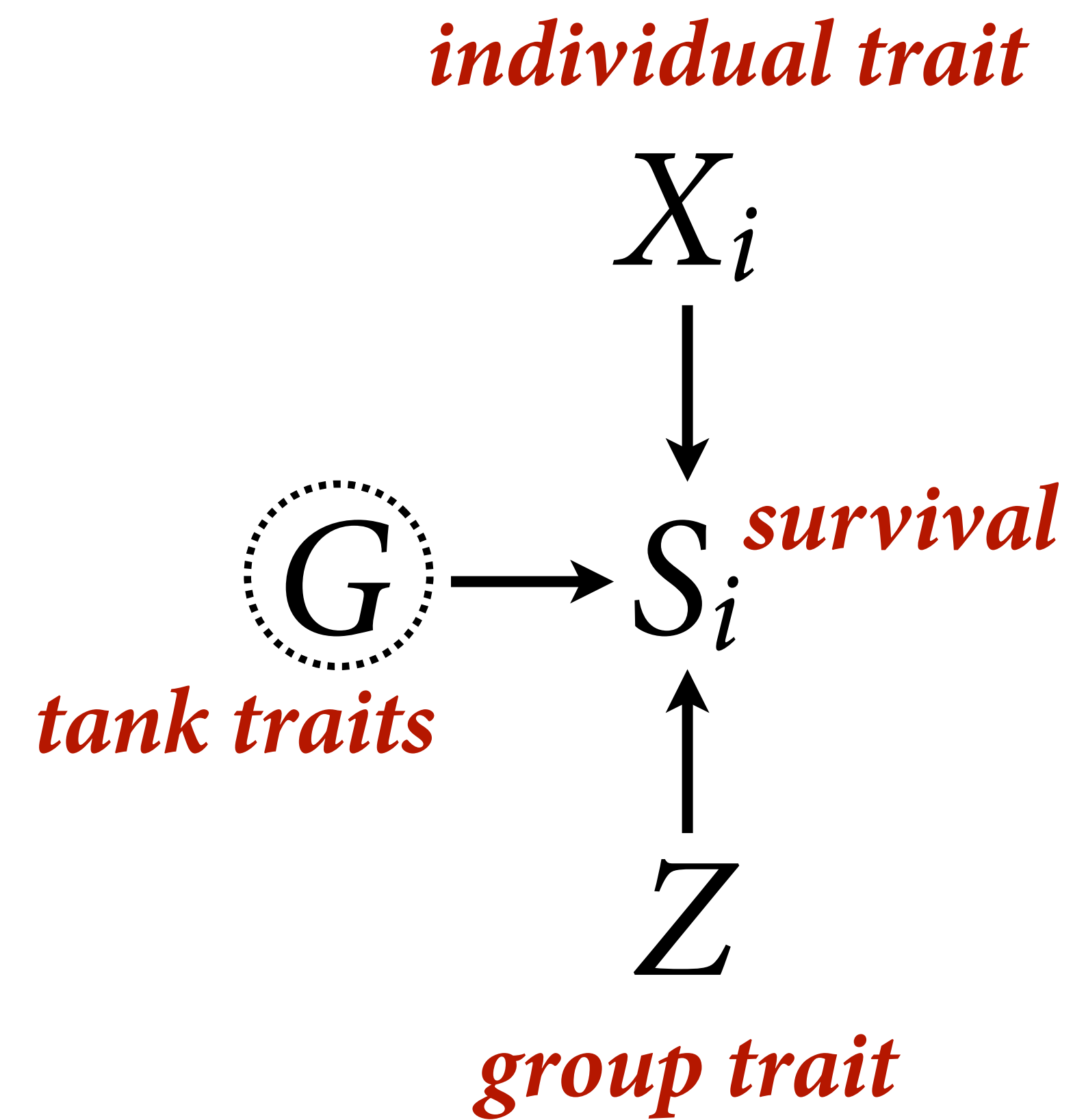
Dizzying terminology: group-level confounding, endogeneity, correlated errors, econometrics

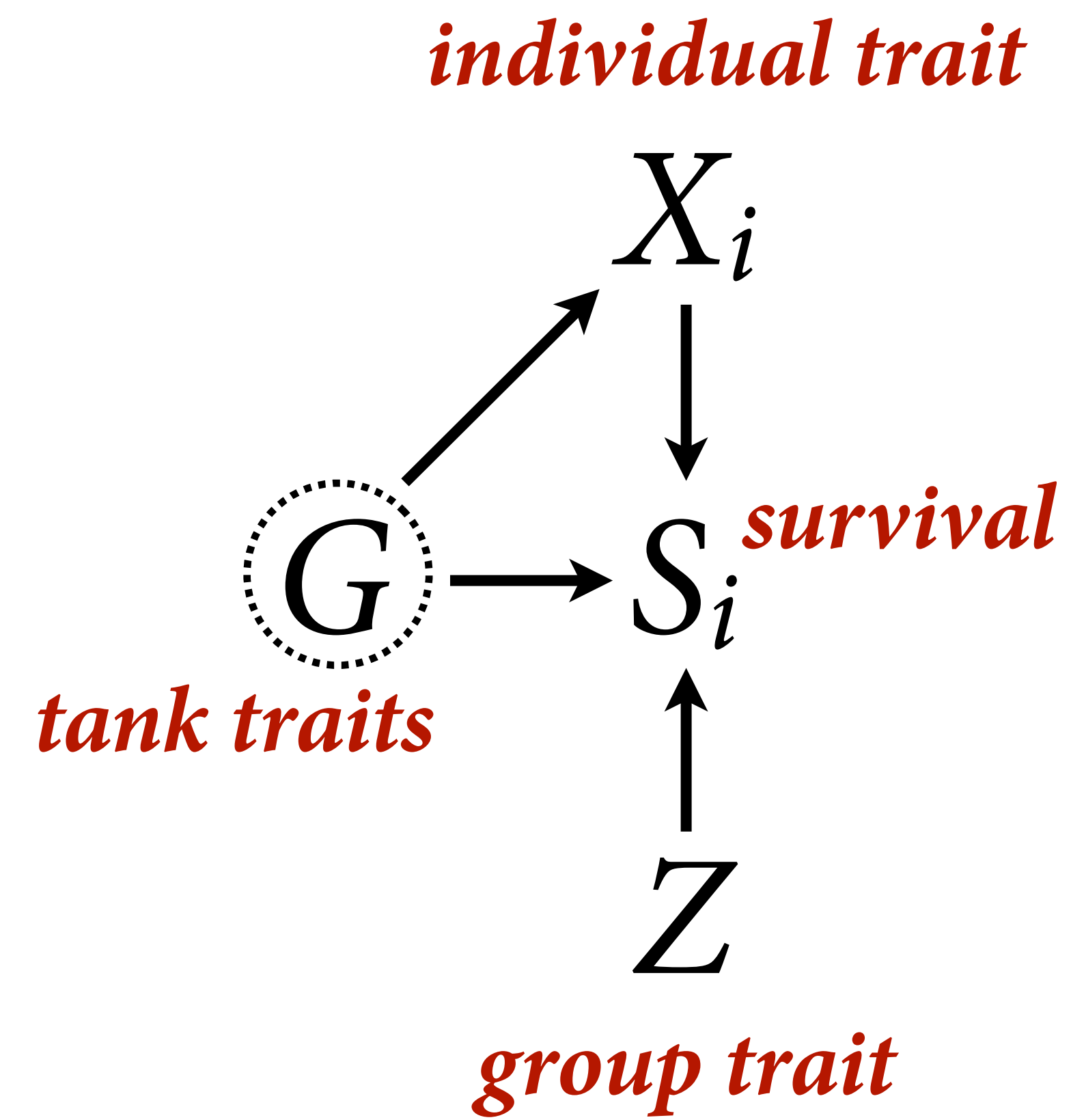
Group-level variables have direct and indirect influences





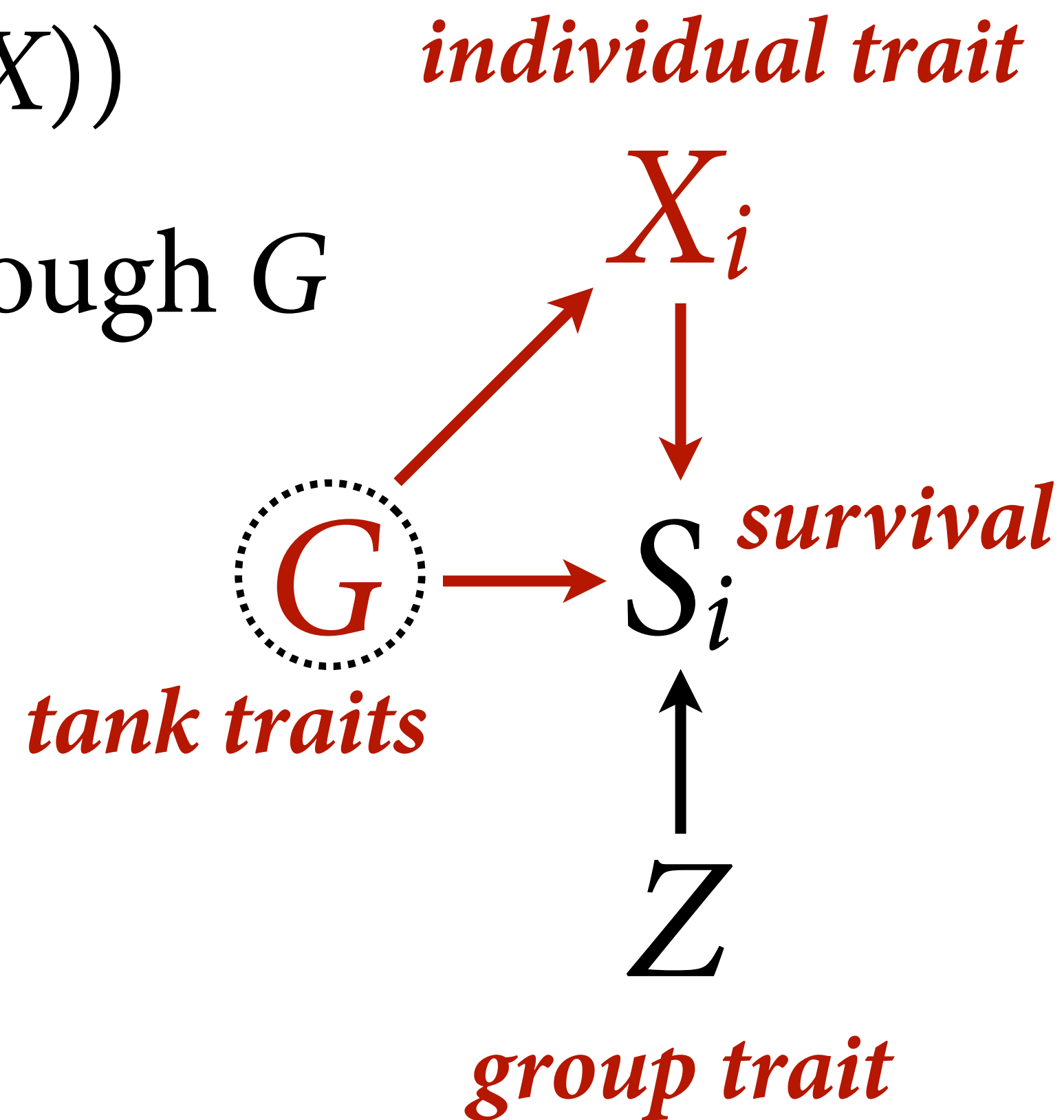


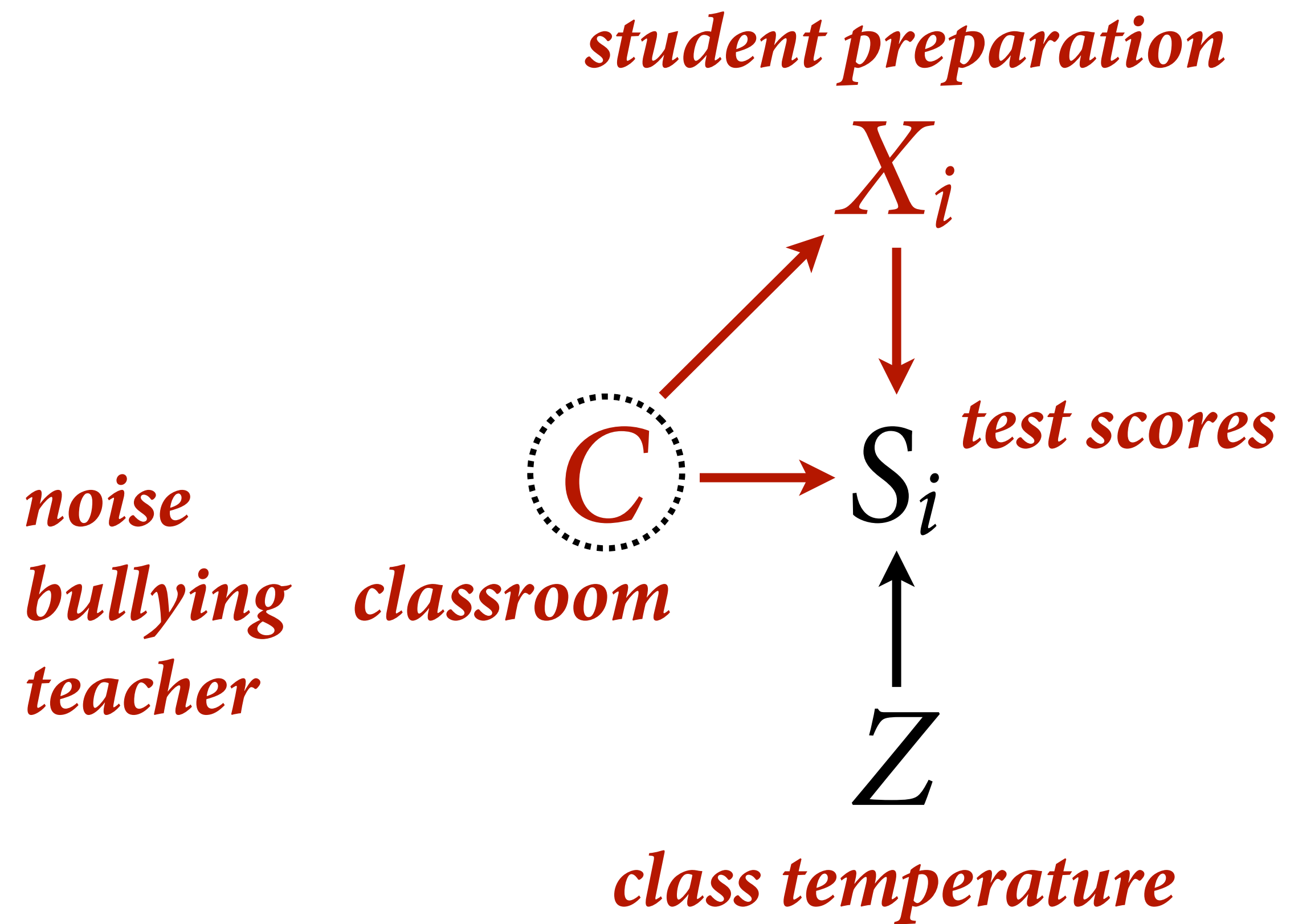


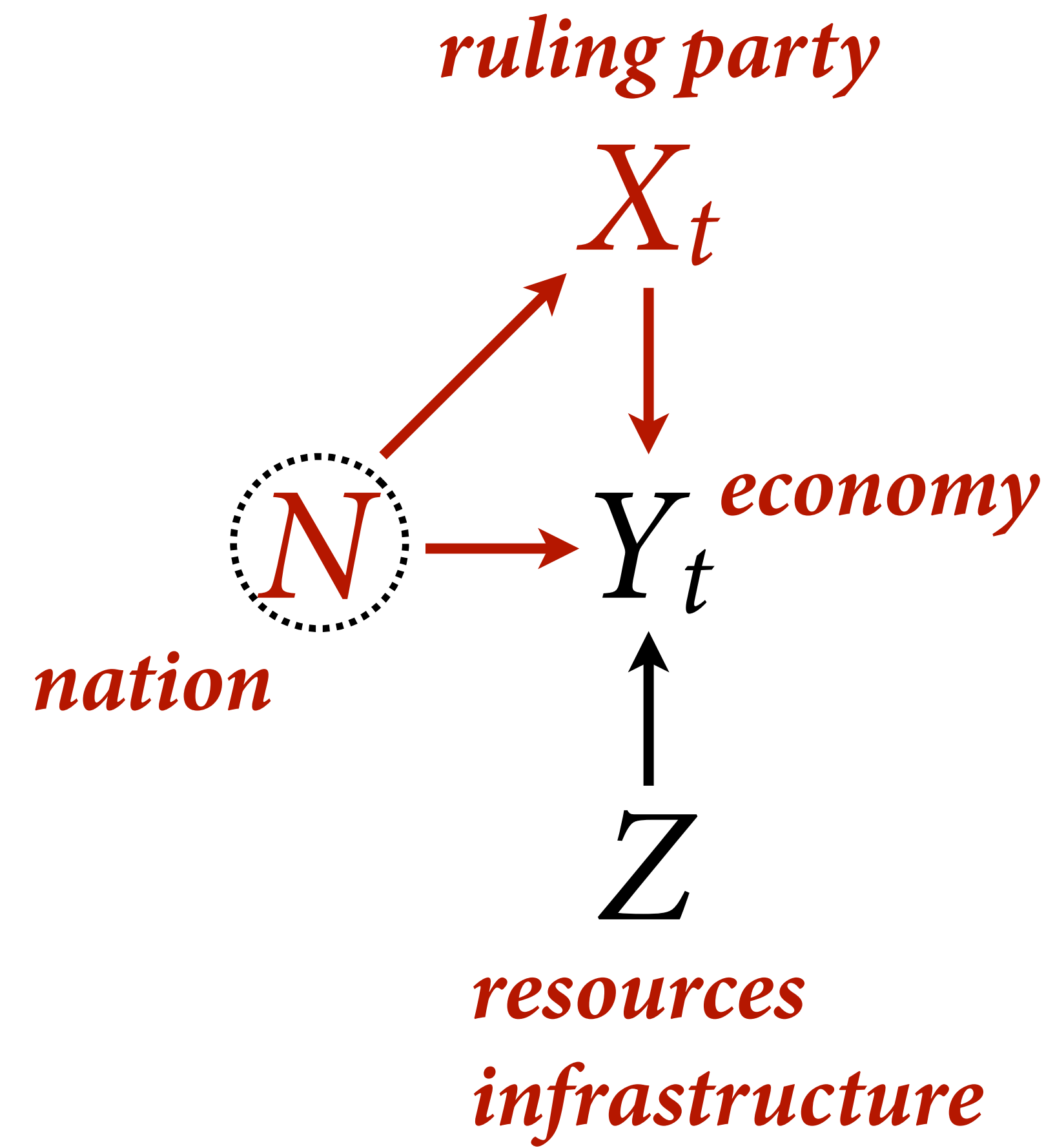


Estimand: $p(S|\text{do}(X))$

Backdoor path through G







Random confounds

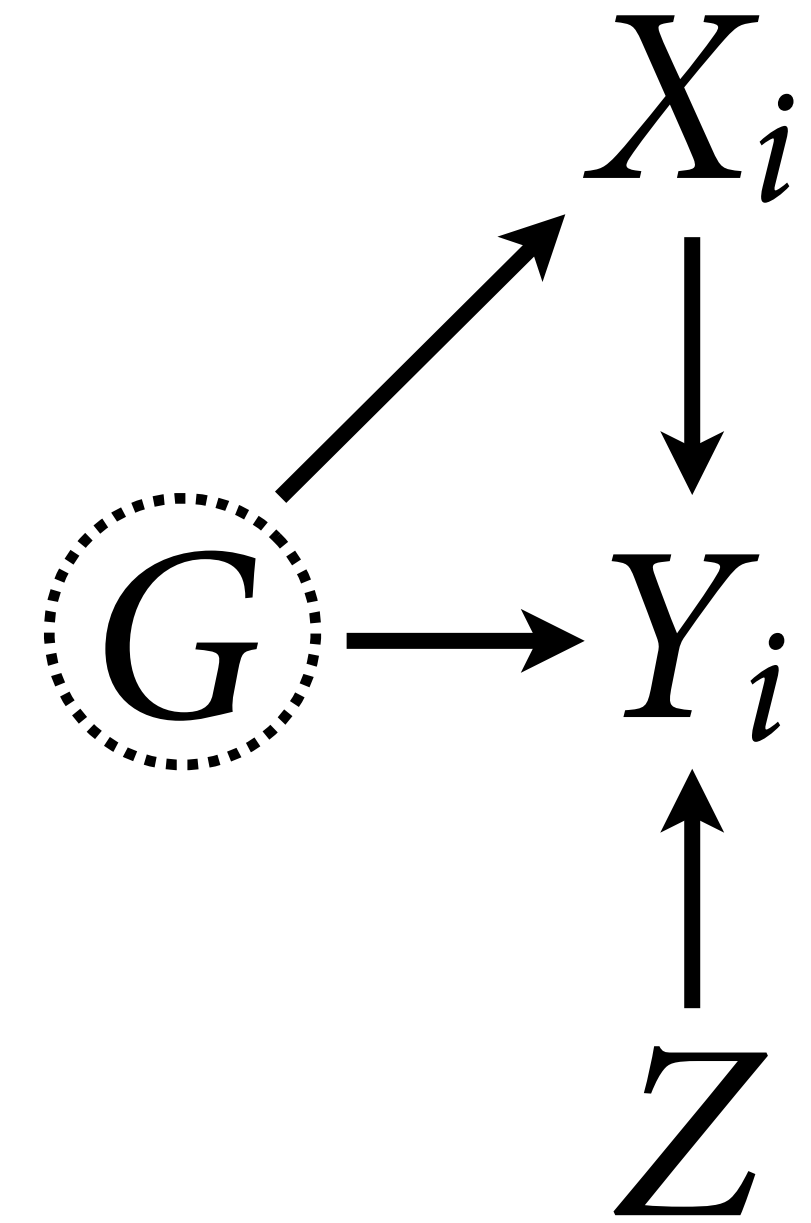
Estimand: Influence of X on Y

Estimator?

(1) **Fixed** effects model

(2) **Multilevel** model

(3) Mundlak Machines

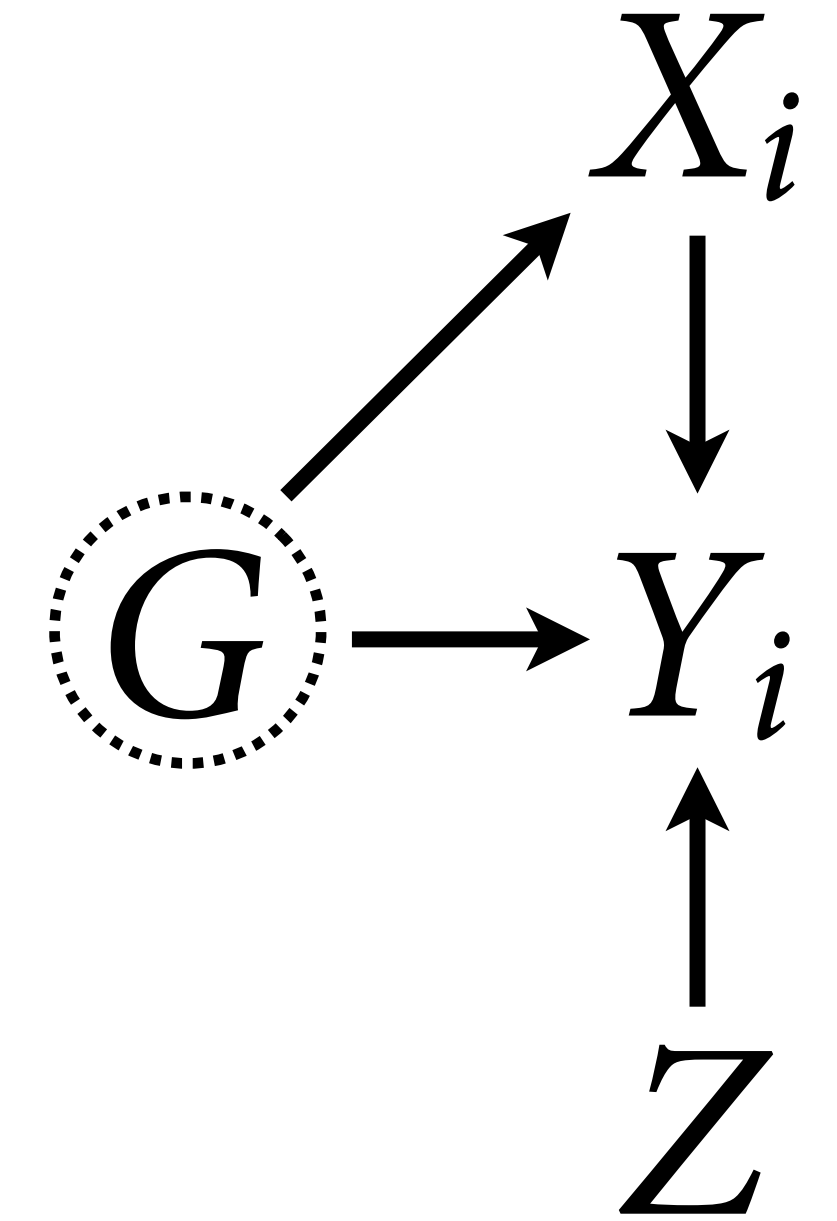


```

N_groups <- 30
N_id <- 200
a0 <- (-2)
bZY <- (-0.5)
g <- sample(1:N_groups, size=N_id, replace=TRUE) # sample into groups
Ug <- rnorm(N_groups, 1.5) # group confounds
X <- rnorm(N_id, Ug[g] ) # individual varying trait
Z <- rnorm(N_groups) # group varying trait (observed)
Y <- rbern(N_id, p=inv_logit( a0 + X + Ug[g] + bZY*Z[g] ) )

table(g)

```



```

> table(g)
g
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
11  5  8  3 11  7  6  4  9  8  4  5  3  7  7  7  4  3  8 10  4  7  6 12  5 12

27 28 29 30
 7  7  5  5

```

Fixed effects model

Estimate a different average rate for each group, **without pooling**

Inefficient, but soaks up group-level (fixed) confounding (G)

Problem: Cannot identify any group-level effects (Z)

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{G[i]} + \beta_X X_i + \beta_Z Z_{G[i]}$$

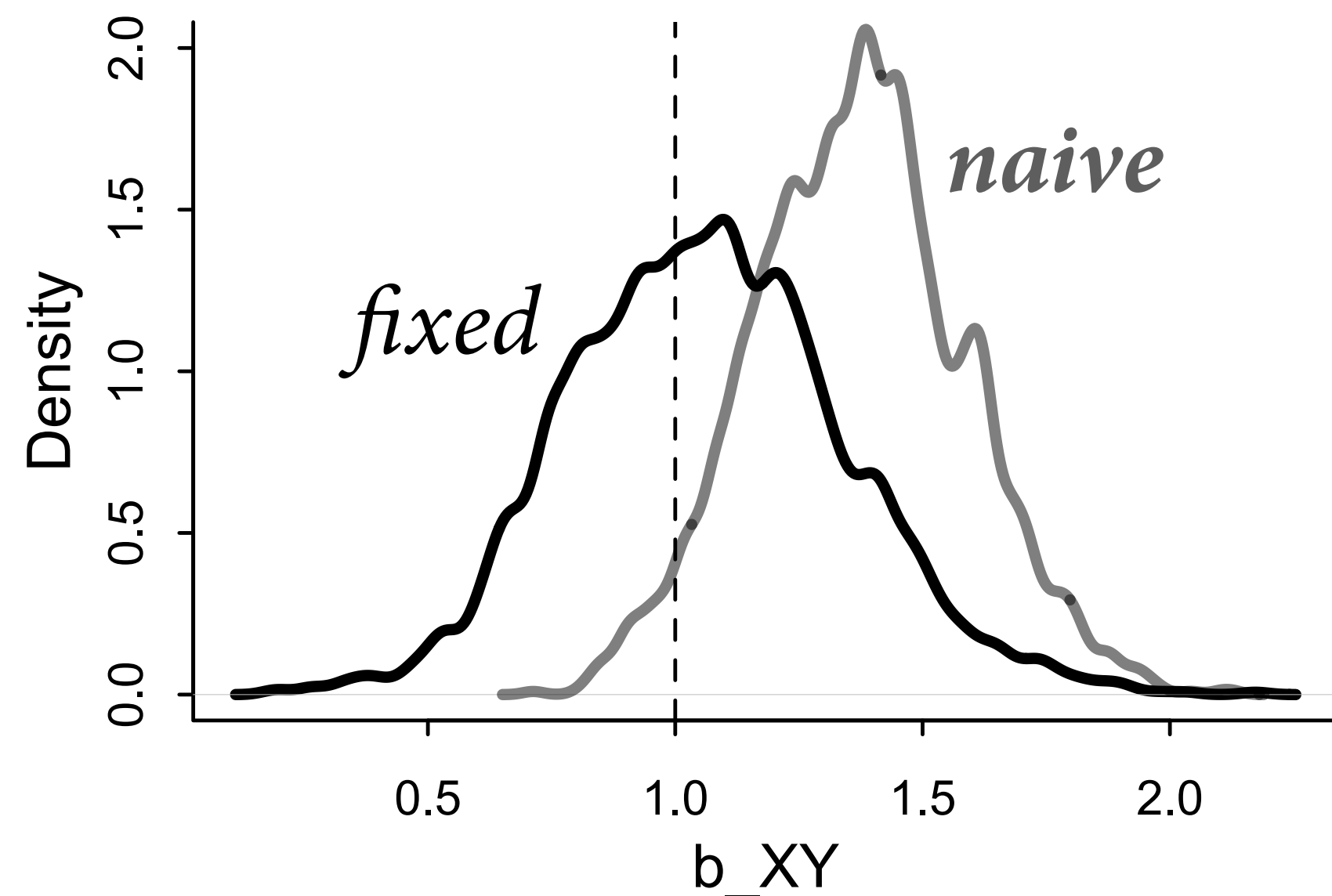
$$\alpha_j \sim \text{Normal}(0, 10)$$

$$\beta_{X,Z} \sim \text{Normal}(0, 1)$$

Fixed effects model

```
dat <- list(Y=Y,X=X,g=g,Ng=N_groups,Z=Z)

# fixed effects
mf <- ulam(
  alist(
    Y ~ bernoulli(p),
    logit(p) <- a[g] + bxy*X + bzy*Z[g],
    a[g] ~ dnorm(0,10),
    c(bxy,bzy) ~ dnorm(0,1)
  ) , data=dat , chains=4 , cores=4 )
```

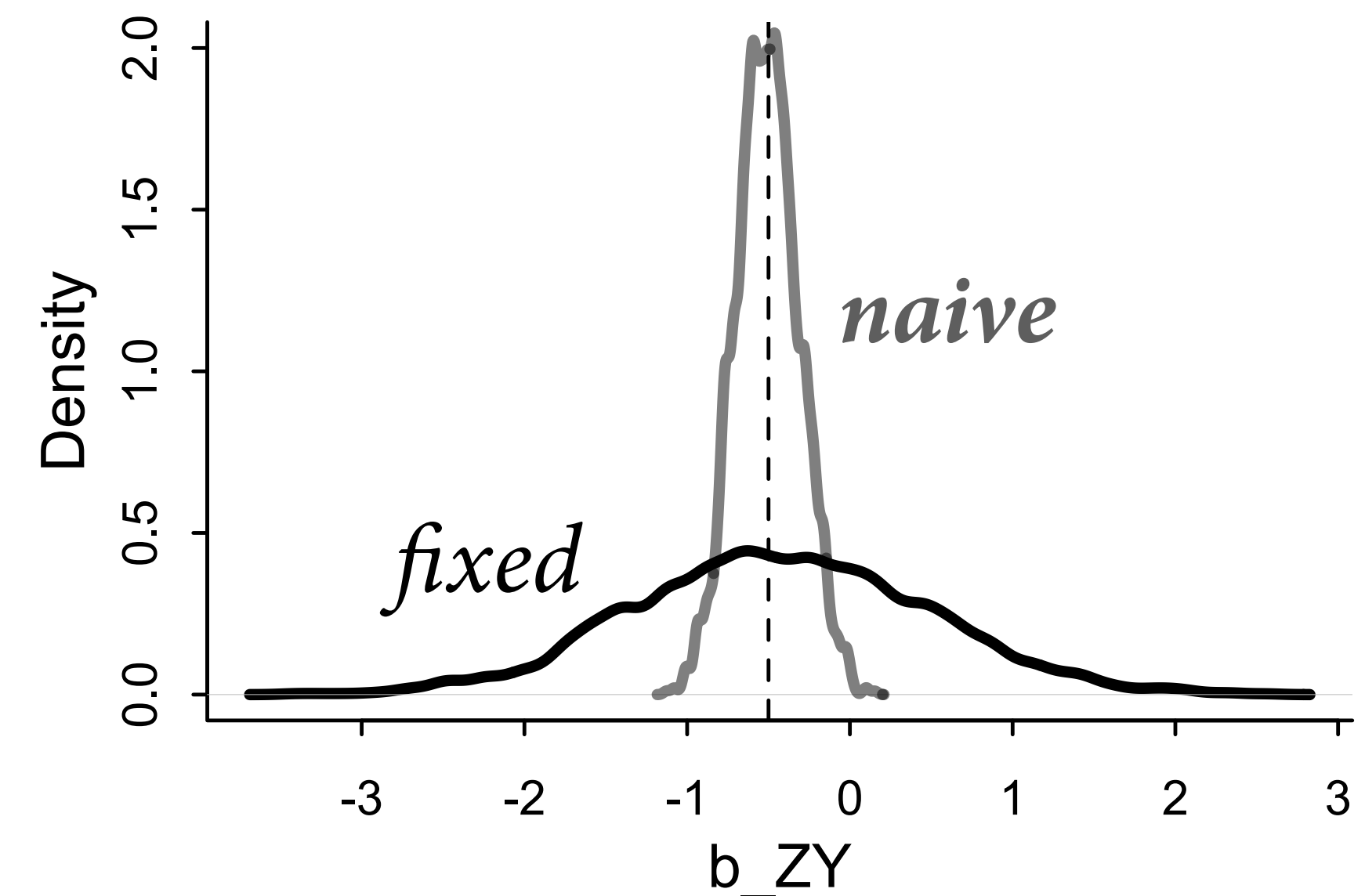


$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{G[i]} + \beta_X X_i + \beta_Z Z_{G[i]}$$

$$\alpha_j \sim \text{Normal}(0,10)$$

$$\beta_{X,Z} \sim \text{Normal}(0,1)$$



Multilevel model

Estimate a different average rate for each group, **partial pooling**

Better estimates for G , worse estimate for X

Bonus: Can identify Z

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha_{G[i]} + \beta_X X_i + \beta_Z Z_{G[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \tau)$$

$$\beta_{X,Z} \sim \text{Normal}(0,1)$$

$$\bar{\alpha} \sim \text{Normal}(0,1)$$

$$\tau \sim \text{Exponential}(1)$$

```
# varying effects (non-centered - next week!)
mr <- ulam(
  alist(
    Y ~ bernoulli(p),
    logit(p) <- a[g] + bxy*X + bzy*Z[g],
    transpars> vector[Ng]:a <-< abar + z*tau,
    z[g] ~ dnorm(0,1),
    c(bxy,bzy) ~ dnorm(0,1),
    abar ~ dnorm(0,1),
    tau ~ dexp(1)
  ), data=dat , chains=4 , cores=4 , sample=TRUE )
```

$$Y_i \sim \text{Bernoulli}(p_i)$$

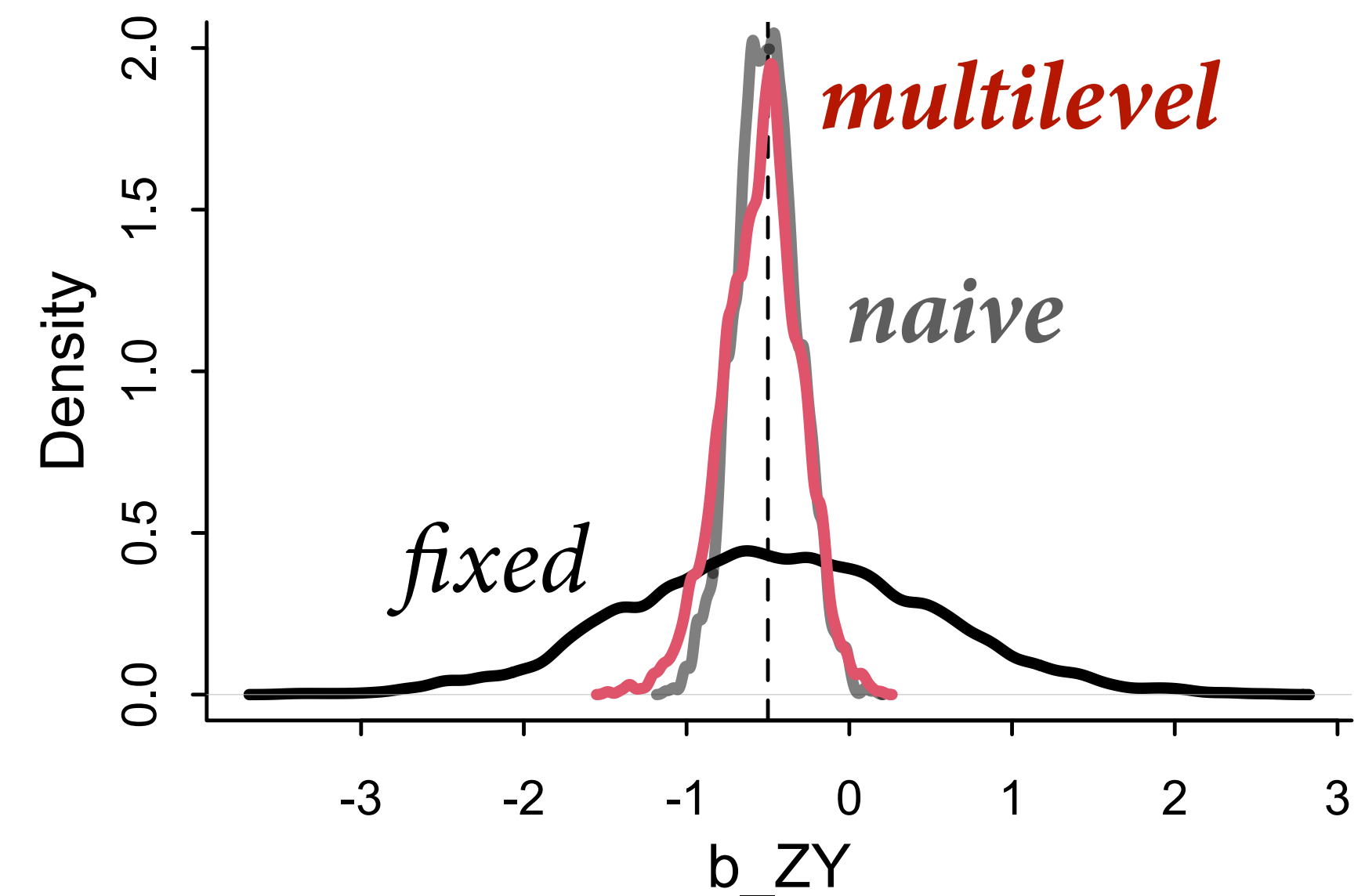
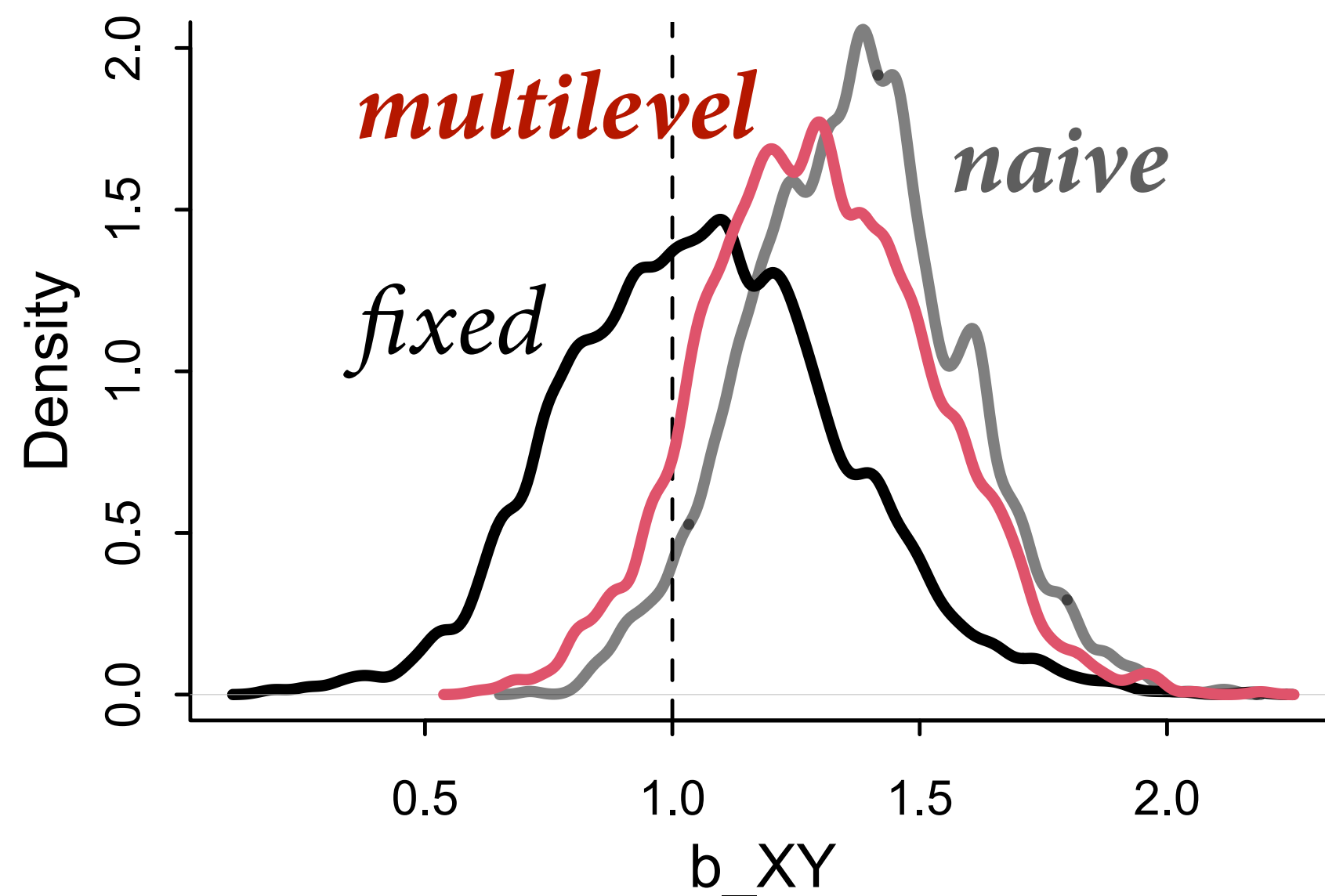
$$\text{logit}(p_i) = \alpha_{G[i]} + \beta_X X_i + \beta_Z Z_{G[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \tau)$$

$$\beta_{X,Z} \sim \text{Normal}(0,1)$$

$$\bar{\alpha} \sim \text{Normal}(0,1)$$

$$\tau \sim \text{Exponential}(1)$$

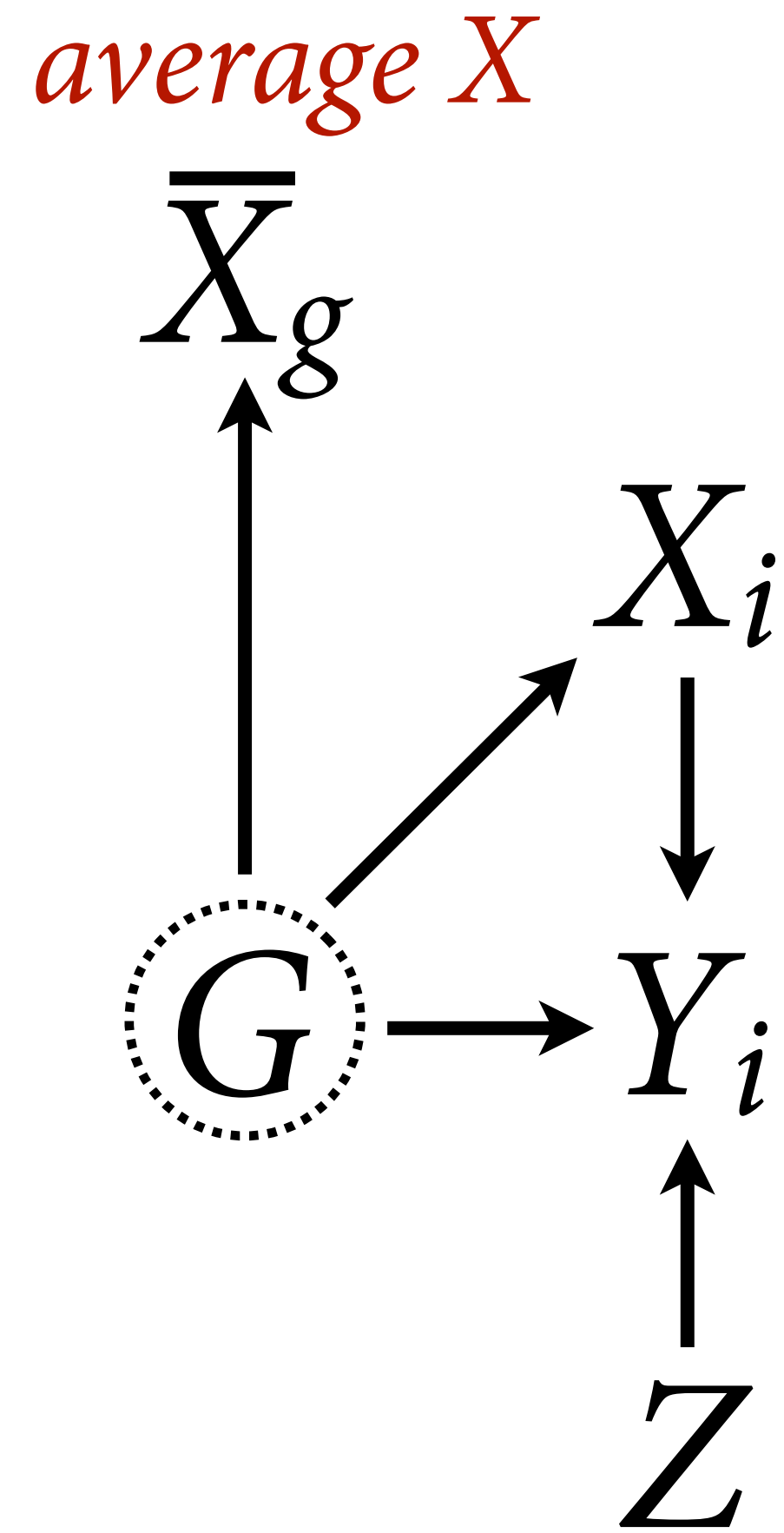


Mundlak machine

Estimate a different average rate for each group, **partial pooling**

Include group average X

Better X , but improper respect for uncertainty in X -bar



$$\text{logit}(p_i) = \alpha_{G[i]} + \beta_X X_i + \beta_Z Z_{G[i]} + \beta_{\bar{X}} \bar{X}_{G[i]}$$

The Mundlak Machine

```
xbar <- sapply( 1:N_groups , function(j) mean(X[g==j]) )
dat$Xbar <- xbar
mrx <- ulam(
  alist(
    Y ~ bernoulli(p),
    logit(p) <- a[g] + bxy*X + bzy*Z[g] + buy*Xbar[g],
    transpars> vector[Ng]:a <- abar + z*tau,
    z[g] ~ dnorm(0,1),
    c(bxy,buy,bzy) ~ dnorm(0,1),
    abar ~ dnorm(0,1),
    tau ~ dexp(1)
  ) , data=dat , chains=4 , cores=4 , sample=TRUE )
```

$$Y_i \sim \text{Bernoulli}(p_i)$$

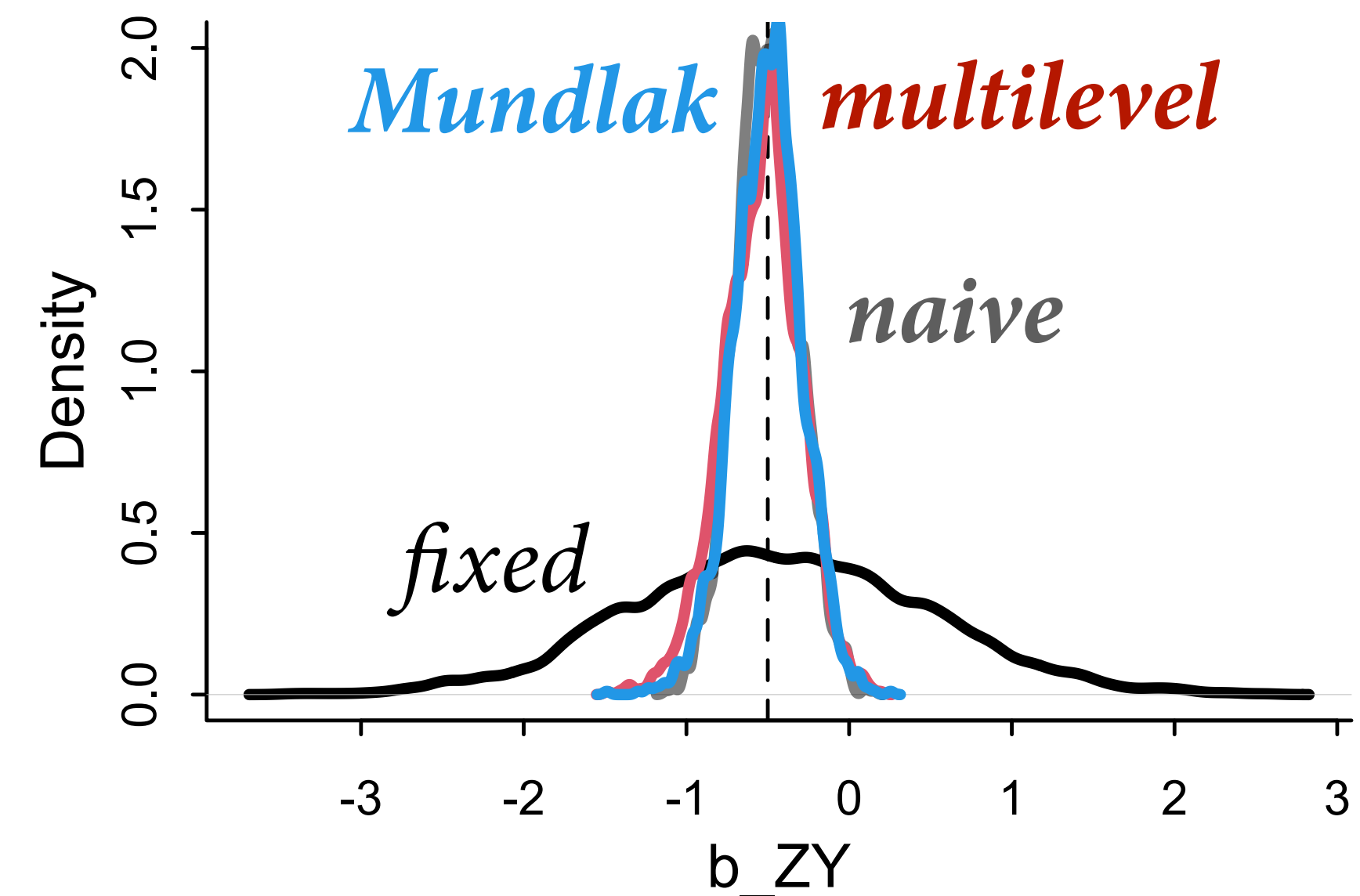
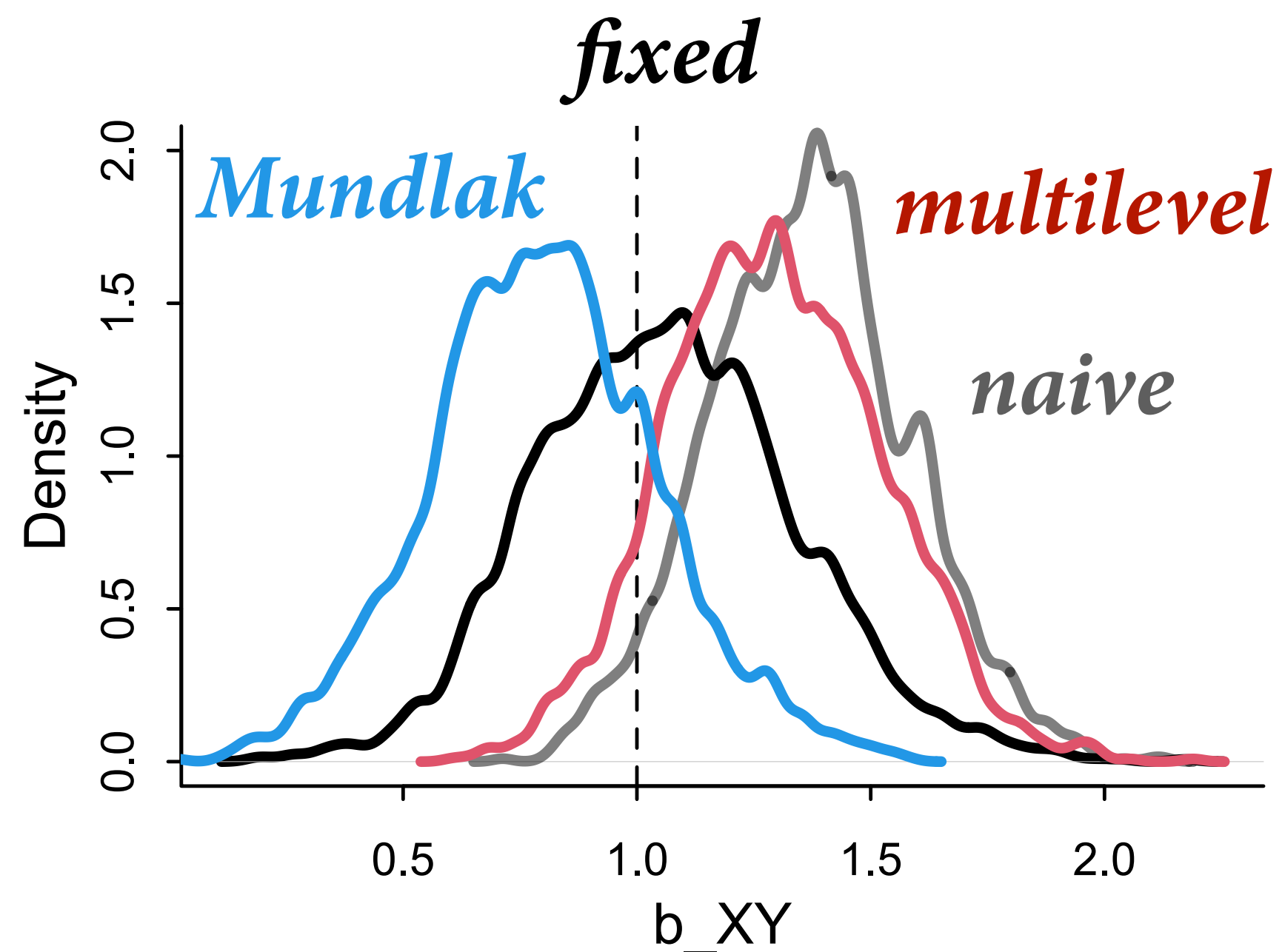
$$\text{logit}(p_i) = \alpha_{G[i]} + \beta_X X_i + \beta_Z Z_{G[i]} + \beta_{\bar{X}} \bar{X}_{G[i]}$$

$$\alpha_j \sim \text{Normal}(\bar{\alpha}, \tau)$$

$$\beta_{X,Z,\bar{X}} \sim \text{Normal}(0,1)$$

$$\bar{\alpha} \sim \text{Normal}(0,1)$$

$$\tau \sim \text{Exponential}(1)$$



Full Luxury Bayes

aka Latent Mundlak Machine

Just express the generative model

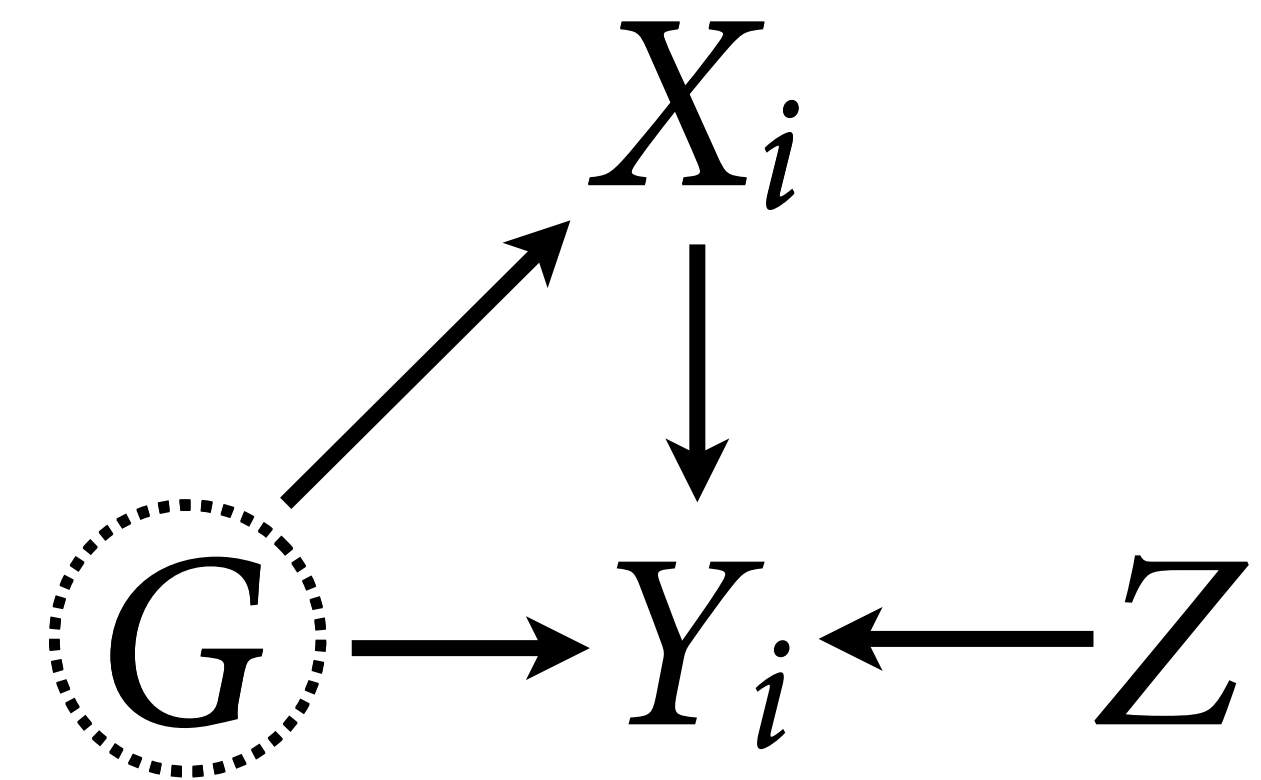
Treat G as unknown and use X_i to estimate

Two simultaneous regressions

(1) Estimate $X_i | \text{do}(G)$

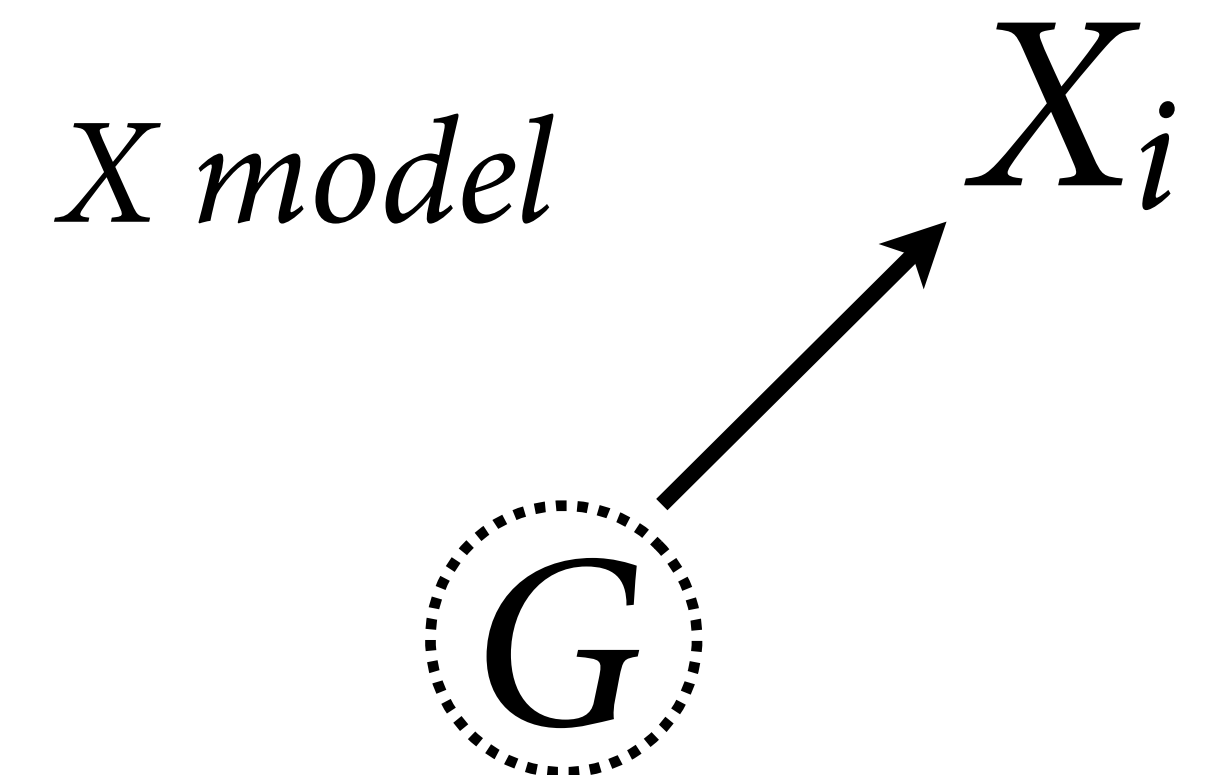
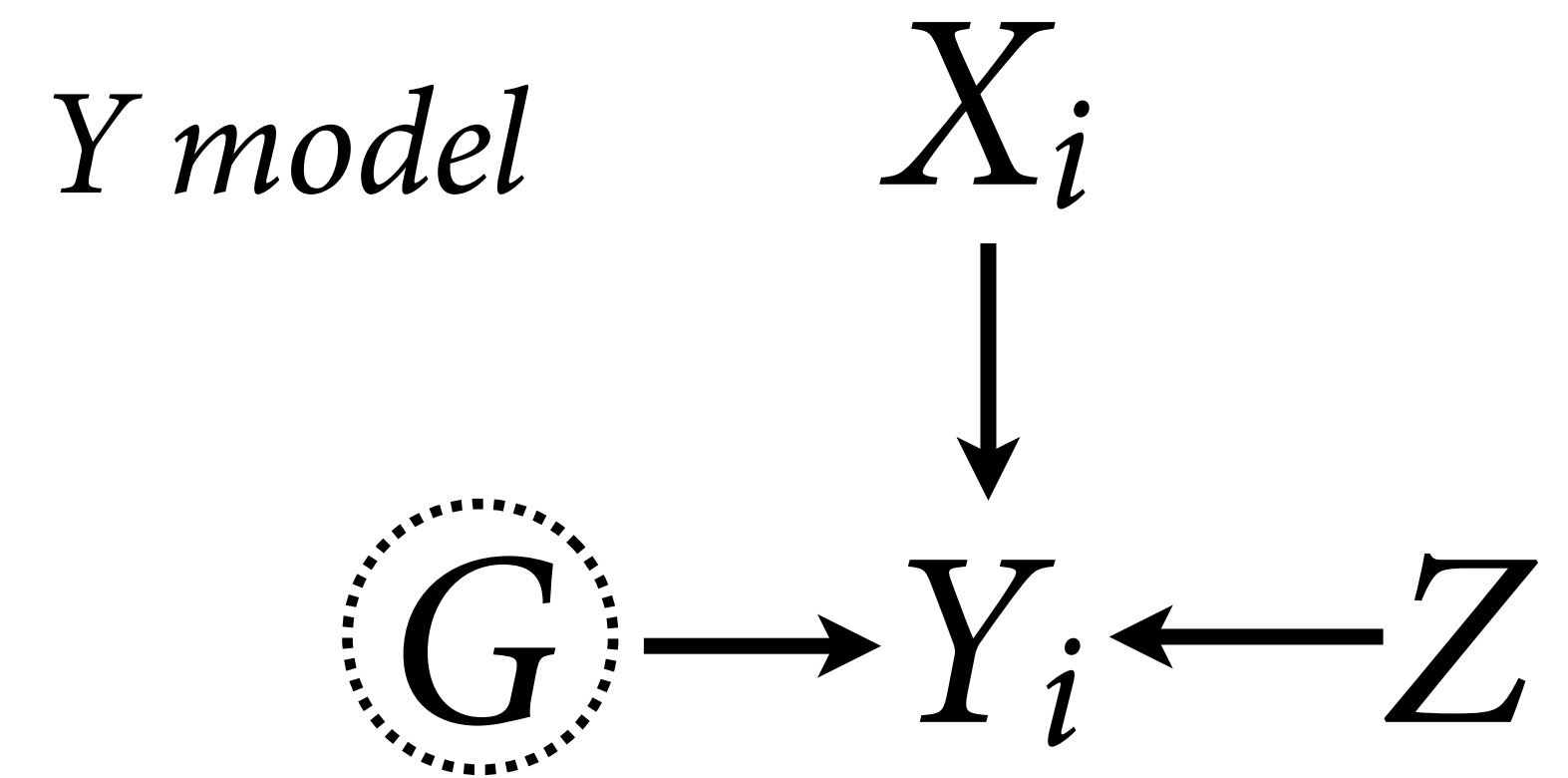
(2) Estimate $Y_i | \text{do}(X_i)$

Respects uncertainty in G



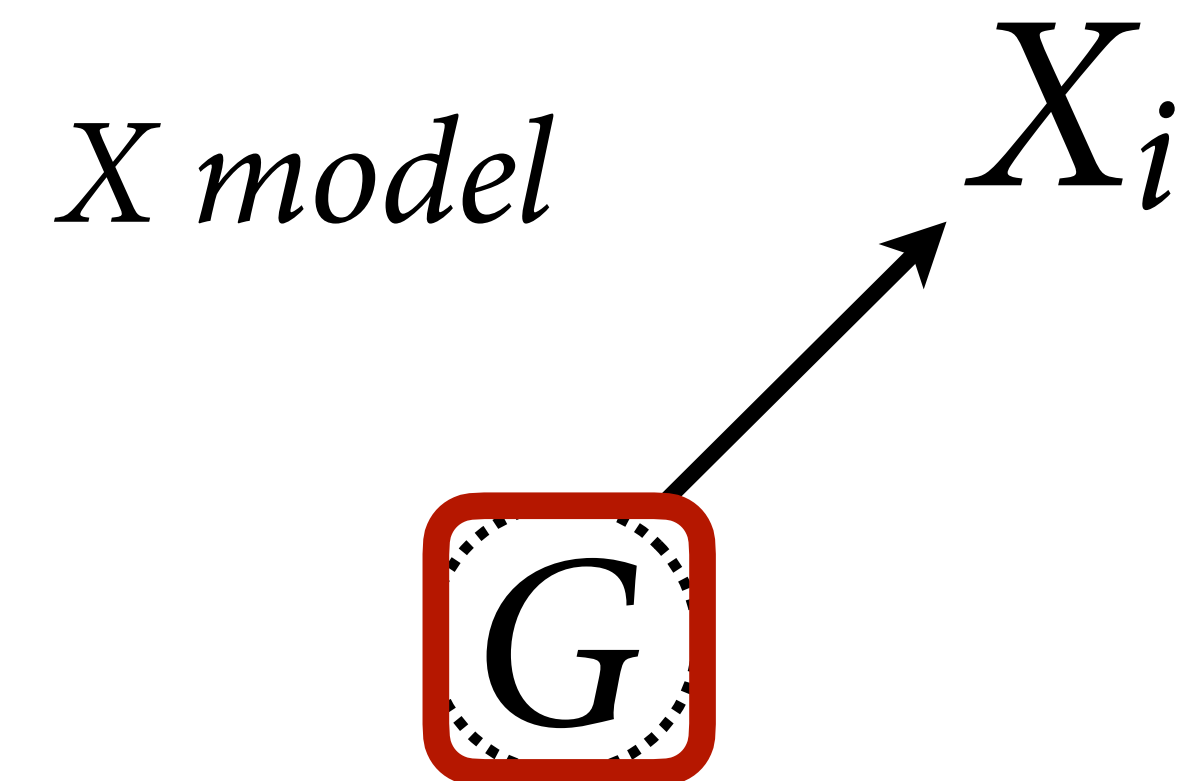
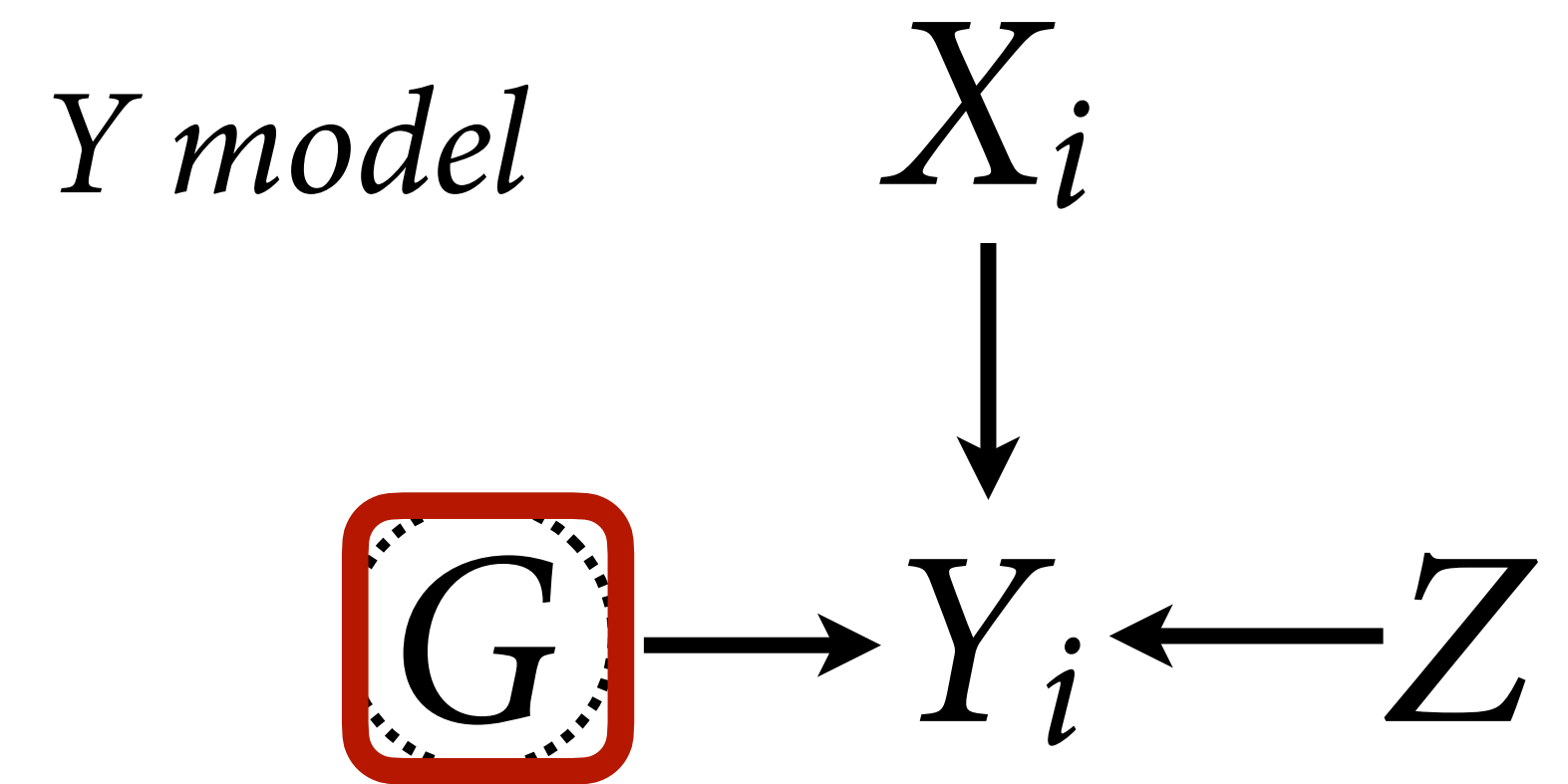
```
# The Latent Mundlak Machine
```

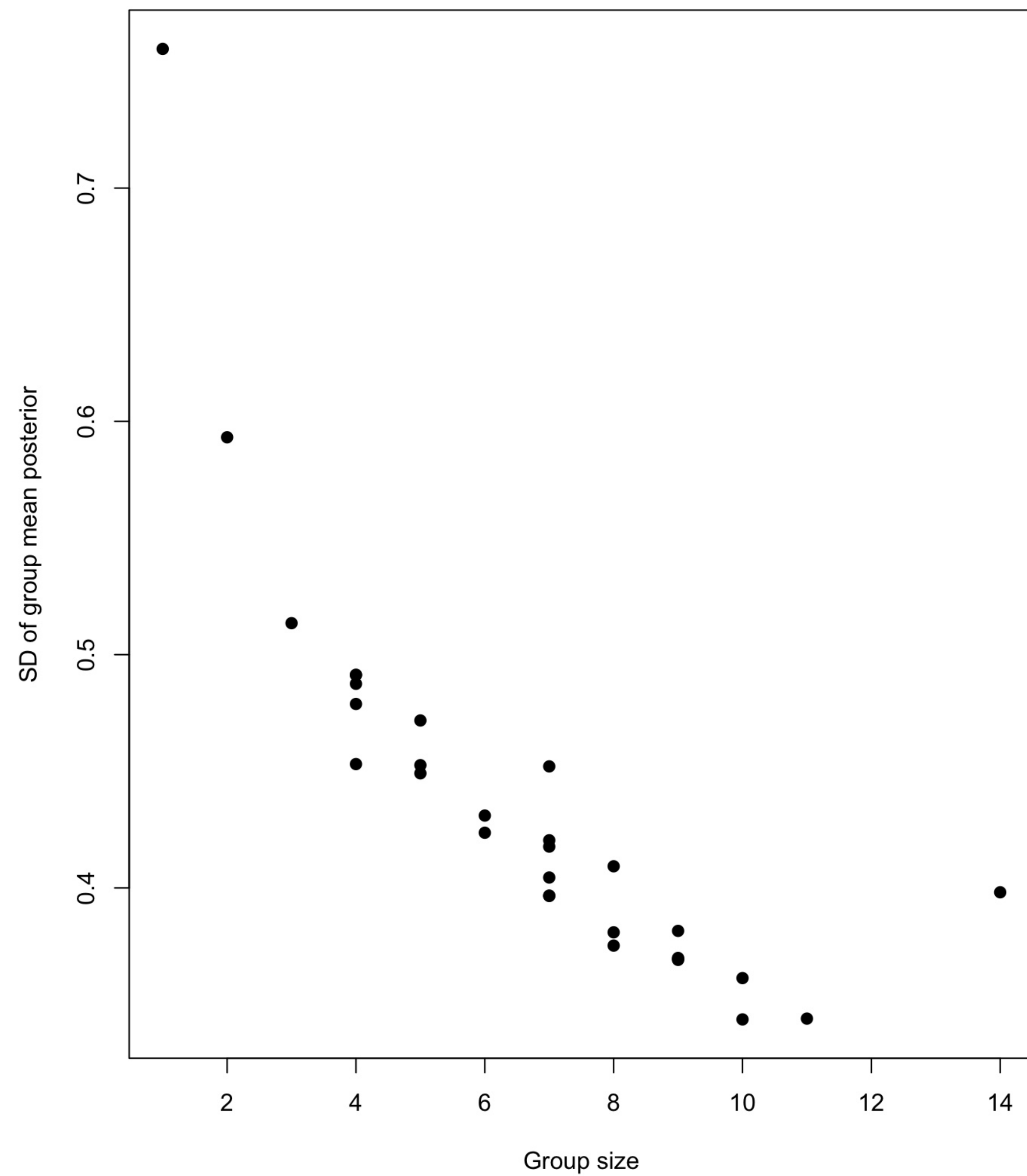
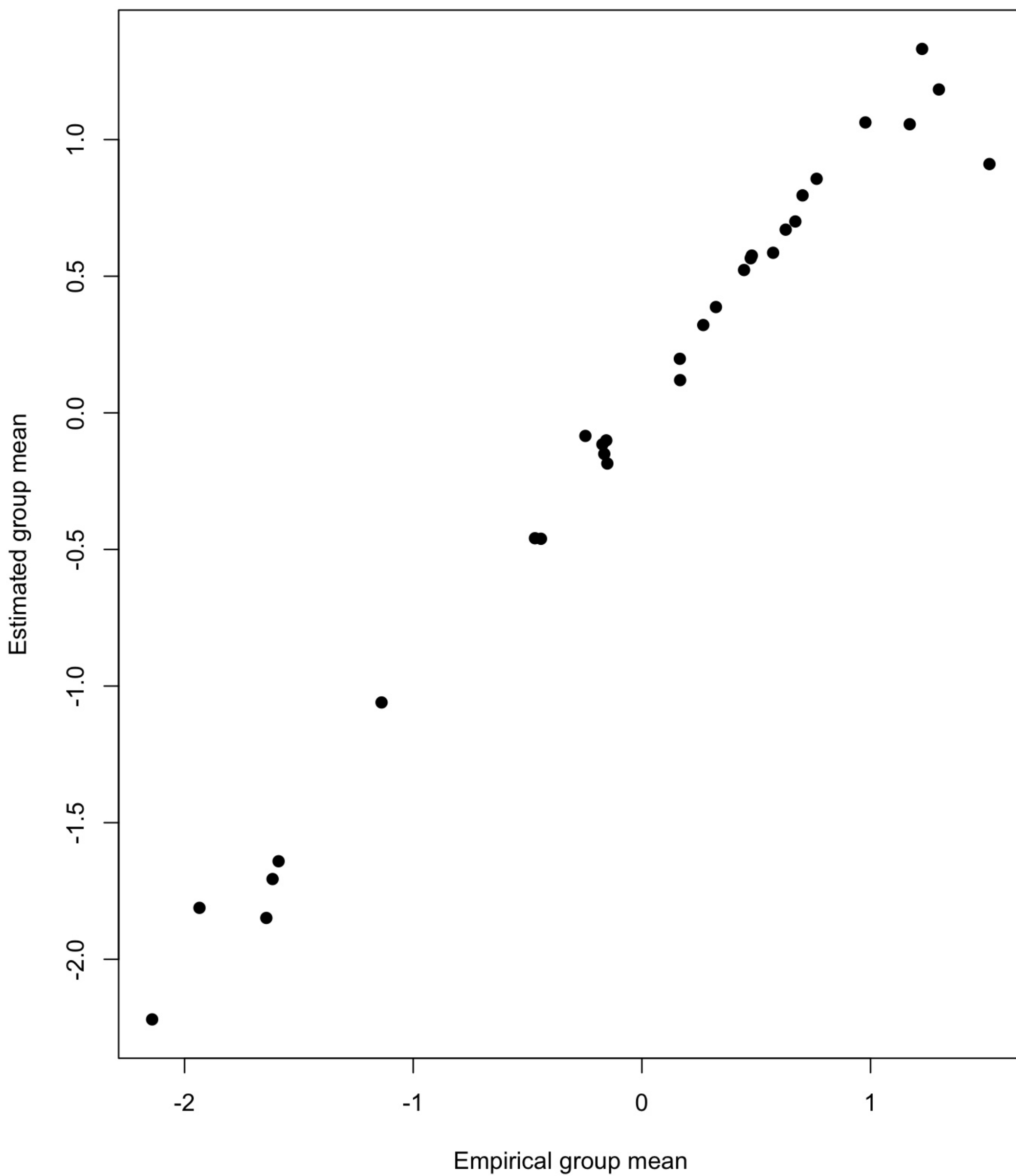
```
mru <- ulam(  
  alist(  
    # Y model  
    Y ~ bernoulli(p),  
    logit(p) <- a[g] + bxy*X + bzy*Z[g] + buy*u[g],  
    transpars> vector[Ng]:a <- abar + z*tau,  
  
    # X model  
    X ~ normal(mu,sigma),  
    mu <- aX + bux*u[g],  
    vector[Ng]:u ~ normal(0,1),  
  
    # priors  
    z[g] ~ dnorm(0,1),  
    c(aX,bxy,buy,bzy) ~ dnorm(0,1),  
    bux ~ dexp(1),  
    abar ~ dnorm(0,1),  
    tau ~ dexp(1),  
    sigma ~ dexp(1)  
  ), data=dat , chains=4 , cores=4 , sample=TRUE )
```



The Latent Mundlak Machine

```
mru <- ulam(  
  alist(  
    # Y model  
    Y ~ bernoulli(p),  
    logit(p) <- a[g] + bxy*X + bzy*Z[g] + buy*u[g],  
    transpars> vector[Ng]:a <- abar + z*tau,  
  
    # X model  
    X ~ normal(mu, sigma),  
    mu <- aX + bux*u[g],  
    vector[Ng]:u ~ normal(0,1),  
  
    # priors  
    z[g] ~ dnorm(0,1),  
    c(aX,bxy,buy,bzy) ~ dnorm(0,1),  
    bux ~ dexp(1),  
    abar ~ dnorm(0,1),  
    tau ~ dexp(1),  
    sigma ~ dexp(1)  
  ), data=dat , chains=4 , cores=4 , sample=TRUE )
```





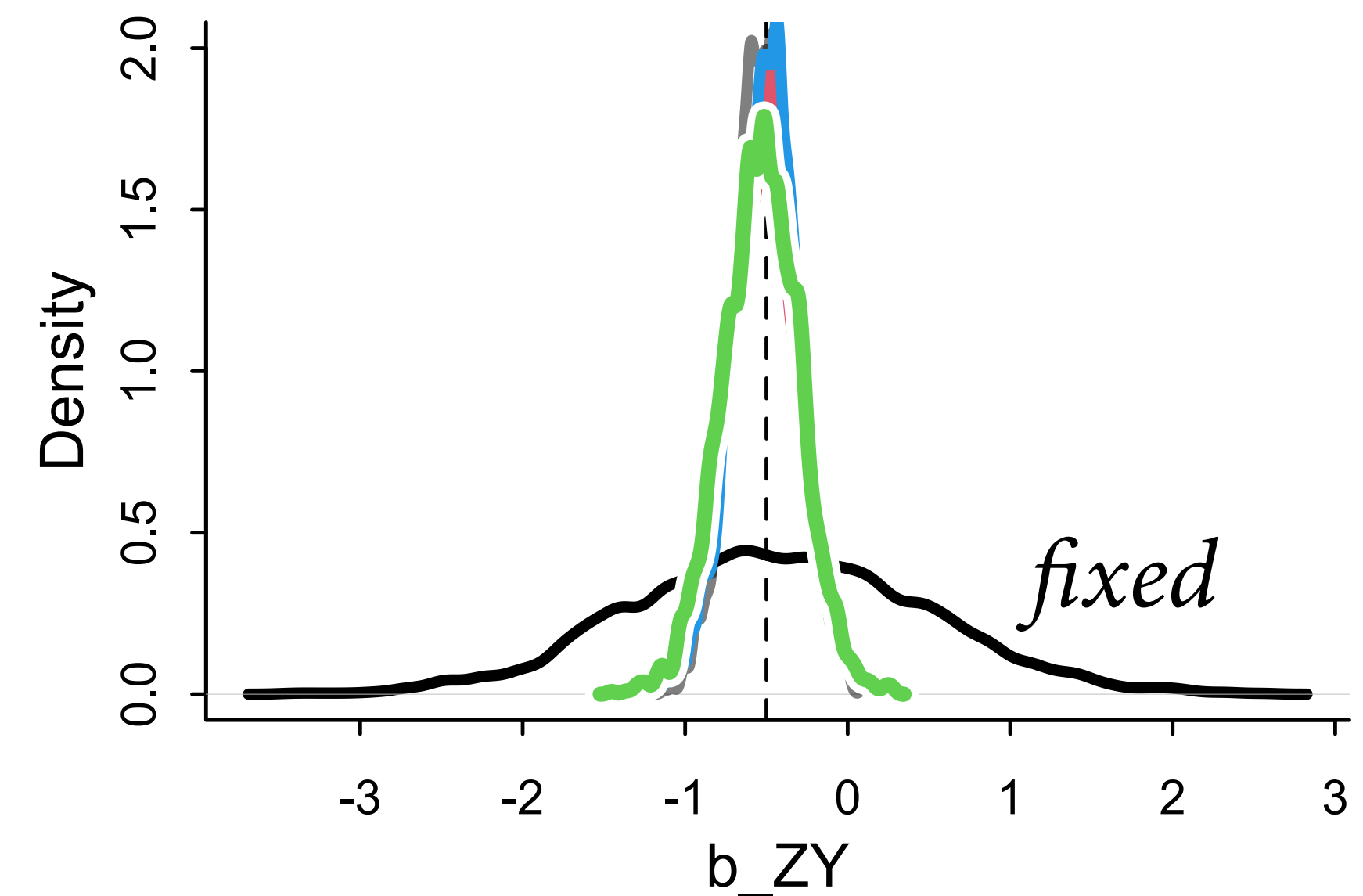
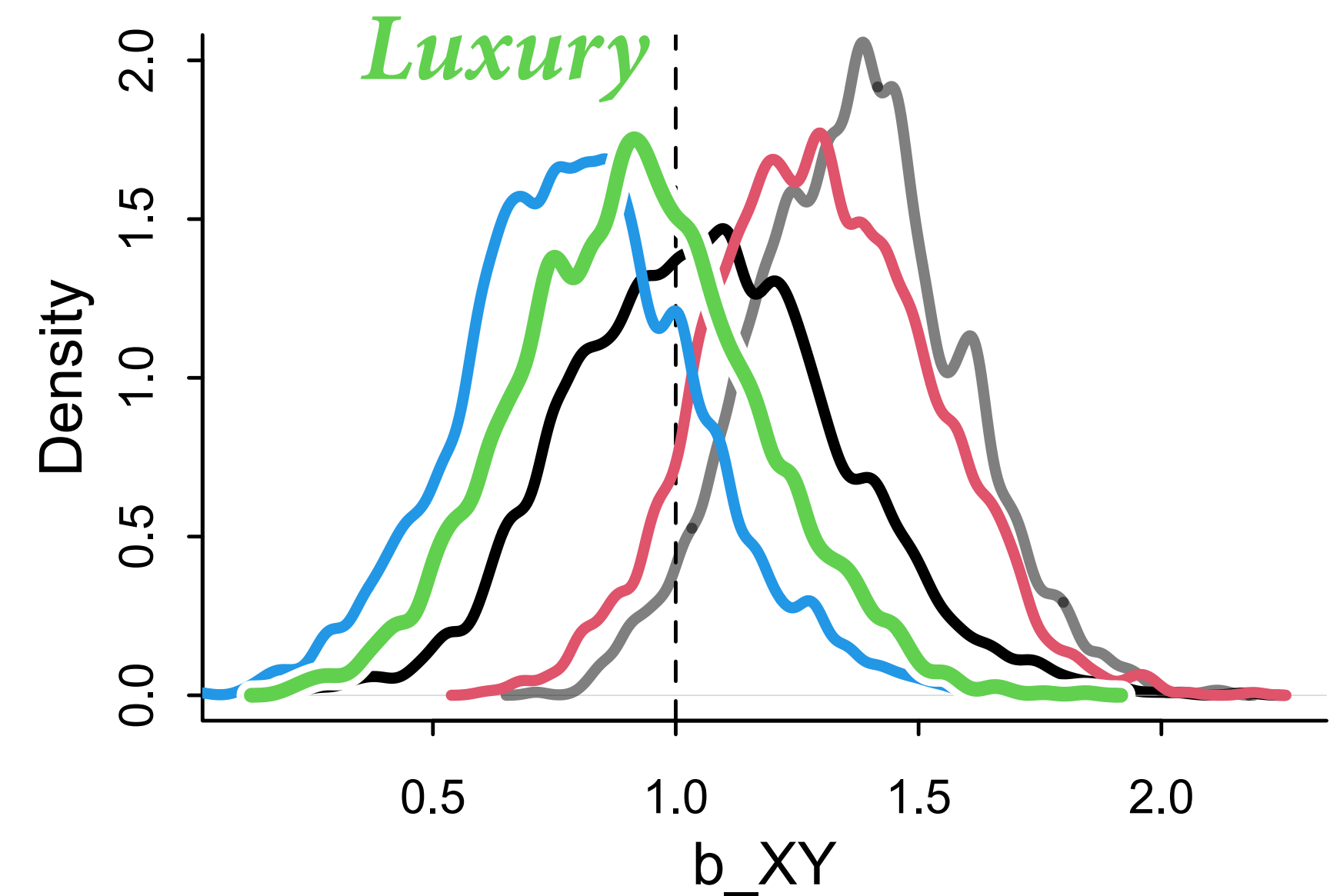
```

# The Latent Mundlak Machine
mru <- ulam(
  alist(
    # Y model
    Y ~ bernoulli(p),
    logit(p) <- a[g] + bxy*X + bzy*Z[g] + buy*u[g],
    transpars> vector[Ng]:a <-< abar + z*tau,

    # X model
    X ~ normal(mu,sigma),
    mu <- aX + bux*u[g],
    vector[Ng]:u ~ normal(0,1),

    # priors
    z[g] ~ dnorm(0,1),
    c(aX,bxy,buy,bzy) ~ dnorm(0,1),
    bux ~ dexp(1),
    abar ~ dnorm(0,1),
    tau ~ dexp(1),
    sigma ~ dexp(1)
  ) , data=dat , chains=4 , cores=4 , sample=TRUE )

```



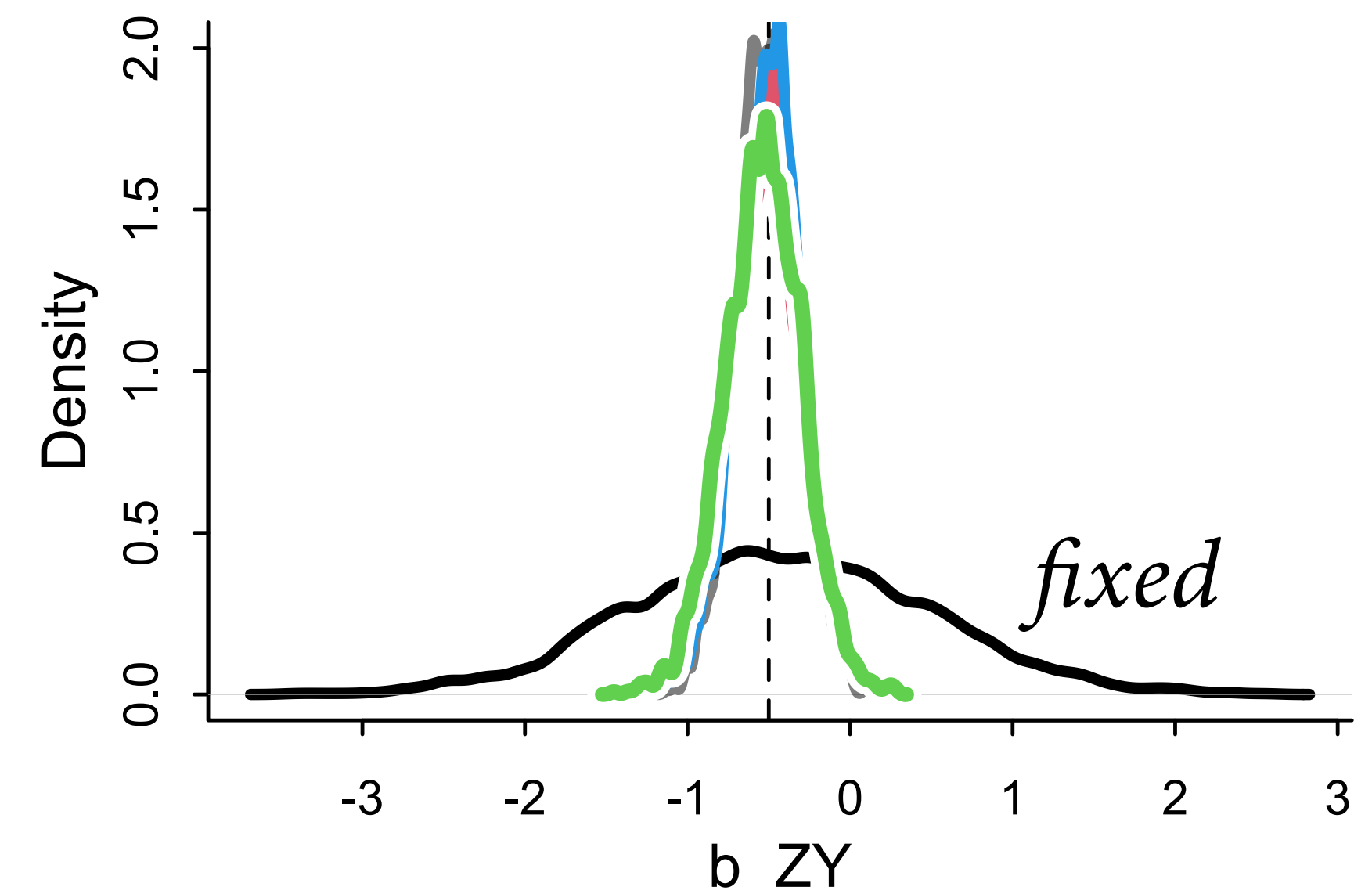
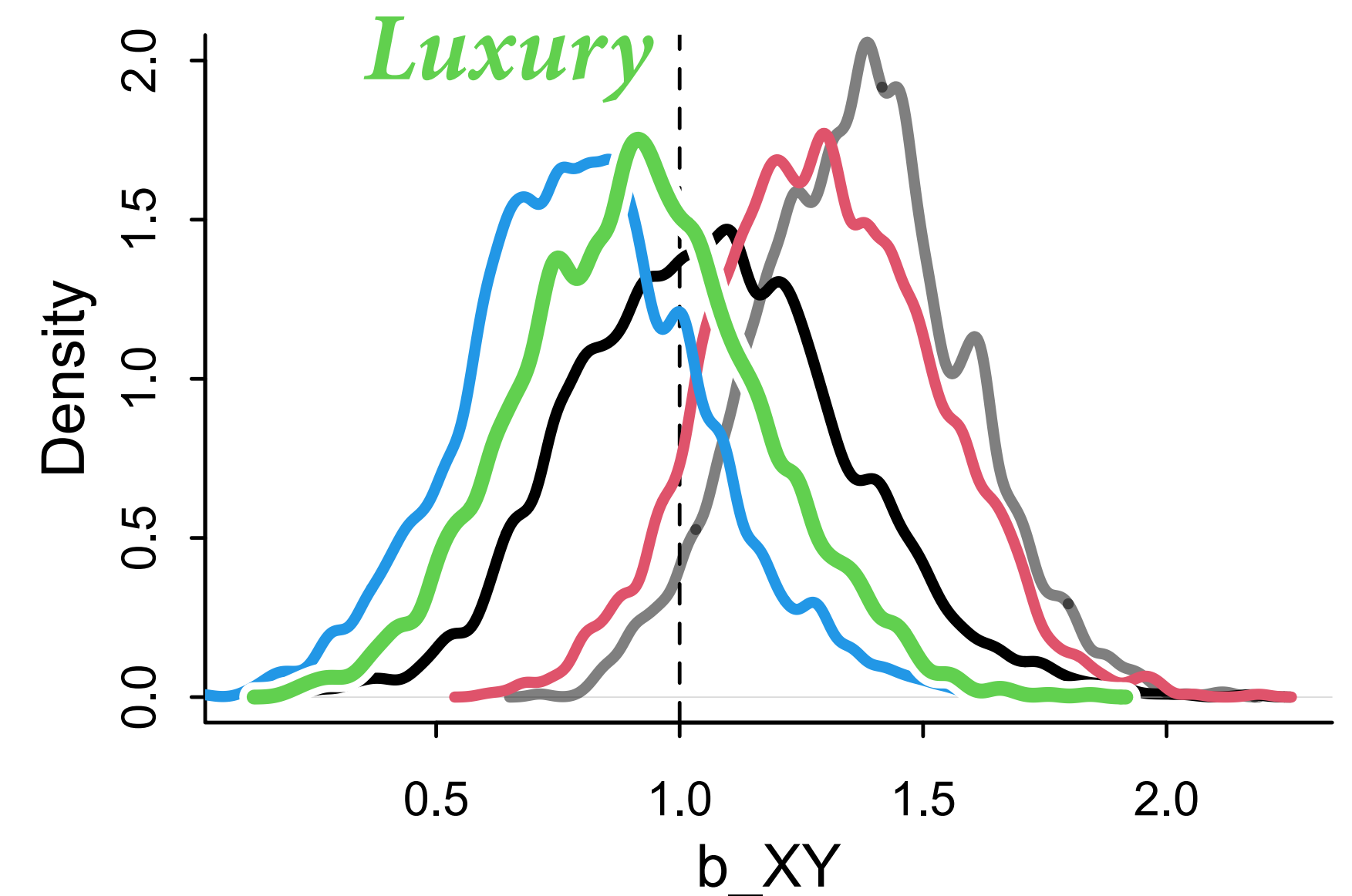
Random confounds

Should you use fixed effects?

Should you include average X ?

Use a generative model, model the confound

Confounds also vary at individual level — no single solution



What Motivates Sharing?

data(KosterLeckie)

Year of food transfers among 25 households
in Arang Dak

$25!/(2!(25-2)!) = 300$ dyads

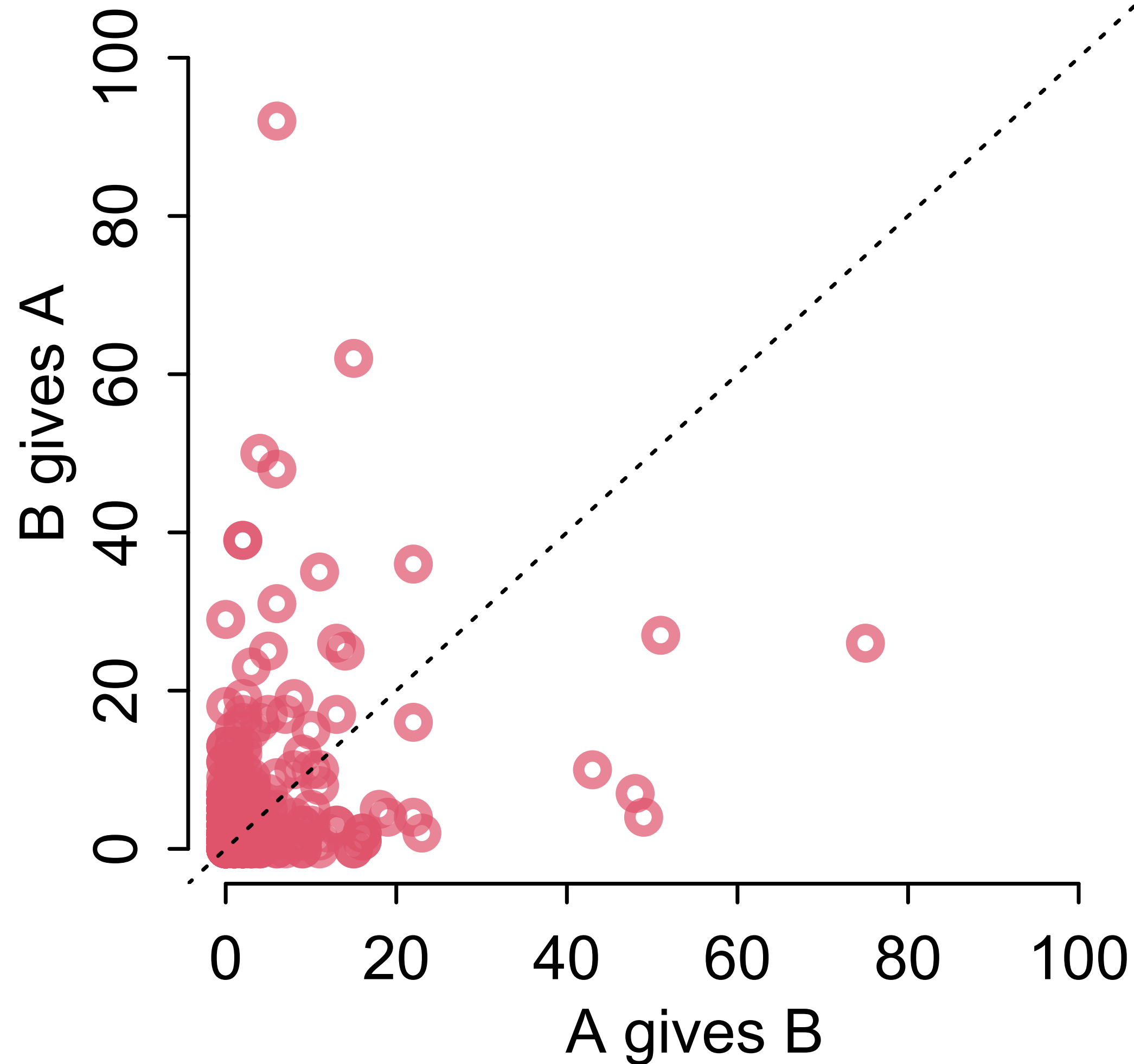
2871 observed transfers between households

How much sharing explained by reciprocity?
How much by generalized giving?



What Motivates Sharing?

A simple correlation of counts is not a good measure of reciprocity

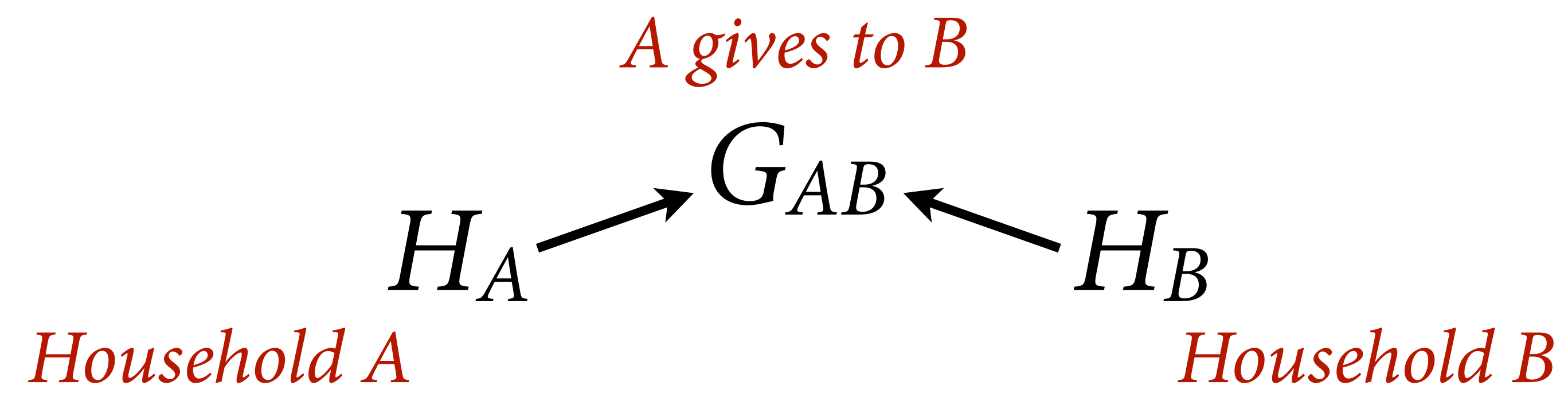


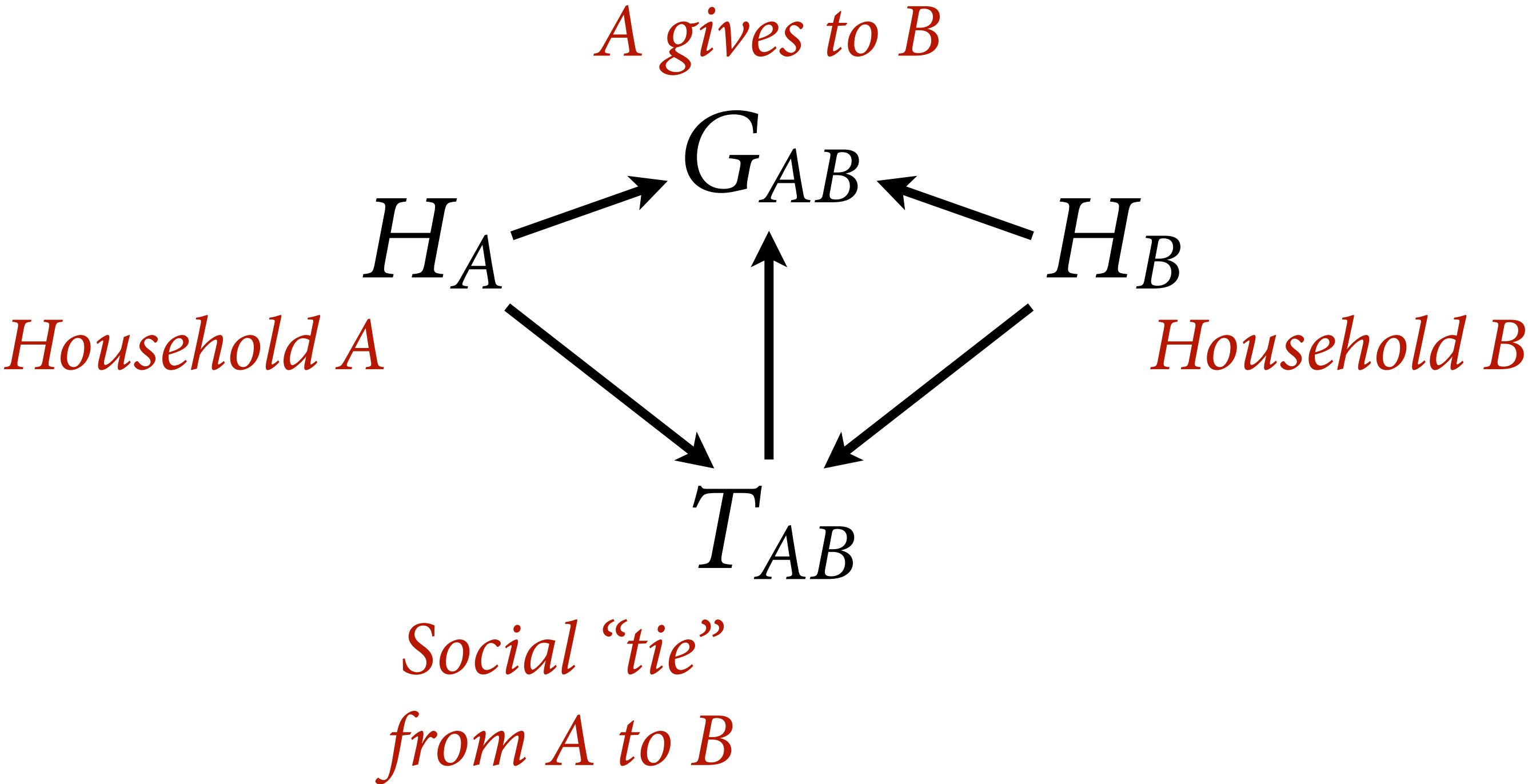
How to draw an owl

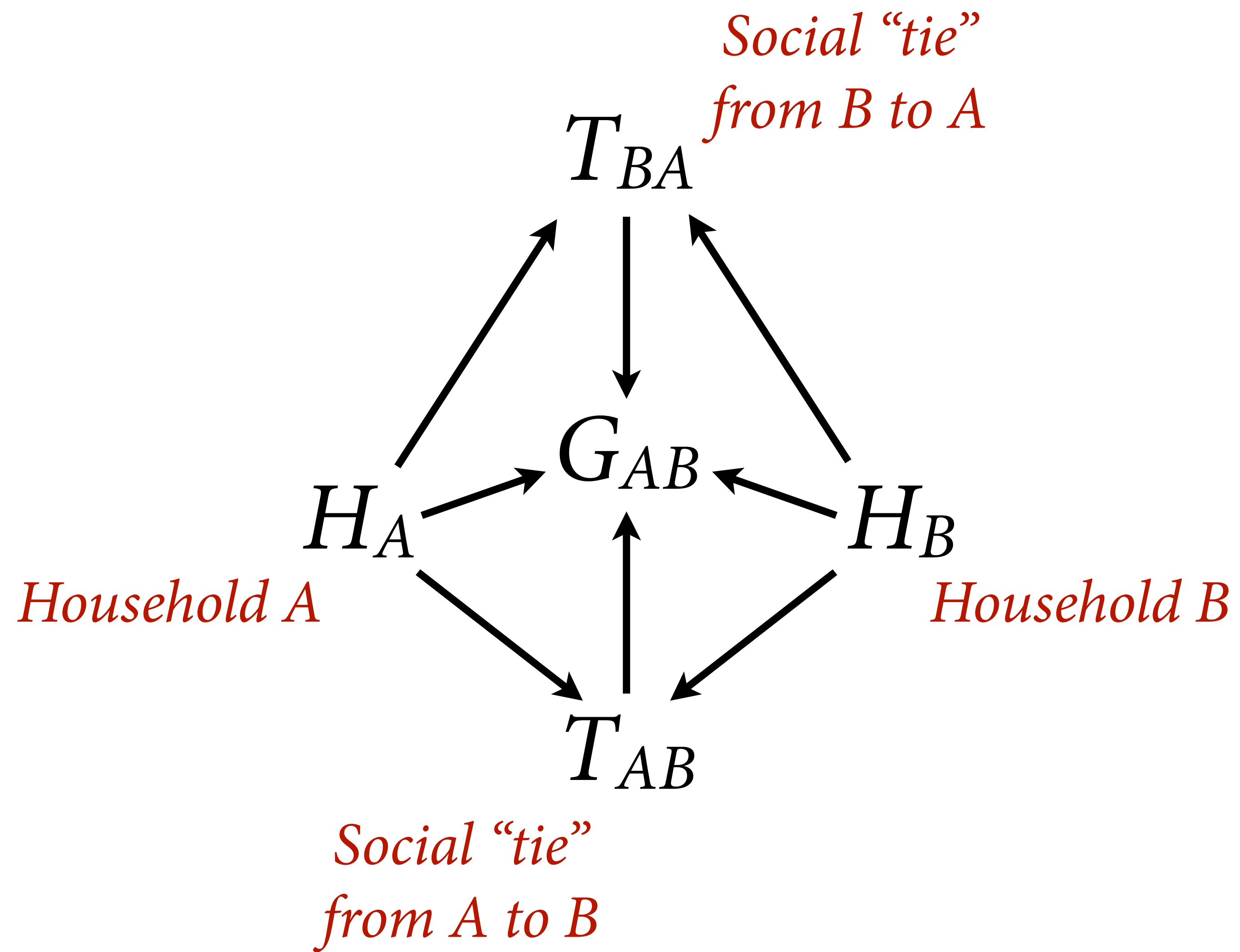
1.

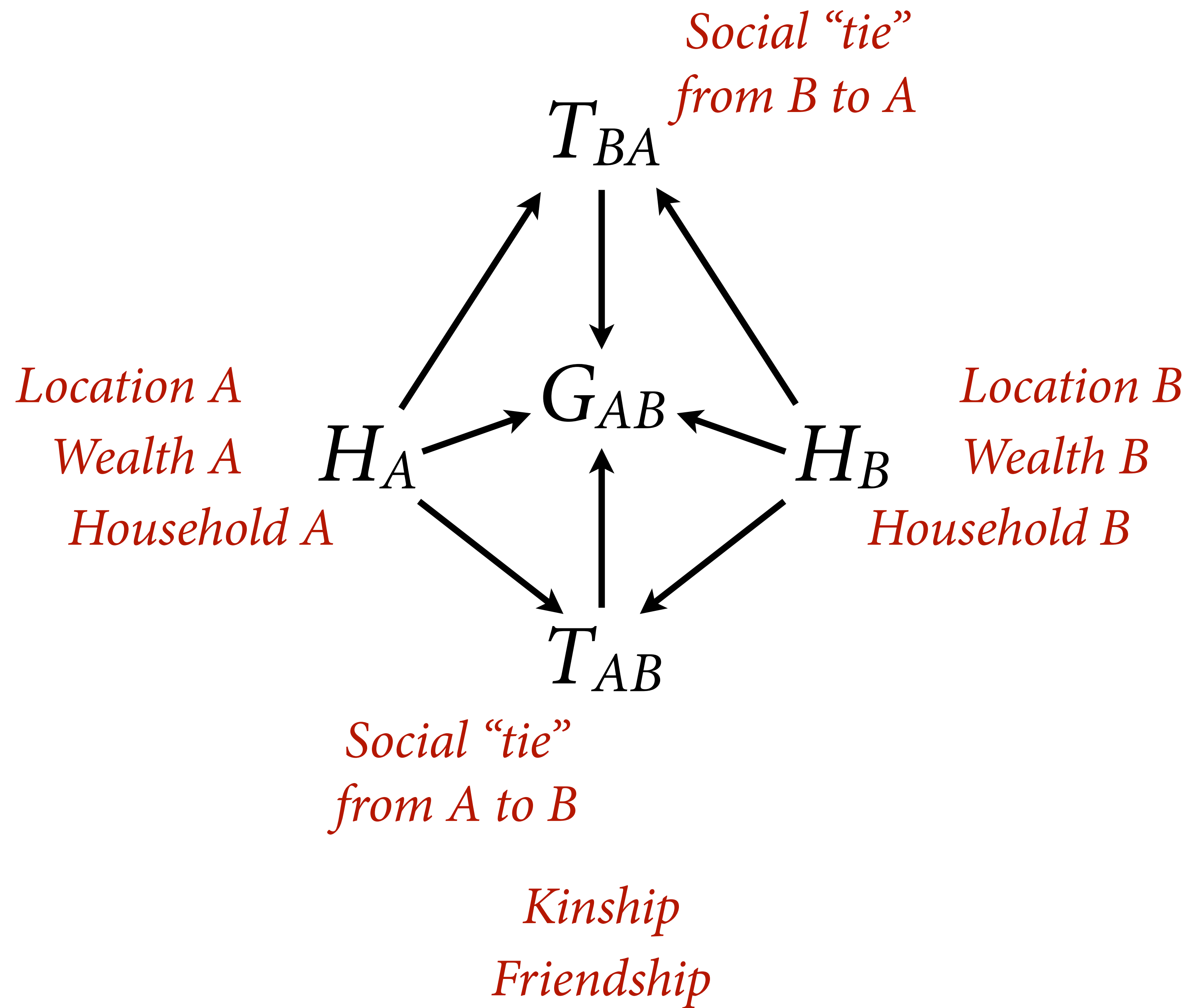


1. Draw some circles









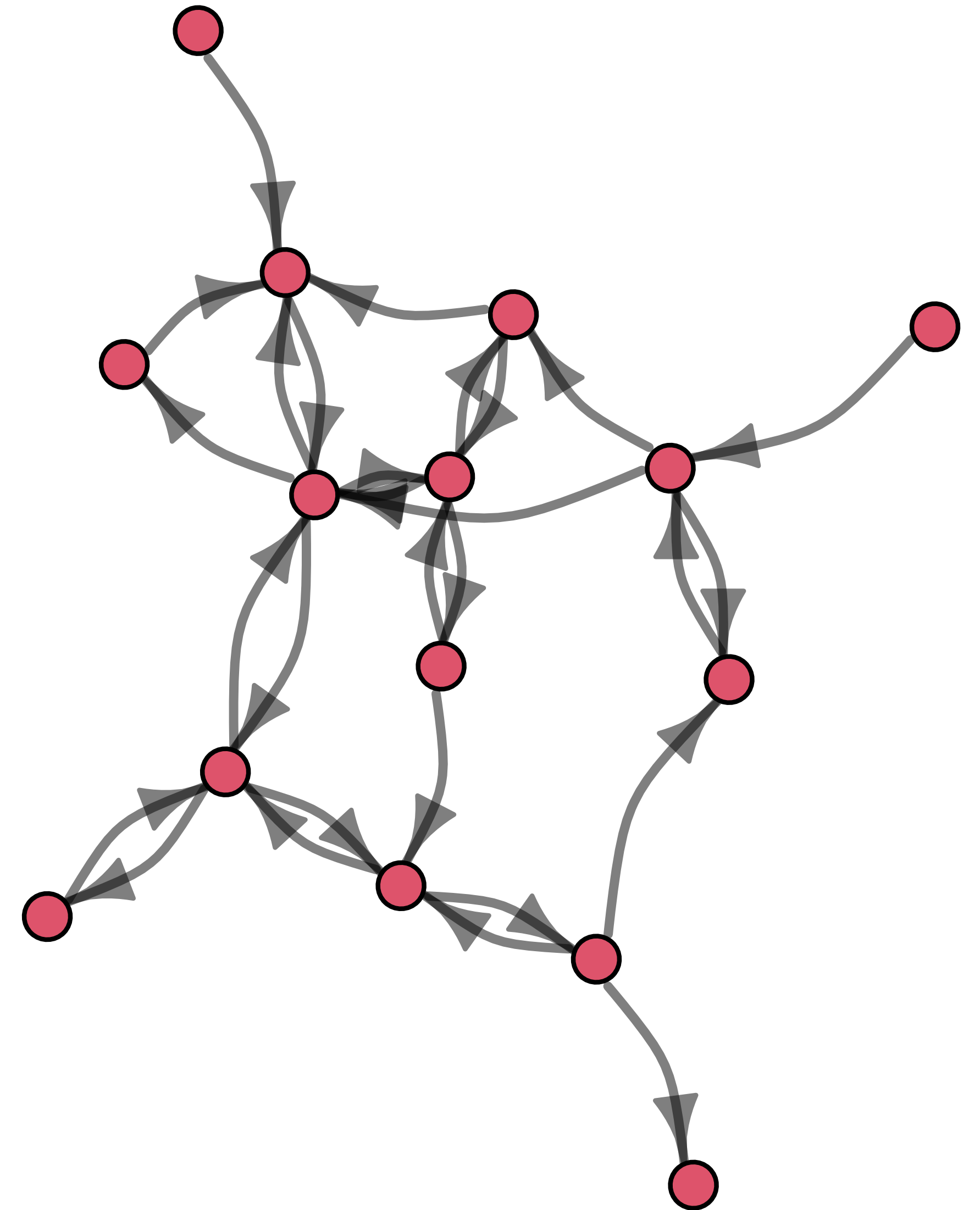
What Motivates Sharing?

T_{AB} and T_{BA} are not observable

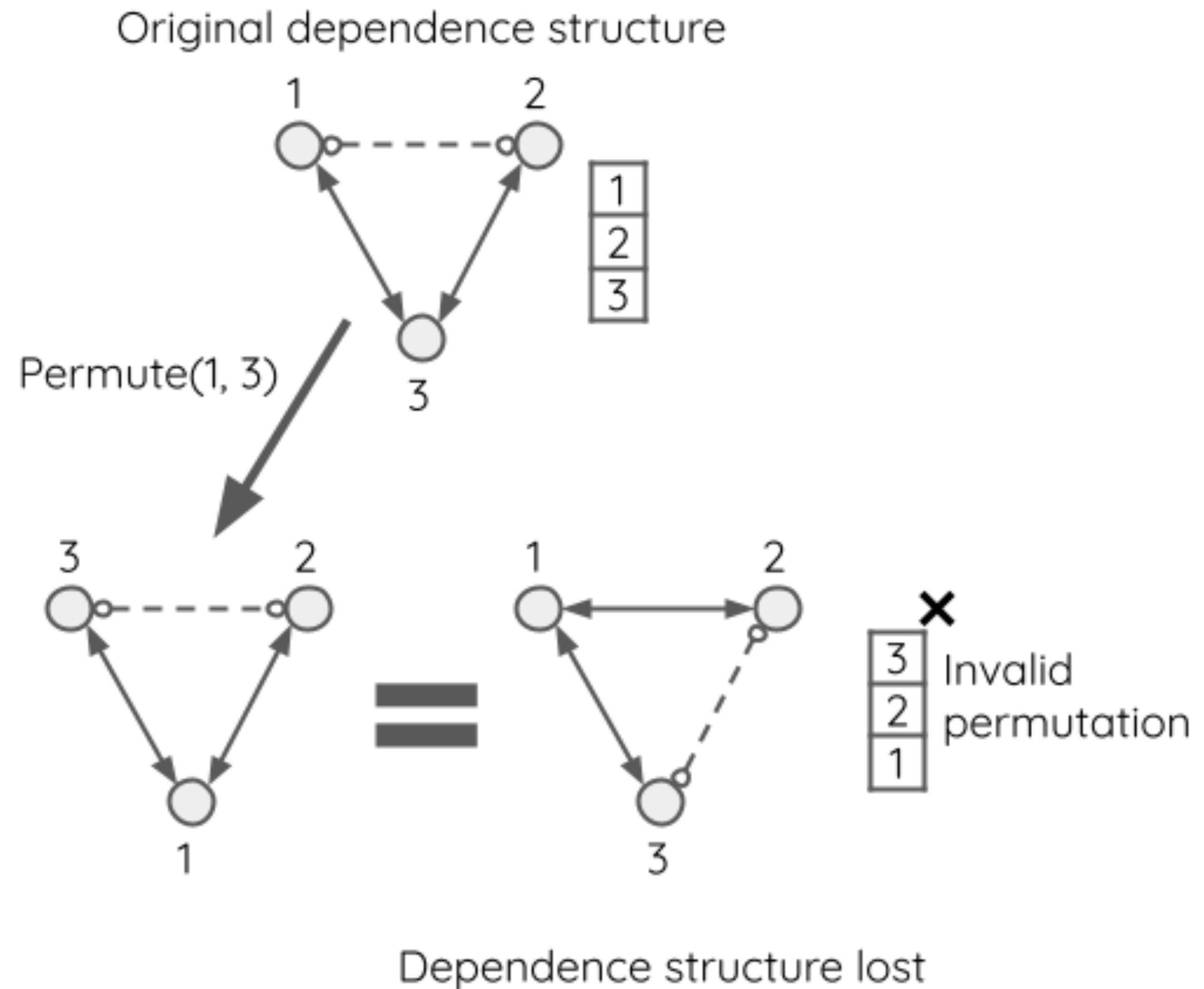
Social network: Pattern of directed exchange

Social networks are **abstractions**,
are **not data**

What is a principled approach?



Resist Adhockery



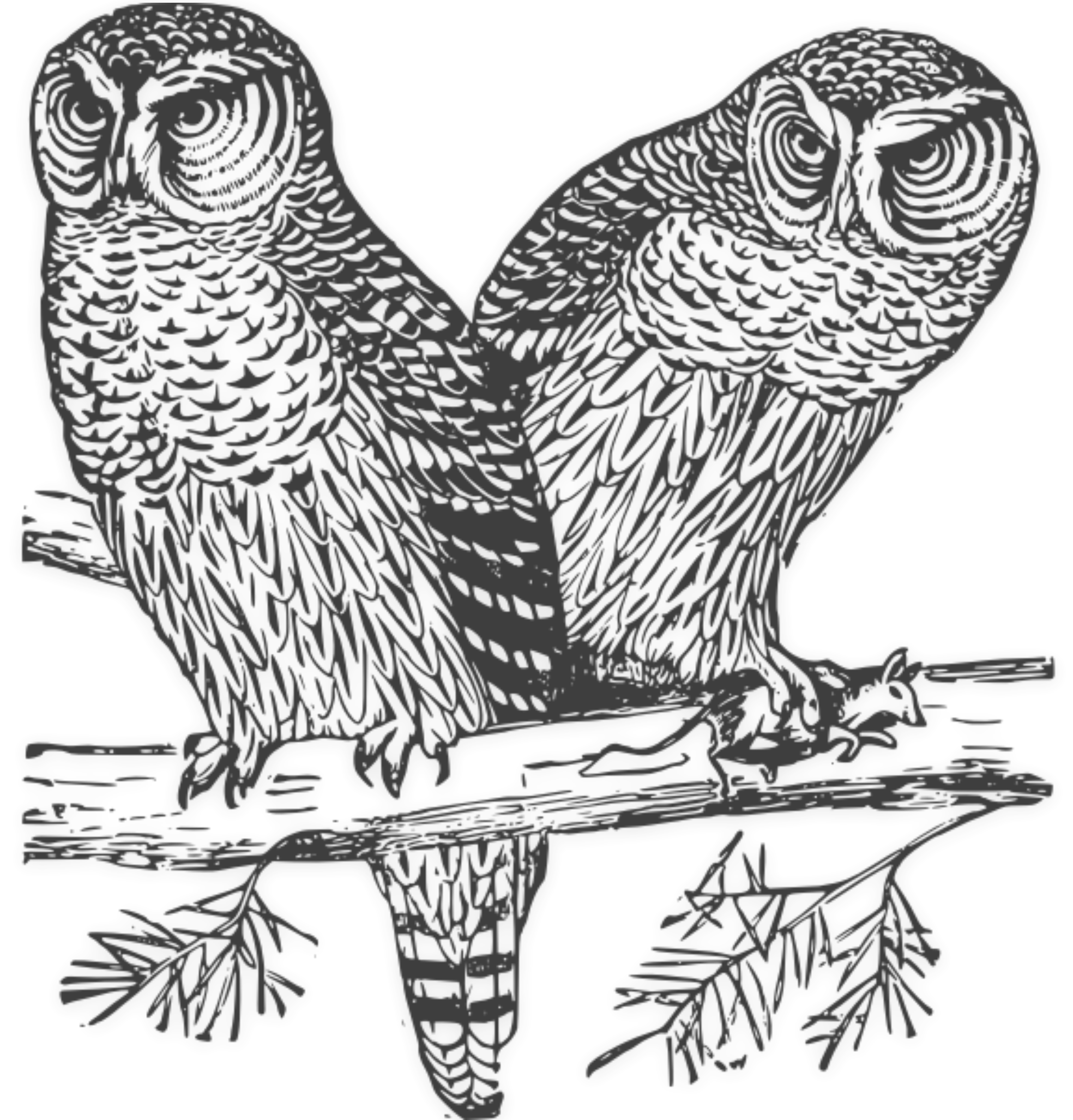
Drawing the Social Owl

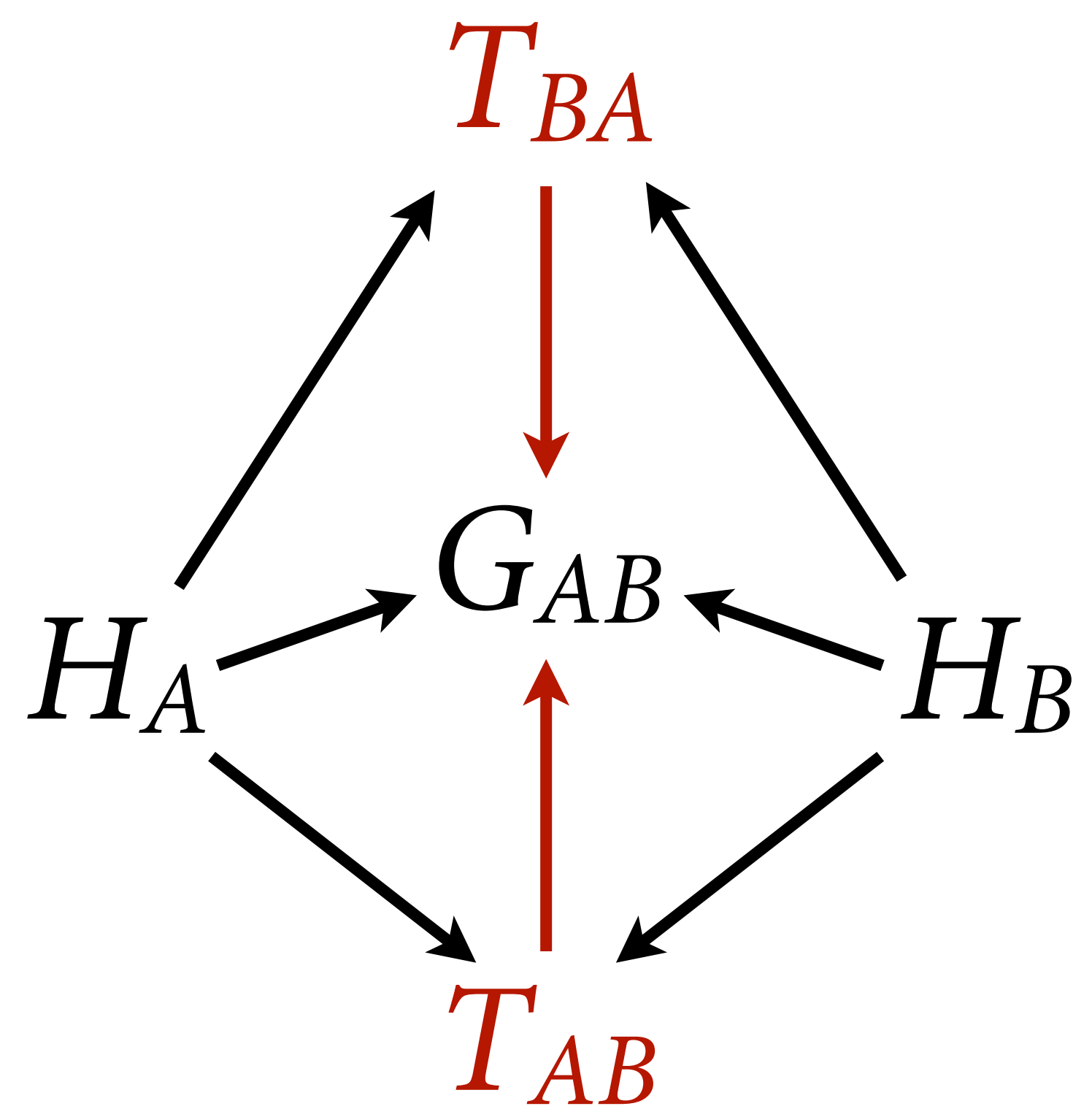
(1) Estimand: Reciprocity & what explains it

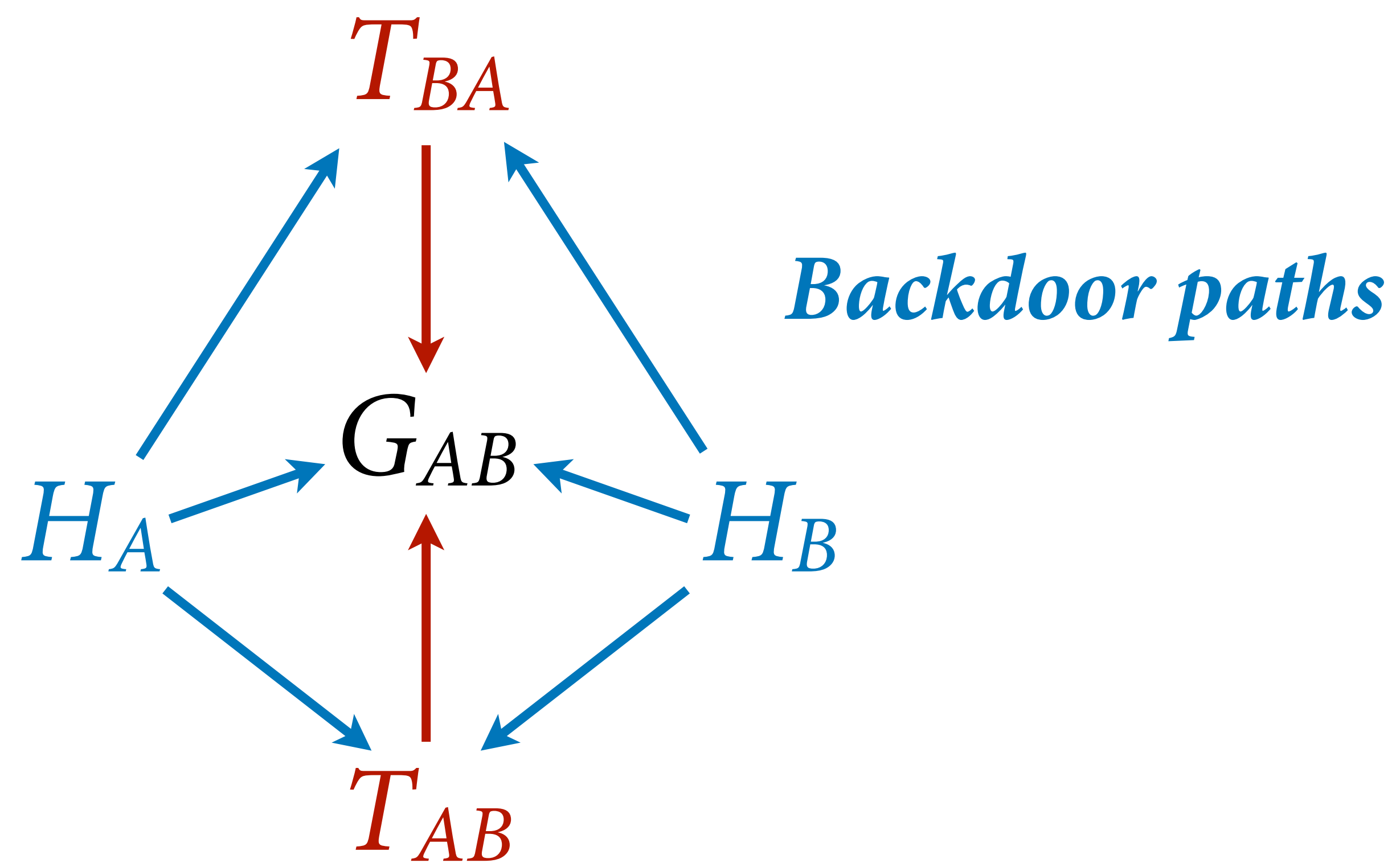
(2) Generative model

(3) Statistical model

(4) Analyze sample







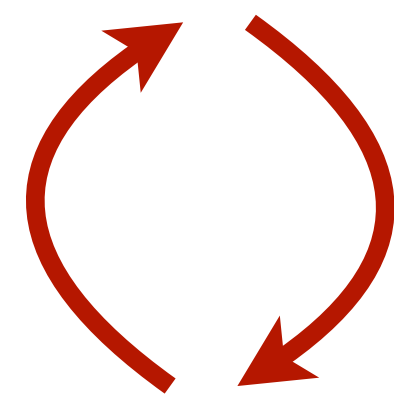
Drawing the Social Owl

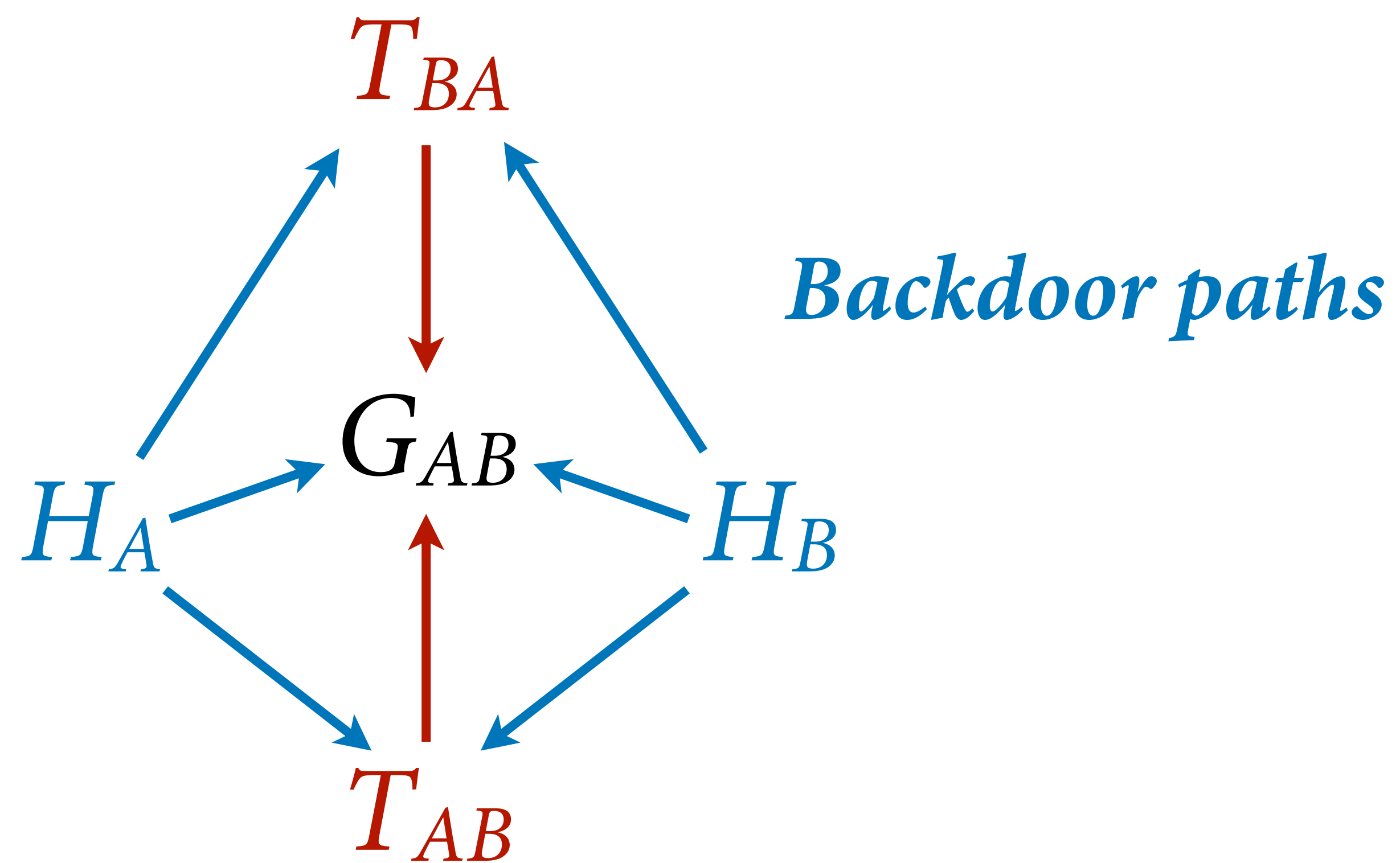
(1) Estimand: Reciprocity & what explains it

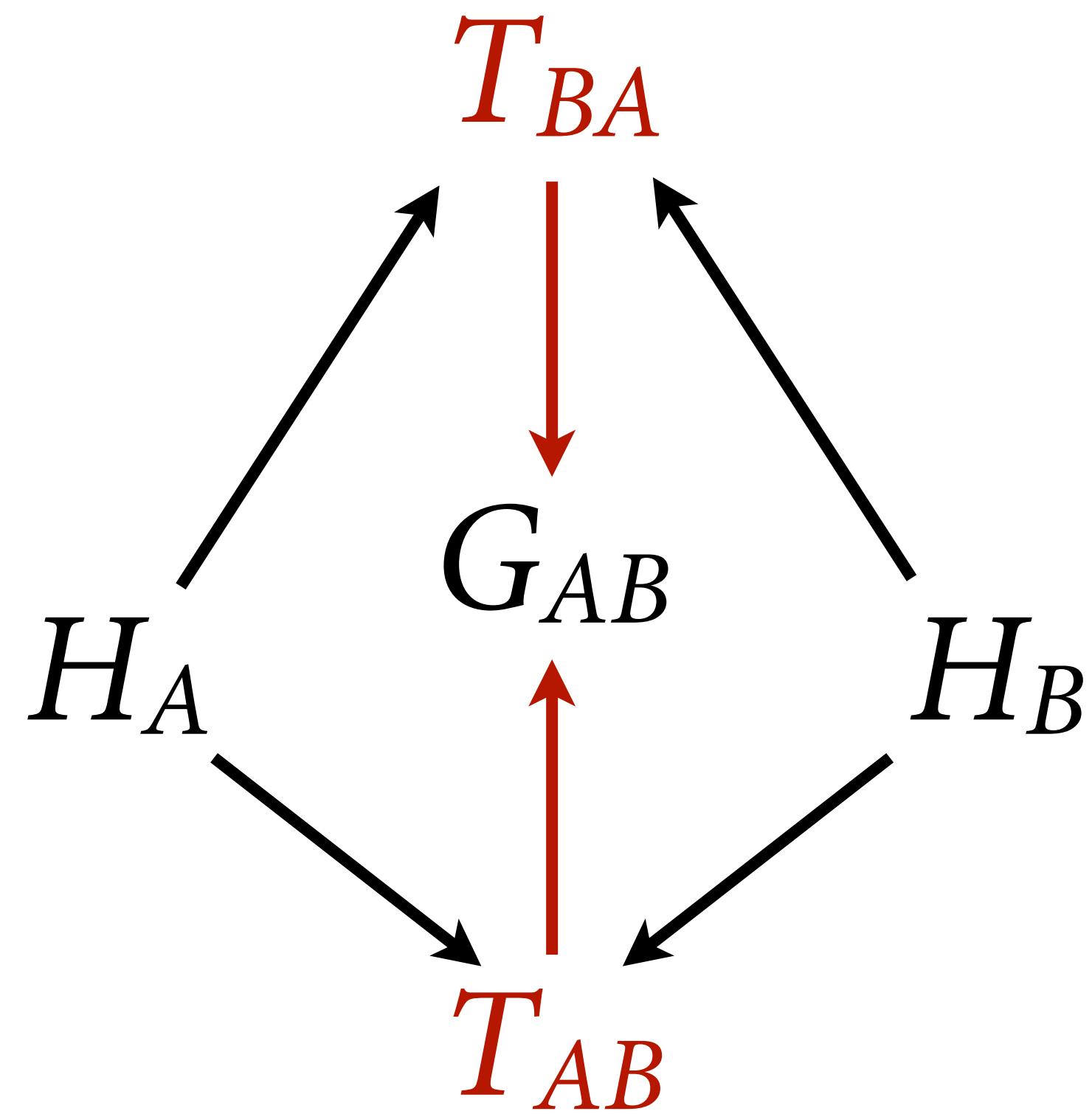
(2) Generative model

(3) Statistical model

(4) Analyze sample







Begin simple, model tie formation

```
# N households  
N <- 25  
dyads <- t(combn(N,2))  
N_dyads <- nrow(dyads)
```

```
# N households
N <- 25
dyads <- t(combn(N,2))
N_dyads <- nrow(dyads)
```

```
> dyads
      [,1] [,2]
[1,]    1    2
[2,]    1    3
[3,]    1    4
[4,]    1    5
[5,]    1    6
[6,]    1    7
[7,]    1    8
[8,]    1    9
[9,]    1   10
[10,]   1   11
[11,]   1   12
[12,]   1   13
[13,]   1   14
[14,]   1   15
[15,]   1   16
[16,]   1   17
[17,]   1   18
[18,]   1   19
[19,]   1   20
[20,]   1   21
[21,]   1   22
[22,]   1   23
[23,]   1   24
[24,]   1   25
[25,]   2    3
[26,]   2    4
[27,]   2    5
[28,]   2    6
[29,]   2    7
```

```
[30,]    2    8
[31,]    2    9
[32,]    2   10
[33,]    2   11
[34,]    2   12
[35,]    2   13
[36,]    2   14
[37,]    2   15
[38,]    2   16
[39,]    2   17
[40,]    2   18
[41,]    2   19
[42,]    2   20
[43,]    2   21
[44,]    2   22
[45,]    2   23
[46,]    2   24
[47,]    2   25
[48,]    3    4
[49,]    3    5
[50,]    3    6
[51,]    3    7
[52,]    3    8
[53,]    3    9
[54,]    3   10
[55,]    3   11
[56,]    3   12
[57,]    3   13
[58,]    3   14
[59,]    3   15
[60,]    3   16
```

```
[61,]    3   17
[62,]    3   18
[63,]    3   19
[64,]    3   20
[65,]    3   21
[66,]    3   22
[67,]    3   23
[68,]    3   24
[69,]    3   25
[70,]    4    5
[71,]    4    6
[72,]    4    7
[73,]    4    8
[74,]    4    9
[75,]    4   10
[76,]    4   11
[77,]    4   12
[78,]    4   13
[79,]    4   14
[80,]    4   15
[81,]    4   16
[82,]    4   17
[83,]    4   18
[84,]    4   19
[85,]    4   20
[86,]    4   21
[87,]    4   22
[88,]    4   23
[89,]    4   24
[90,]    4   25
[91,]    5    6
```

```
# N households
N <- 25
dyads <- t(combn(N,2))
N_dyads <- nrow(dyads)

# simulate "friendships" in which ties are shared
f <- rbern(N_dyads,0.1) # 10% of dyads are friends
```

```
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# now simulate directed ties for all individuals
# there can be ties that are not reciprocal
alpha <- (-3) # base rate of ties; -3 ~= 0.05
y <- matrix(NA,N,N) # matrix of ties
```

```

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# now simulate directed ties for all individuals
# there can be ties that are not reciprocal
alpha <- (-3) # base rate of ties; -3 ~= 0.05
y <- matrix(NA,N,N) # matrix of ties
for ( i in 1:N ) for ( j in 1:N ) {
  if ( i != j ) {
    # directed tie from i to j
    ids <- sort( c(i,j) )
    the_dyad <- which( dyads[,1]==ids[1] & dyads[,2]==ids[2] )
    p_tie <- f[the_dyad] + (1-f[the_dyad])*inv_logit( alpha )
    y[i,j] <- rbern( 1 , p_tie )
  }
}#ij

```

```

# N households
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    # directed tie from i to j
    ids <- sort( c(i,j) )
    the_dyad <- which( dyads[,1]==ids[1] & dyads[,2]==ids[2] )
    p_tie <- f[the_dyad] + (1-f[the_dyad])*inv_logit( alpha )
    y[i,j] <- rbern( 1 , p_tie )
  }
} #ij

```

*friends
share ties*

```
# now simulate gifts
giftsAB <- rep(NA,N_dyads)
giftsBA <- rep(NA,N_dyads)
lambda <- log(c(0.5,2)) # rates of giving for y=0,y=1
for ( i in 1:N_dyads ) {
  A <- dyads[i,1]
  B <- dyads[i,2]
  giftsAB[i] <- rpois( 1 , exp( lambda[1+y[A,B]] ) )
  giftsBA[i] <- rpois( 1 , exp( lambda[1+y[B,A]] ) )
}
```

```

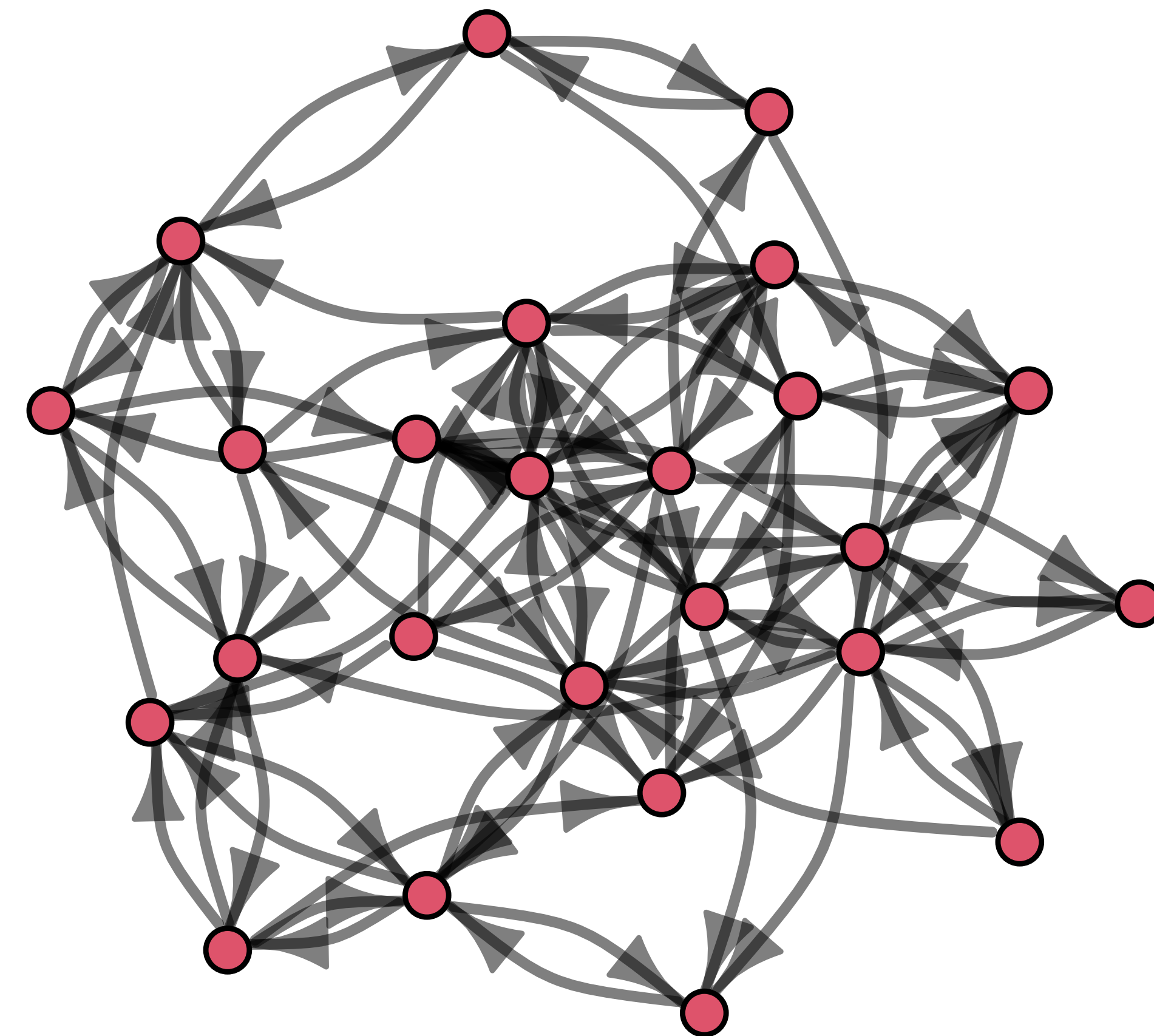
# now simulate gifts
giftsAB <- rep(NA,N_dyads)
giftsBA <- rep(NA,N_dyads)
lambda <- log(c(0.5,2)) # rates of giving for y=0,y=1
for ( i in 1:N_dyads ) {
  A <- dyads[i,1]
  B <- dyads[i,2]
  giftsAB[i] <- rpois( 1 , exp( lambda[1+y[A,B]] ) )
  giftsBA[i] <- rpois( 1 , exp( lambda[1+y[B,A]] ) )
}

```

```

# draw network
library(igraph)
sng <- graph_from_adjacency_matrix(y)
lx <- layout_nicely(sng)
vcol <- "#DE536B"
plot(sng , layout=lx , vertex.size=8 ,
edge.arrow.size=0.75 , edge.width=2 ,
edge.curved=0.35 , vertex.color=vcol ,
edge.color=grau() , asp=0.9 , margin = -0.05 ,
vertex.label=NA )

```



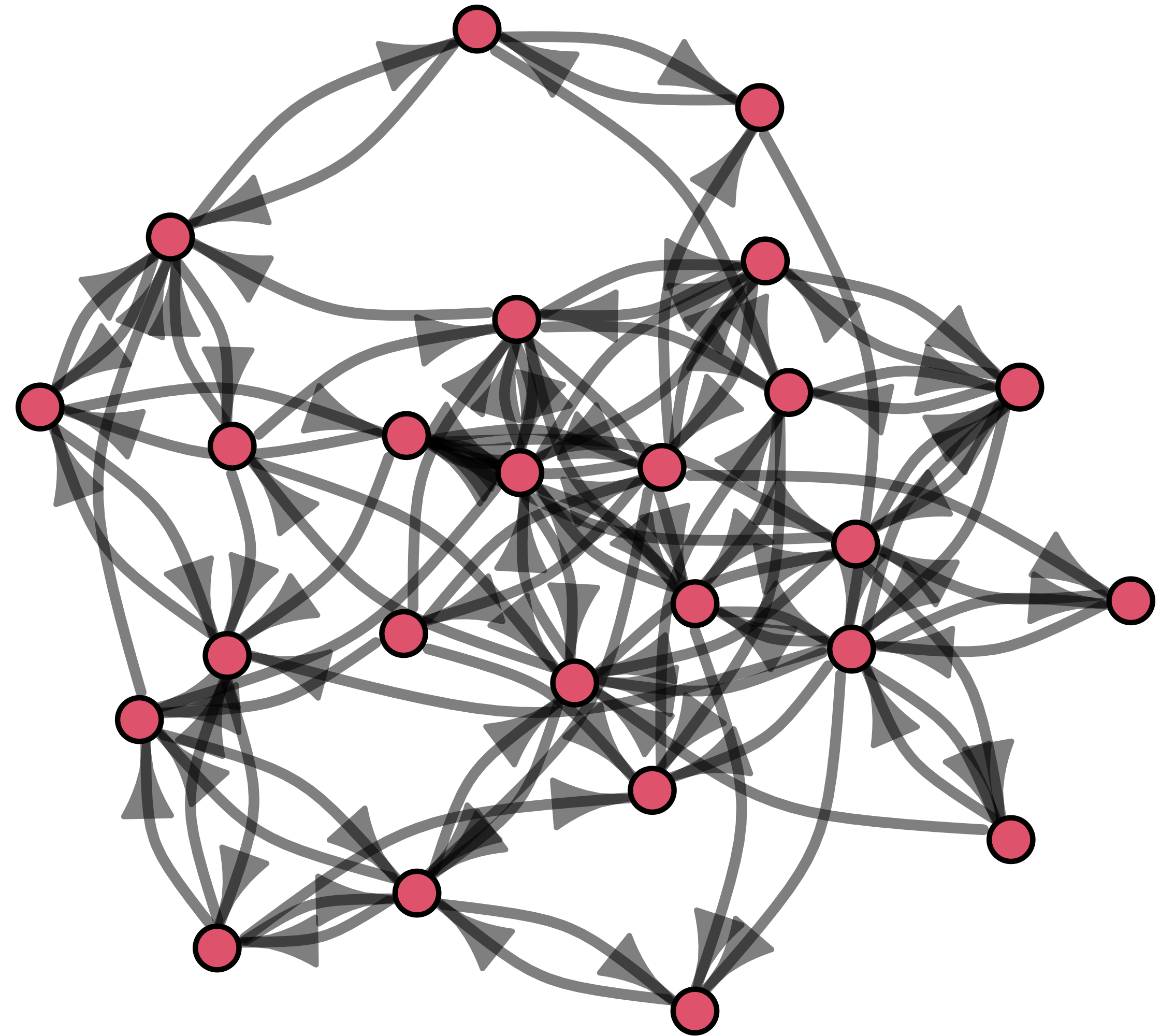
Drawing the Social Owl

(1) Estimand: Reciprocity & what explains it

(2) Generative model

(3) Statistical model

(4) Analyze sample

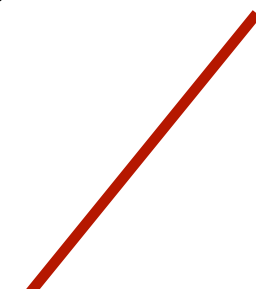


Gifts A to B

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

average



Tie A to B



Gifts A to B

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

average

Tie A to B

Gifts B to A

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

Tie B to A

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

The AB dyad $\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$

*covariance
within dyads*

*variance
among ties*

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

*partial
pooling for
network ties*

$$\left[\begin{array}{l} \begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right) \\ \rho \sim \text{LKJCorr}(2) \\ \sigma \sim \text{Exponential}(1) \\ \alpha \sim \text{Normal}(0,1) \end{array} \right.$$

```

# dyad model
f_dyad <- alist(
  GAB ~ poisson( lambdaAB ),
  GBA ~ poisson( lambdaBA ),
  log(lambdaAB) <- a + T[D,1] ,
  log(lambdaBA) <- a + T[D,2] ,
  a ~ normal(0,1),

  ## dyad effects
  transpars> matrix[N_dyads,2]:T <-
    compose_noncentered( rep_vector(sigma_T,2) , L_Rho_T , Z ),
  matrix[2,N_dyads]:Z ~ normal( 0 , 1 ),
  cholesky_factor_corr[2]:L_Rho_T ~ lkj_corr_cholesky( 2 ),
  sigma_T ~ exponential(1),

  ## compute correlation matrix for dyads
  gq> matrix[2,2]:Rho_T <-<- Chol_to_Corr( L_Rho_T )
)

mGD <- ulam( f_dyad , data=sim_data , chains=4 , cores=4 , iter=2000 )

```

```
# dyad model
```

```
f_dyad <- alist(  
  GAB ~ poisson( lambdaAB ),  
  GBA ~ poisson( lambdaBA ),  
  log(lambdaAB) <- a + T[D,1] ,  
  log(lambdaBA) <- a + T[D,2] ,  
  a ~ normal(0,1),
```

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

```
## dyad effects
```

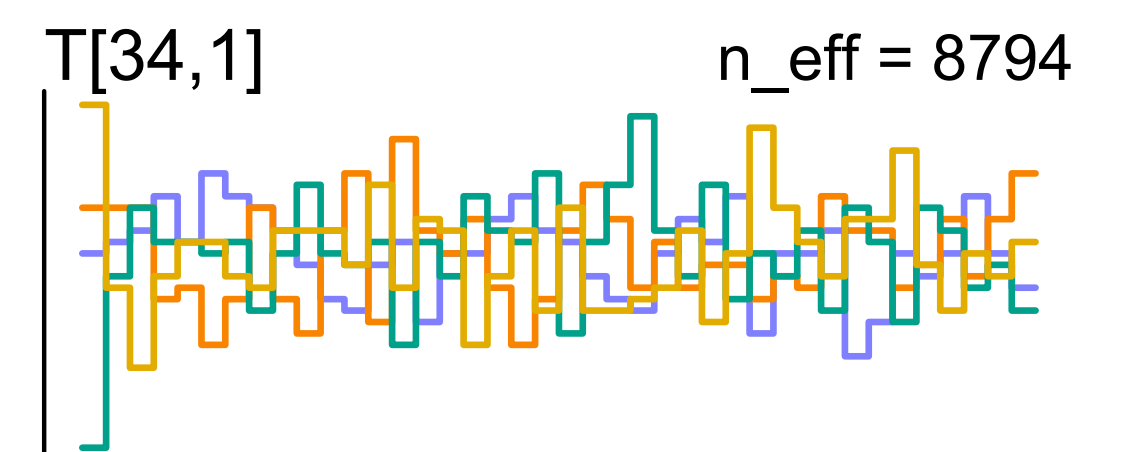
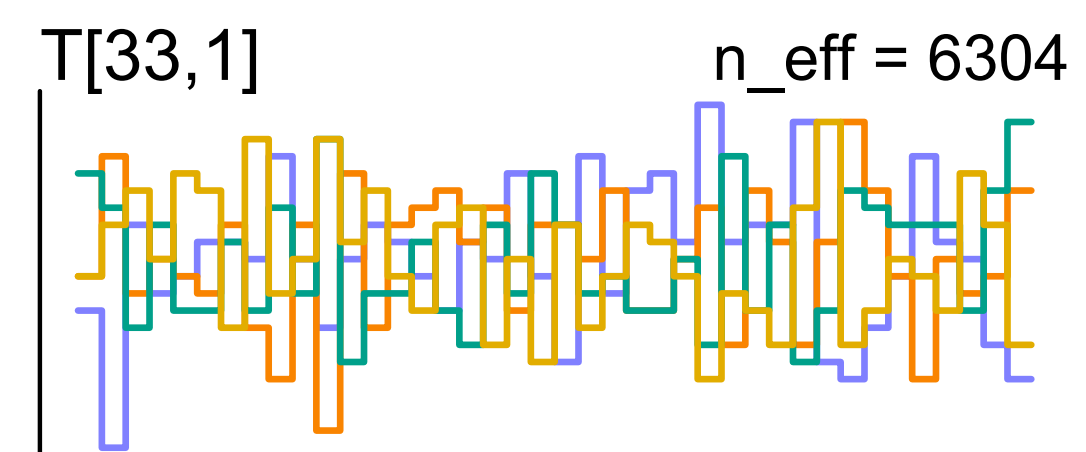
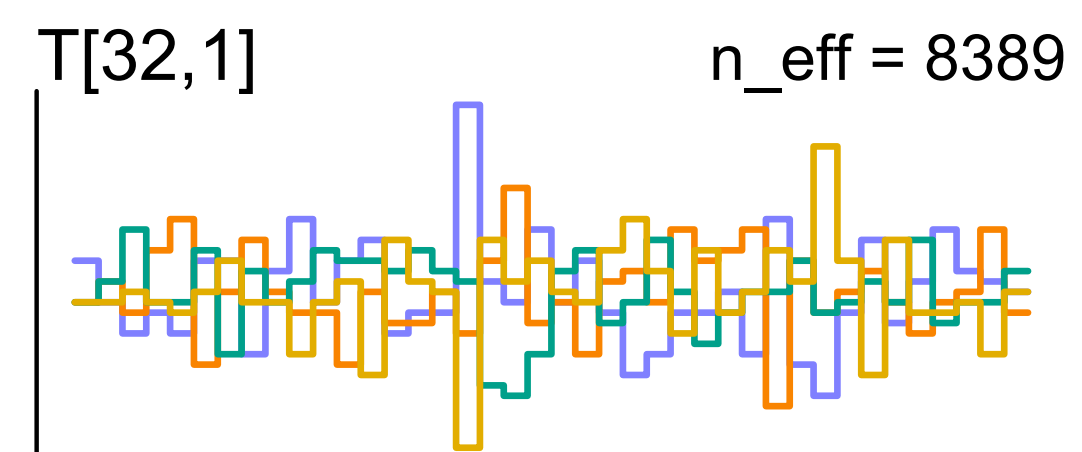
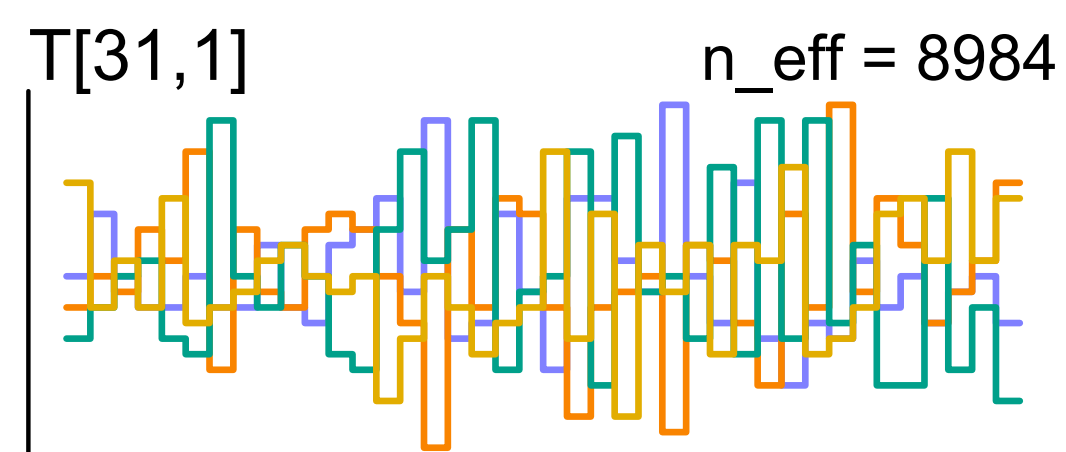
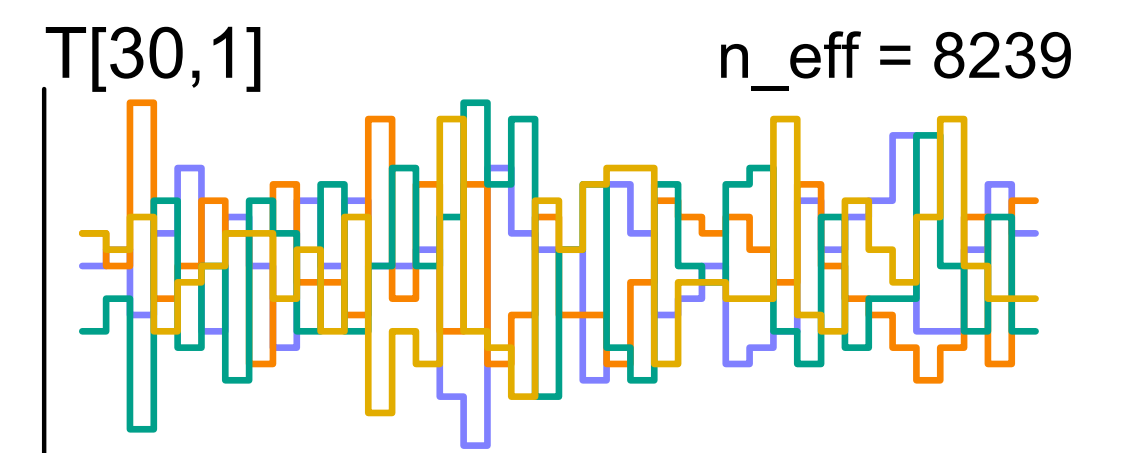
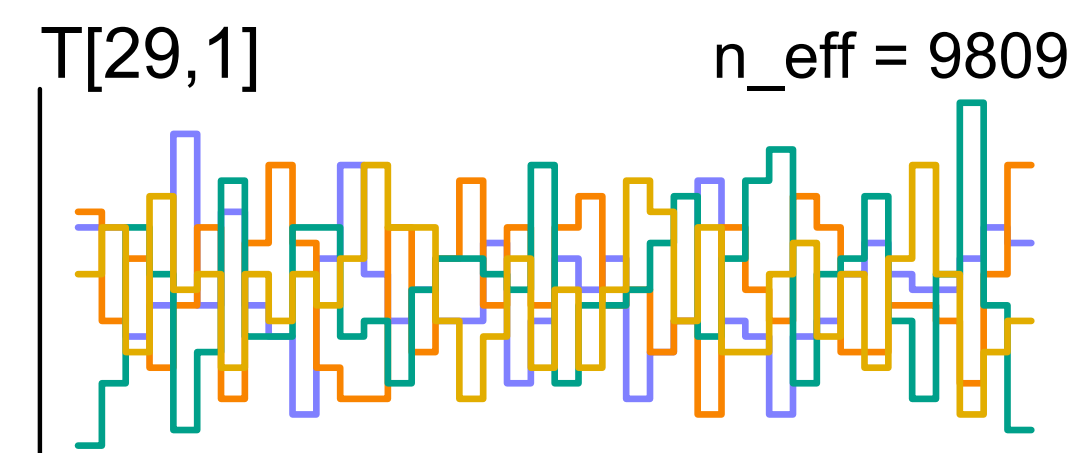
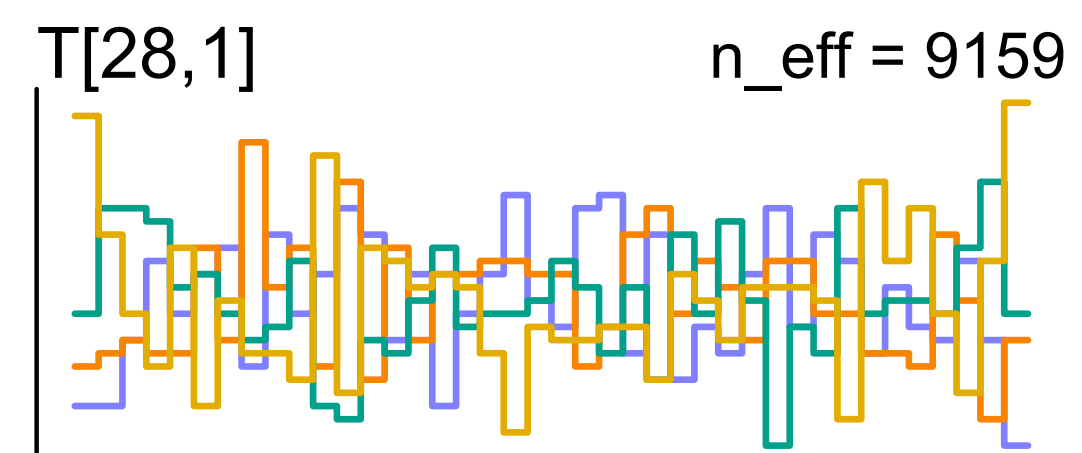
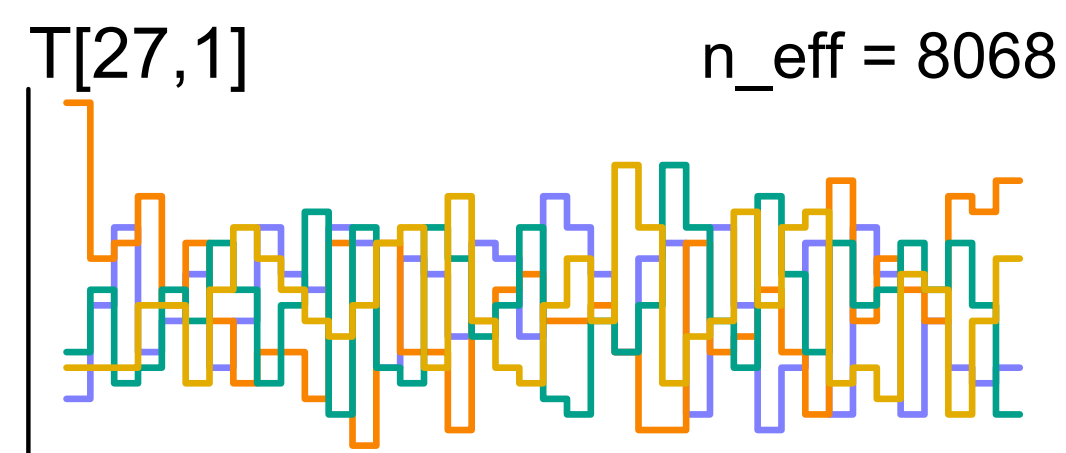
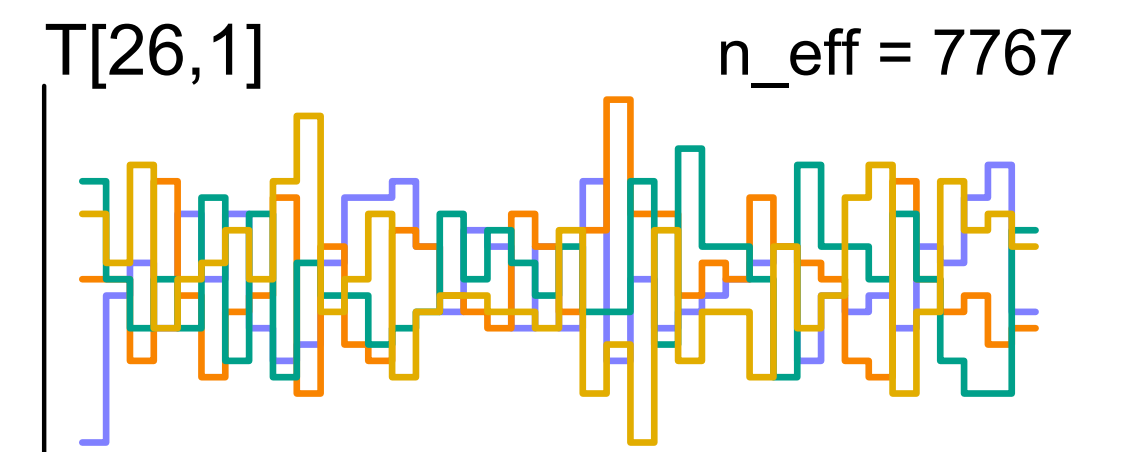
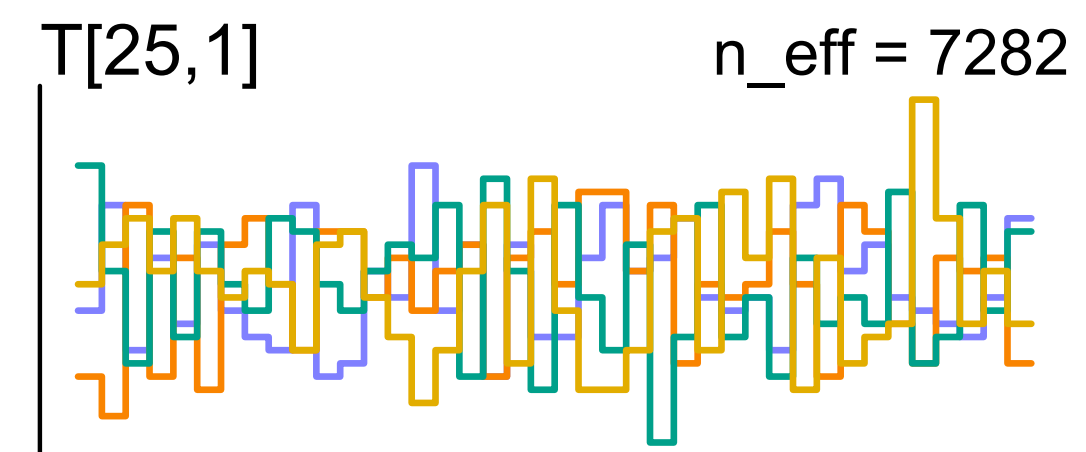
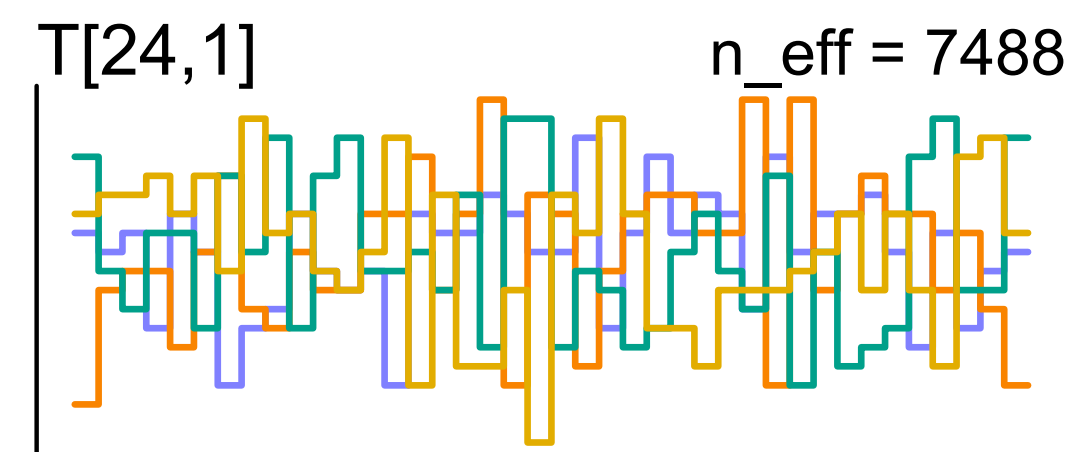
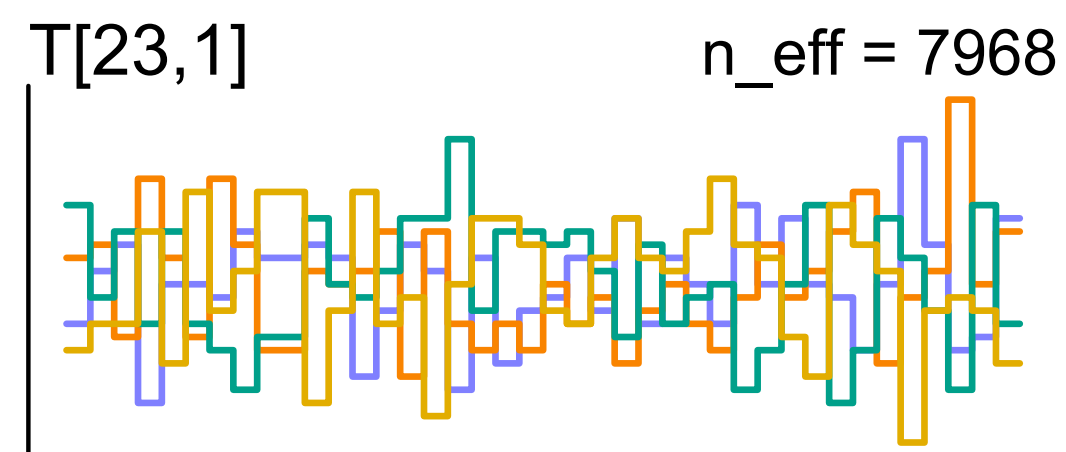
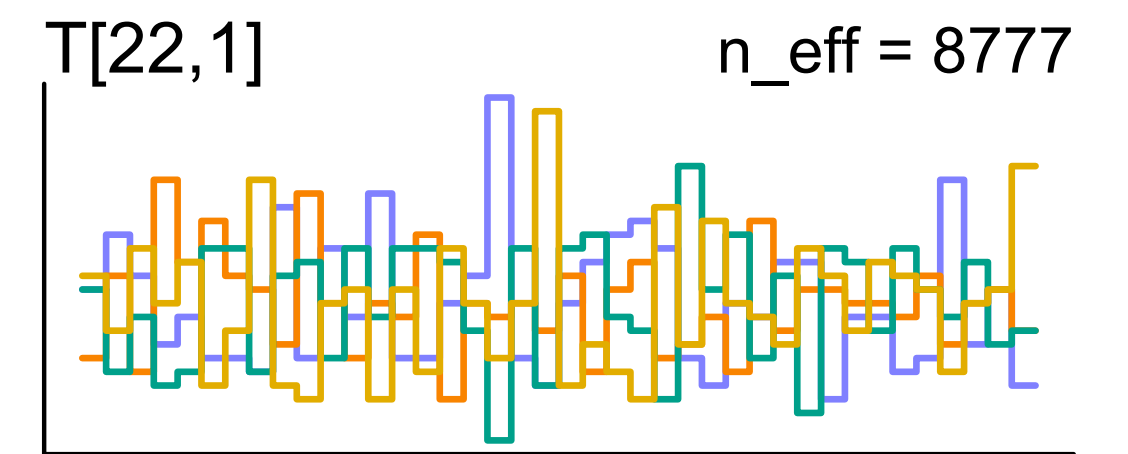
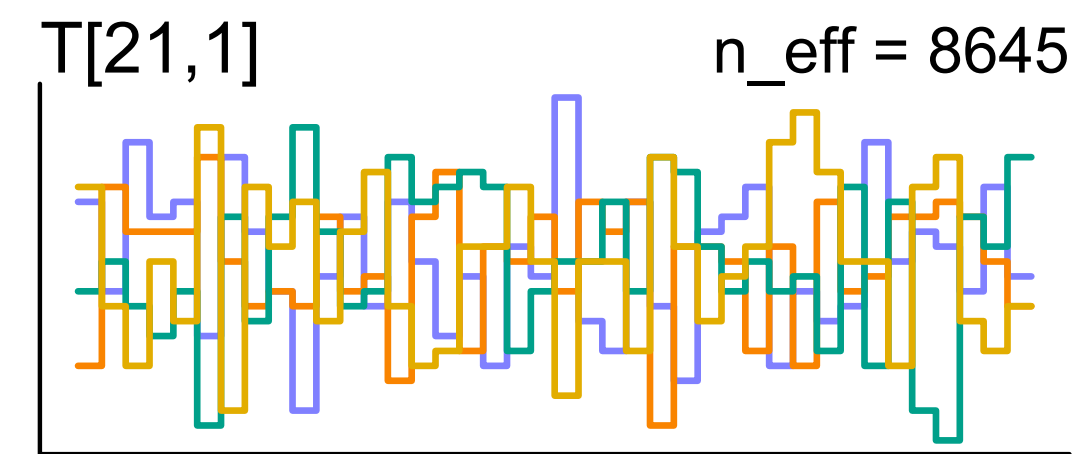
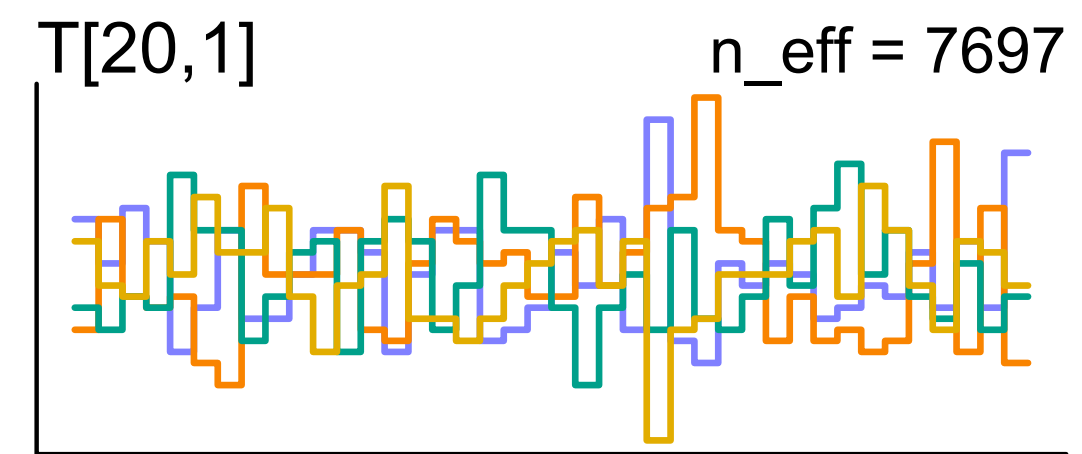
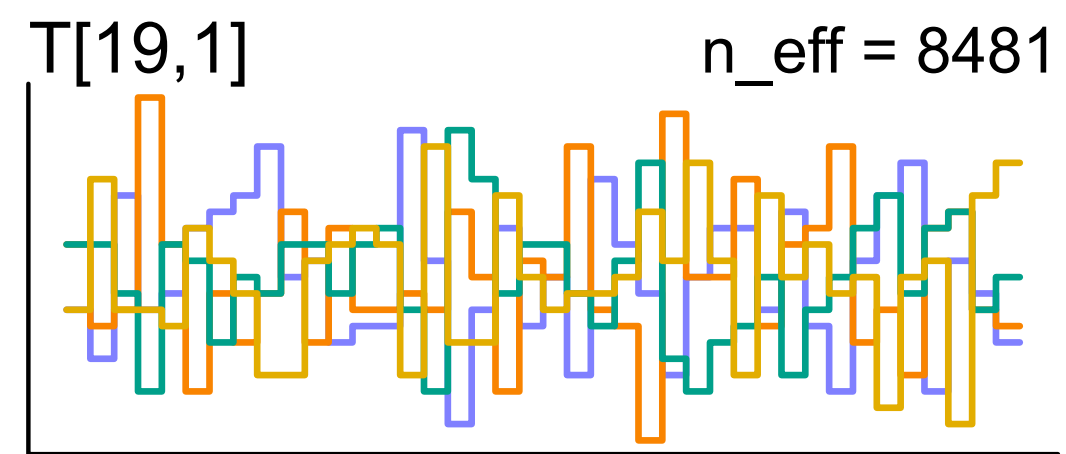
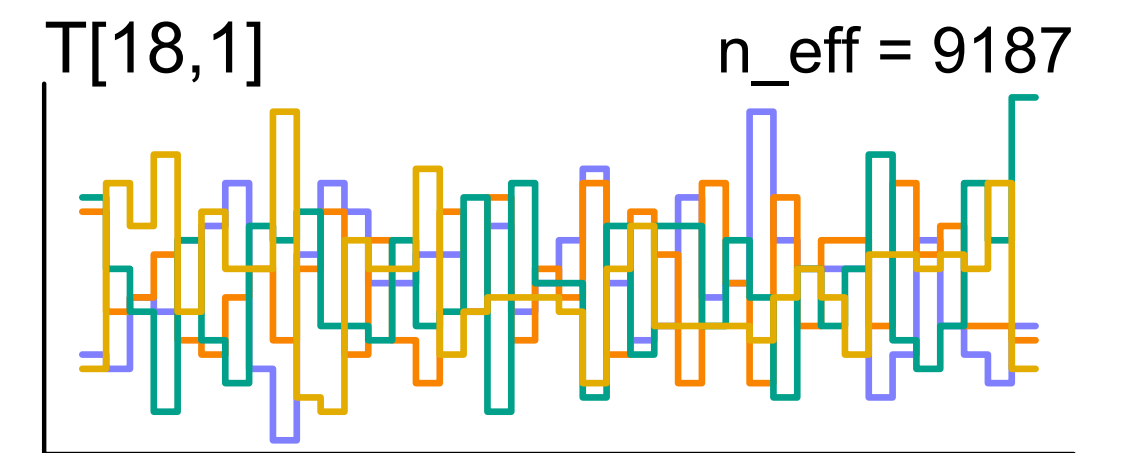
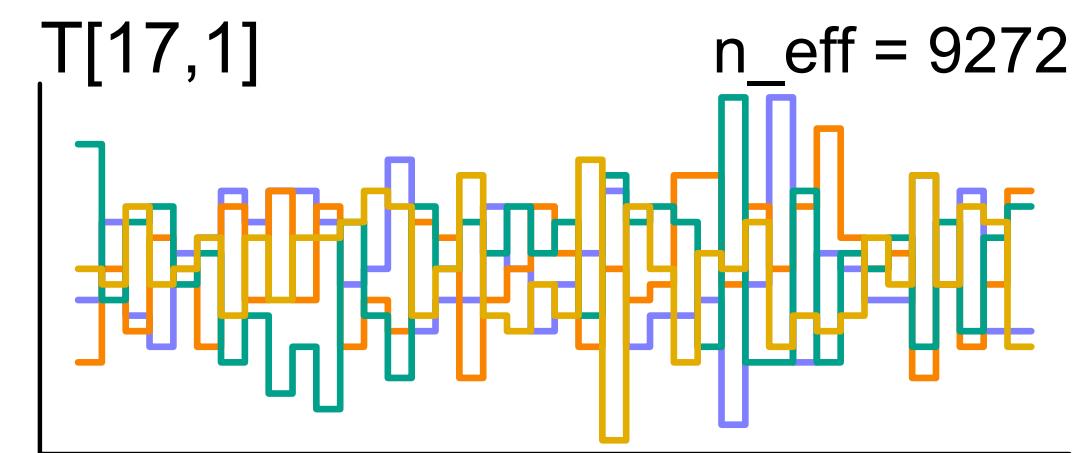
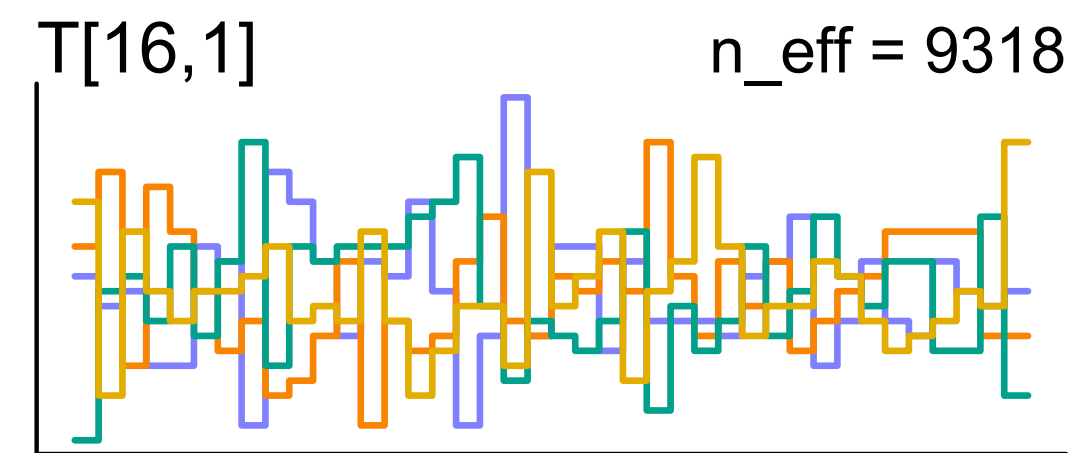
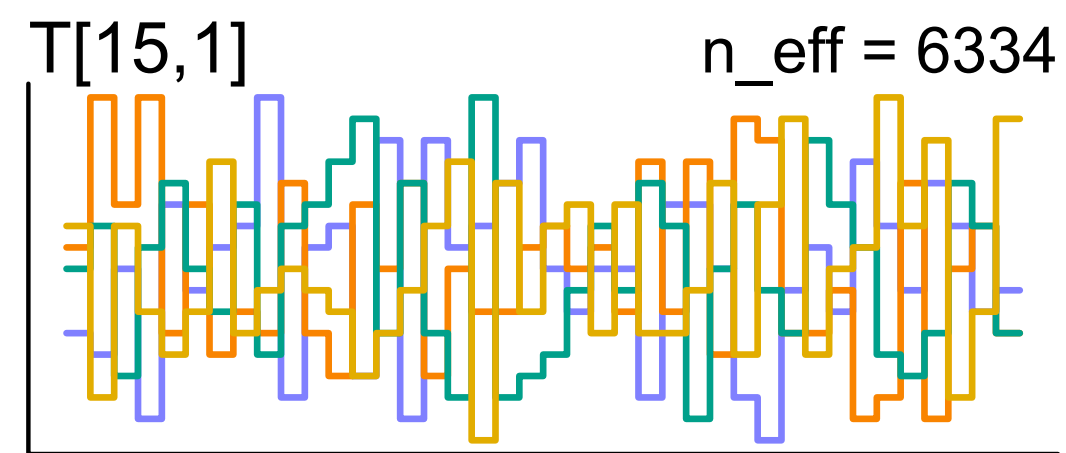
```
transpars> matrix[N_dyads,2]:T <-  
  compose_noncentered( rep_vector(sigma_T,2) , L_Rho_T , Z ),  
matrix[2,N_dyads]:Z ~ normal( 0 , 1 ),  
cholesky_factor_corr[2]:L_Rho_T ~ lkj_corr_cholesky( 2 ),  
sigma_T ~ exponential(1),
```

```
## compute correlation matrix for dyads
```

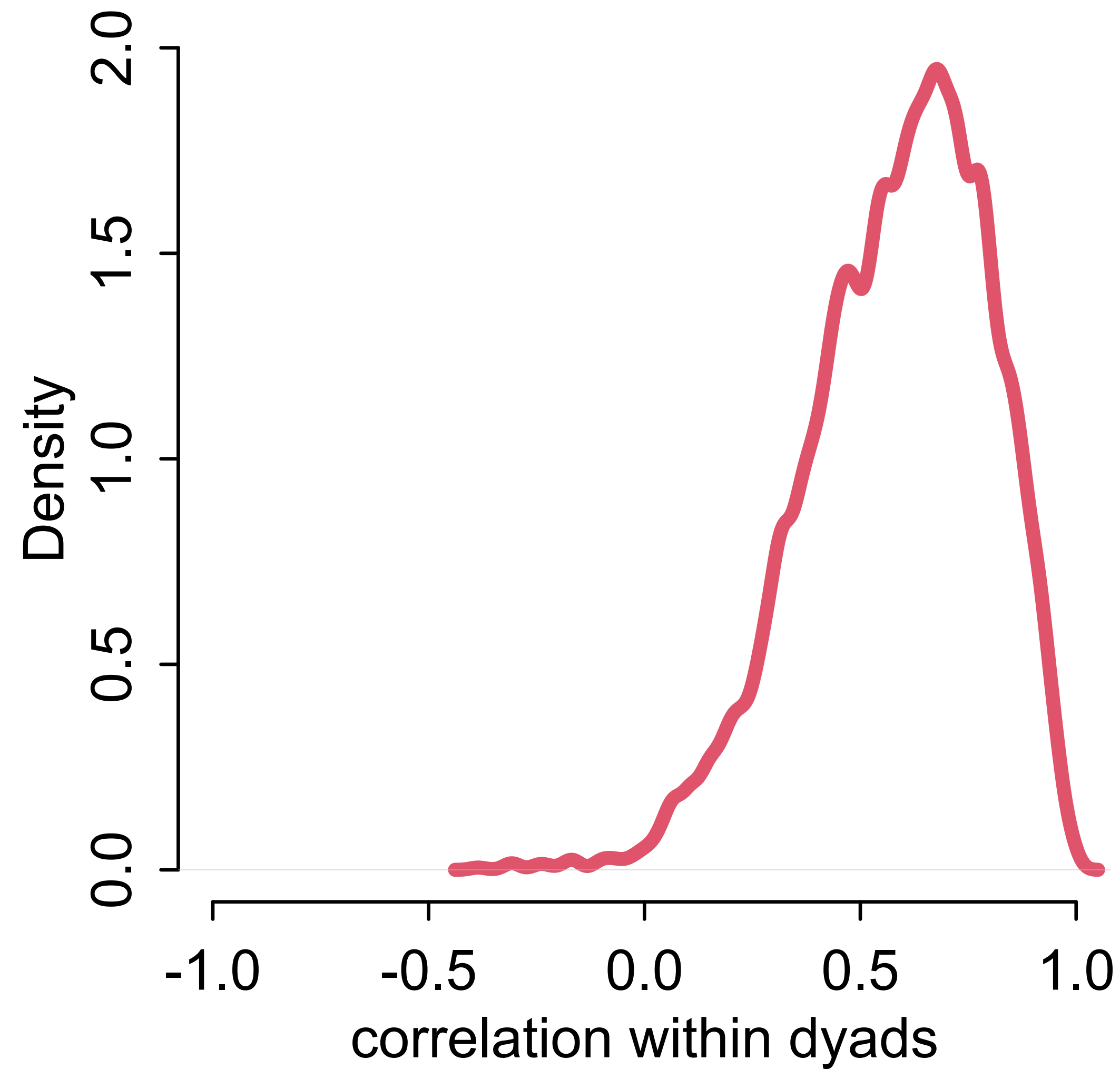
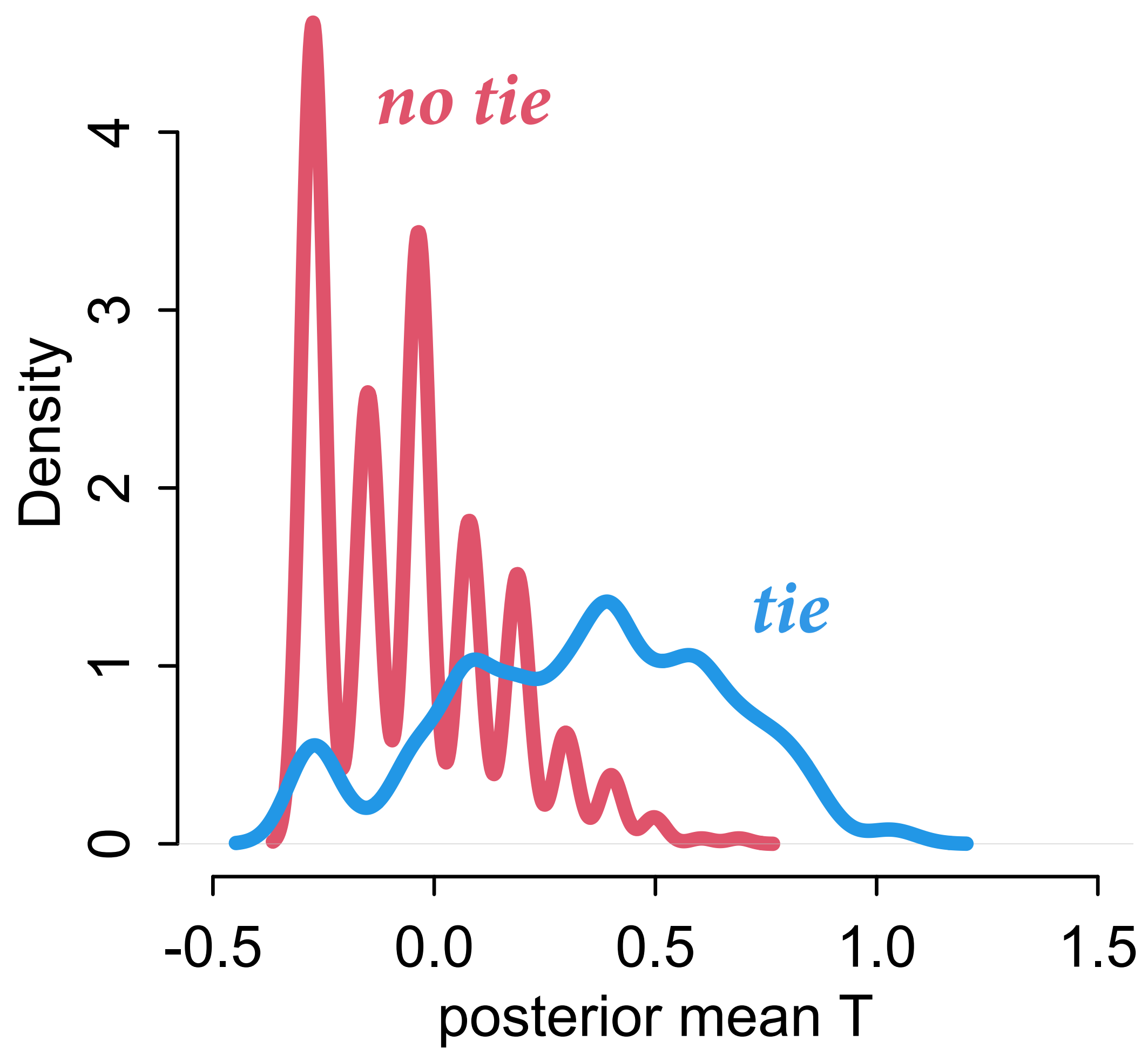
```
gq> matrix[2,2]:Rho_T <-<- Chol_to_Corr( L_Rho_T )
```

```
)
```

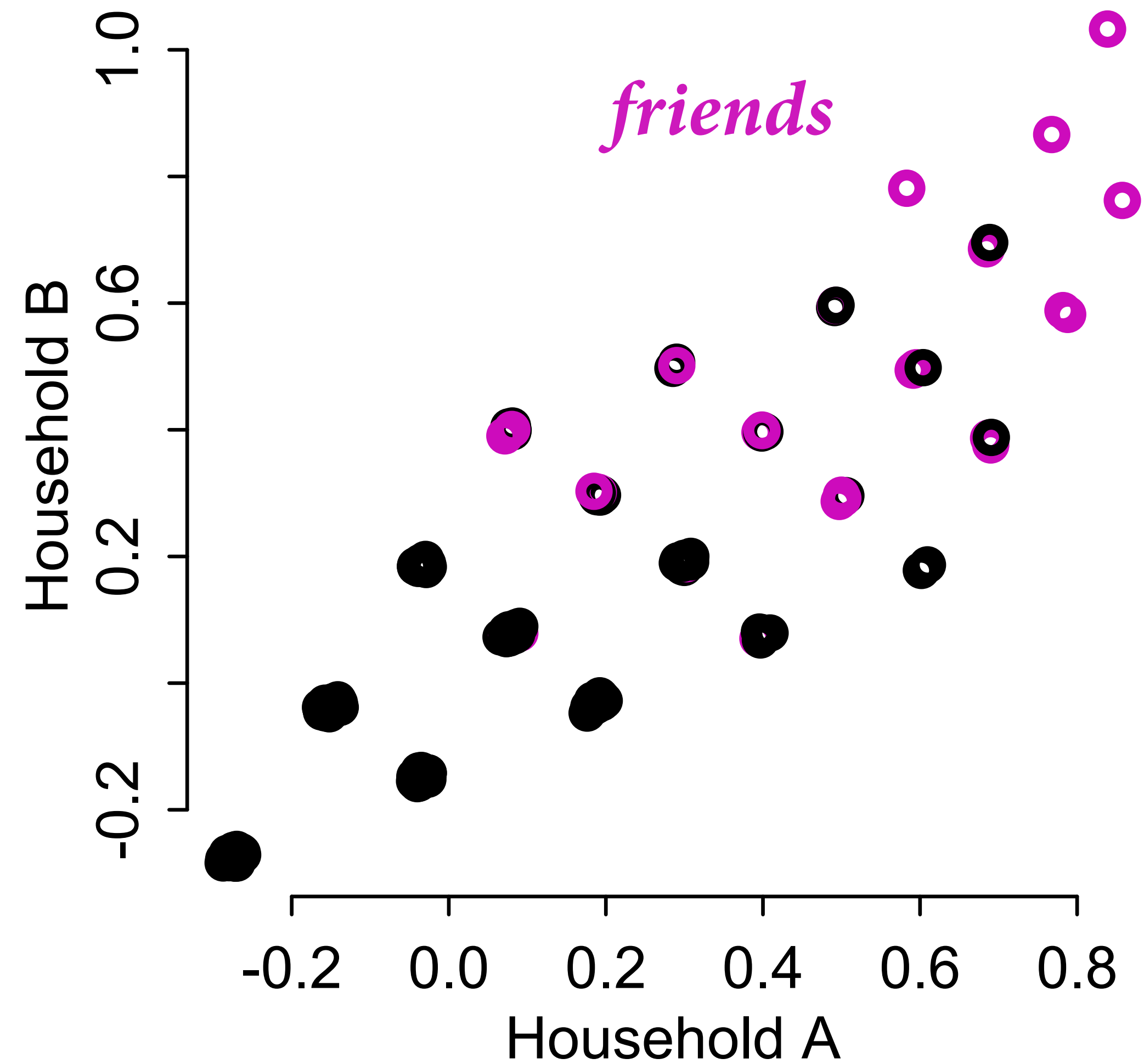
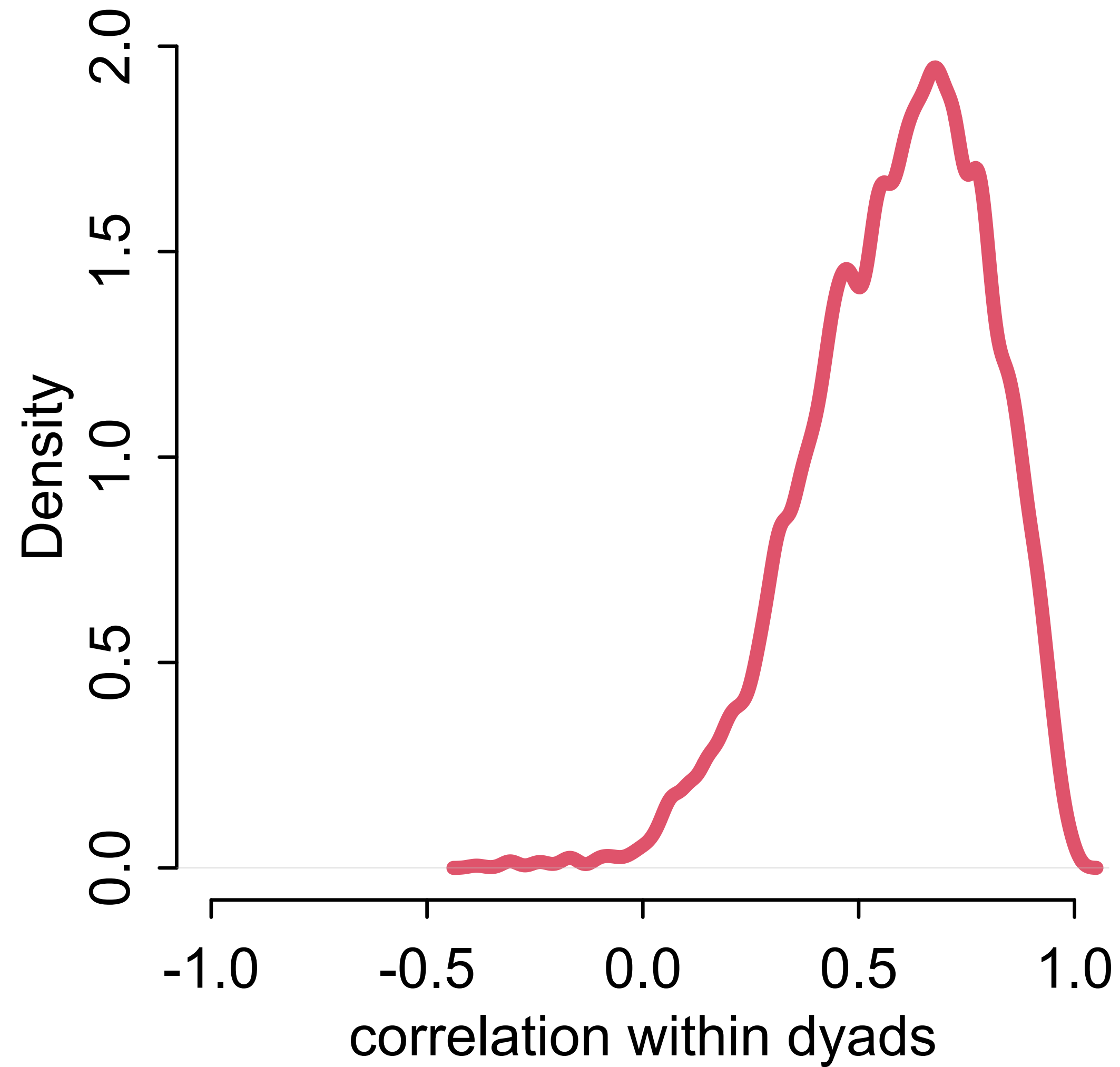
```
mGD <- ulam( f_dyad , data=sim_data , chains=4 , cores=4 , iter=2000 )
```



Posterior ties



Posterior ties



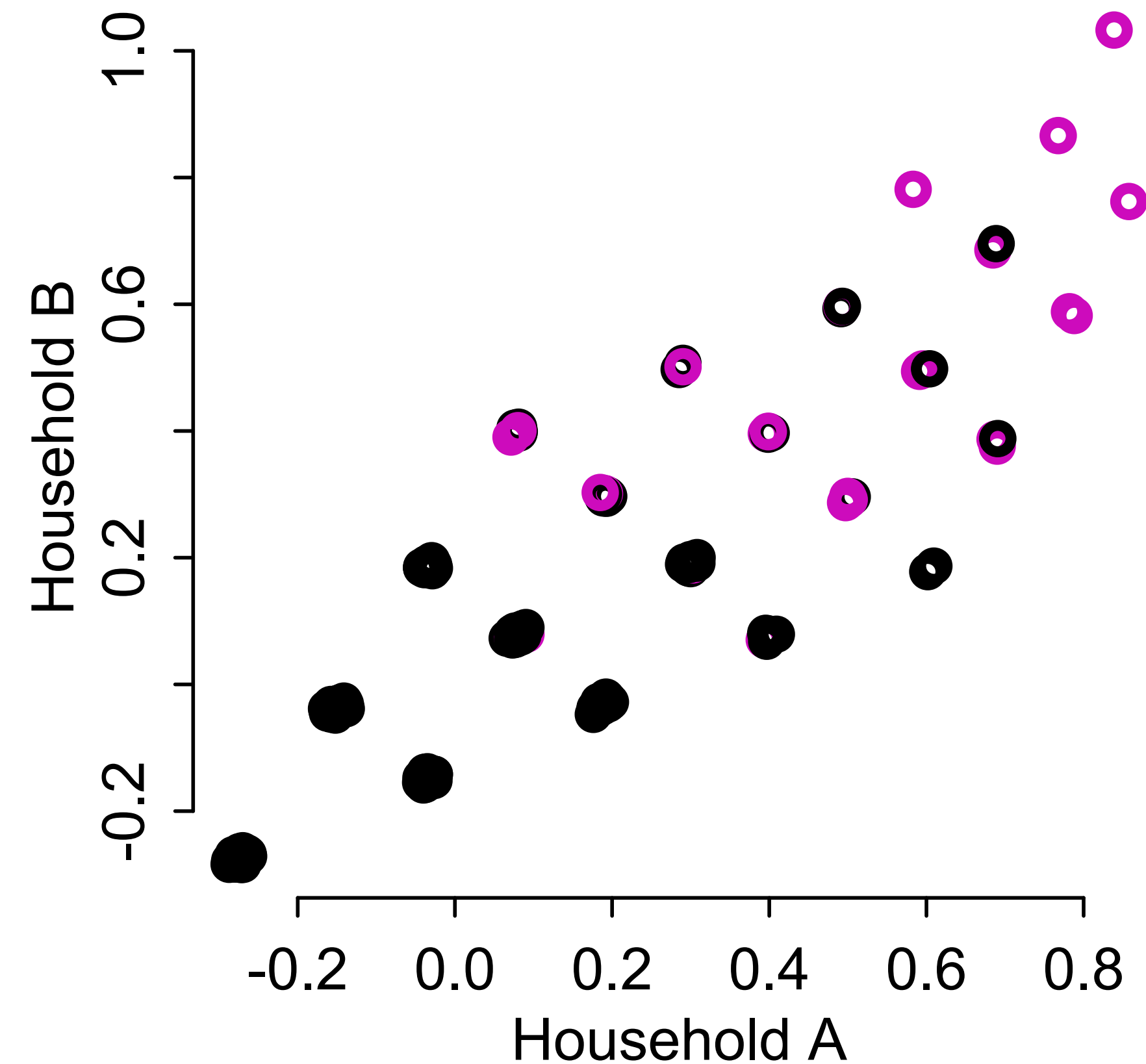
Drawing the Social Owl

(1) Estimand: Reciprocity & what explains it

(2) Generative model

(3) Statistical model

(4) Analyze sample



```
# analyze sample
kl_data <- list(
  N_dyads = nrow(kl_dyads),
  N_households = max(kl_dyads$hidB),
  D = 1:nrow(kl_dyads),
  HA = kl_dyads$hidA,
  HB = kl_dyads$hidB,
  GAB = kl_dyads$giftsAB,
  GBA = kl_dyads$giftsBA )

mGDkl <- ulam( f_dyad , data=kl_data , chains=4 , cores=4 , iter=2000 )

precis( mGDkl , depth=3 , pars=c("a","Rho_T","sigma_T") )
```

```

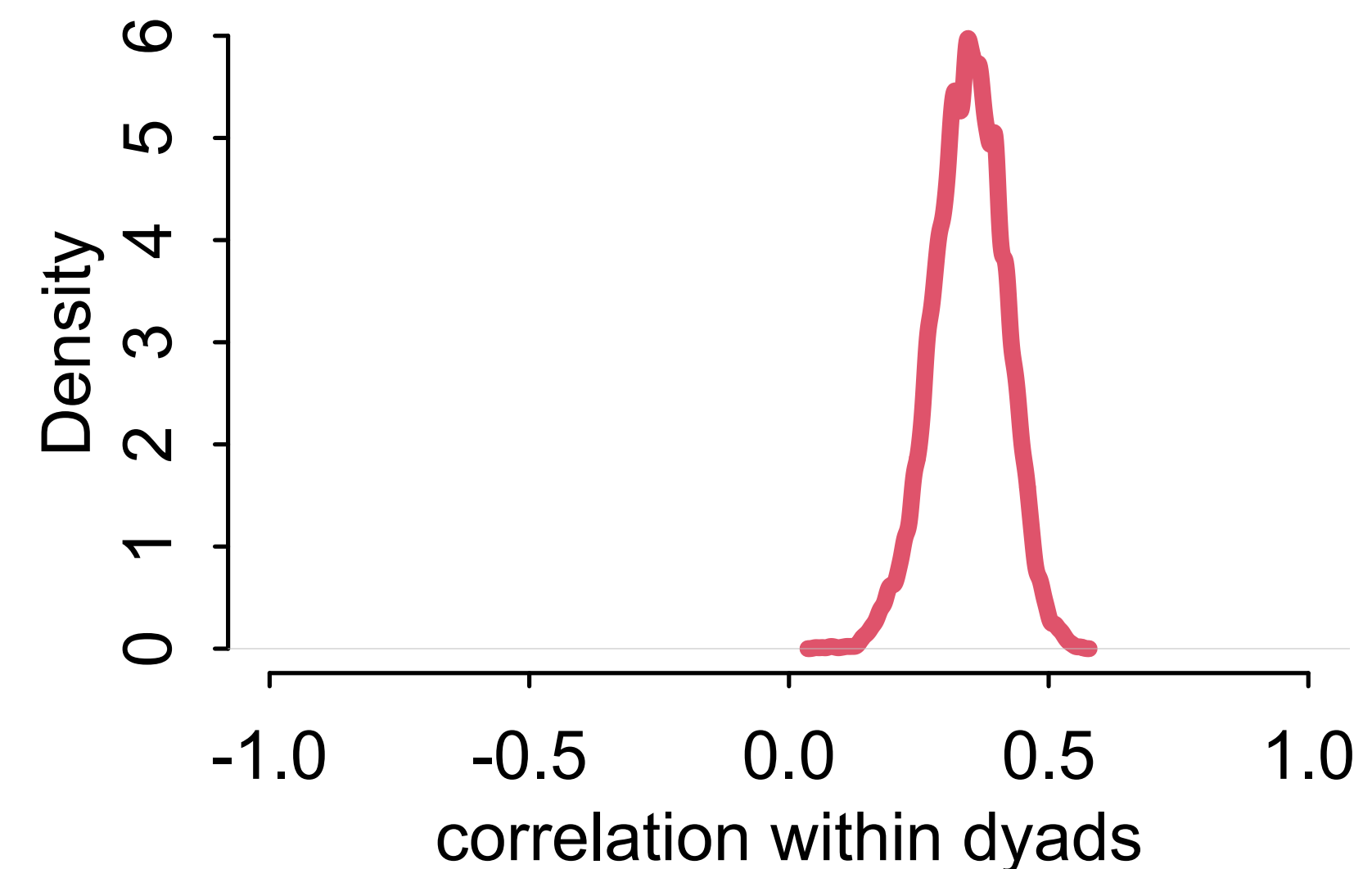
# analyze sample
kl_data <- list(
  N_dyads = nrow(kl_dyads),
  N_households = max(kl_dyads$hidB),
  D = 1:nrow(kl_dyads),
  HA = kl_dyads$hidA,
  HB = kl_dyads$hidB,
  GAB = kl_dyads$giftsAB,
  GBA = kl_dyads$giftsBA )

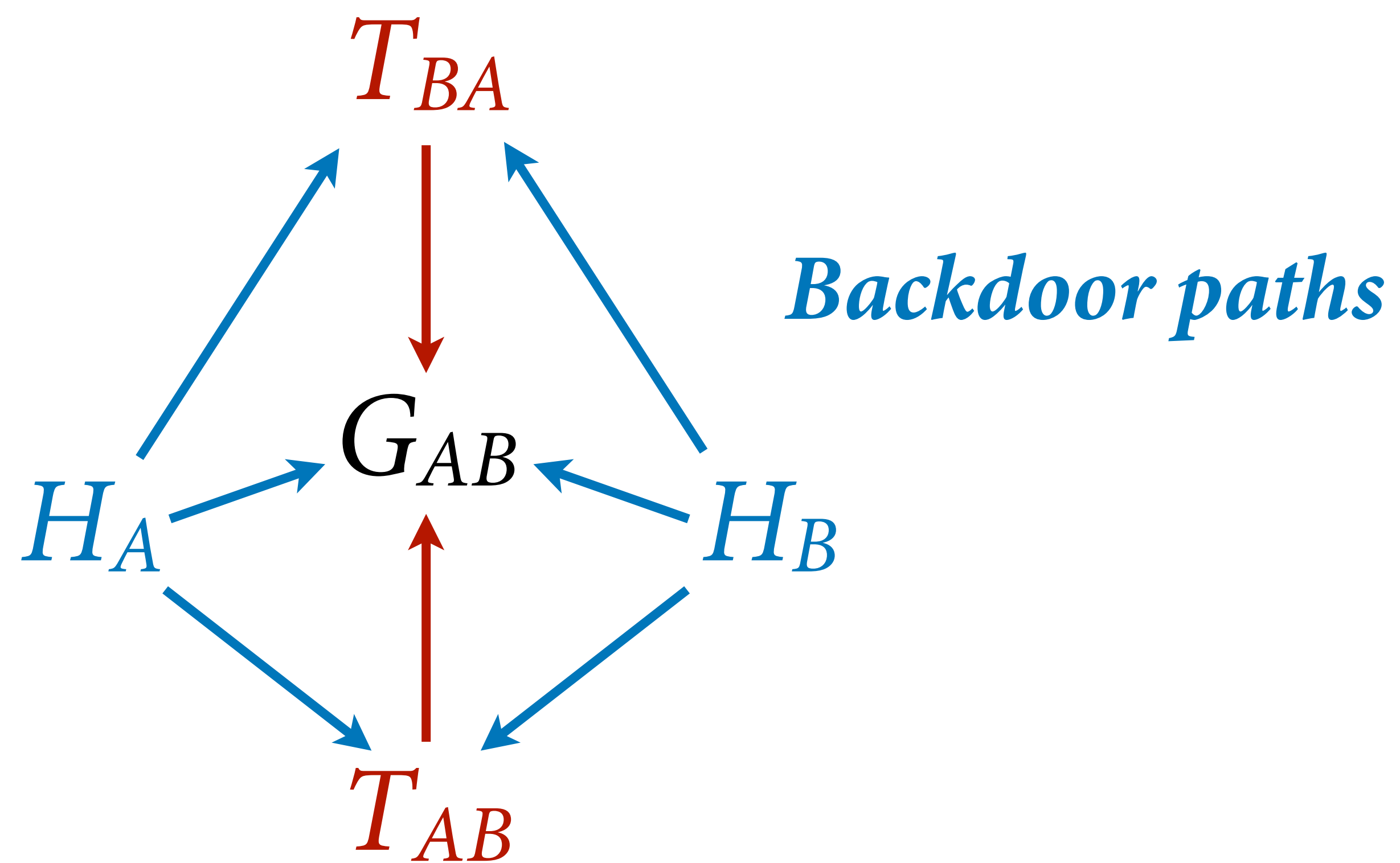
mGDkl <- ulam( f_dyad , data=kl_data , chains=4 , cores=4 , iter=2000 )

precis( mGDkl , depth=3 , pars=c("a","Rho_T","sigma_T") )

```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a	0.55	0.08	0.42	0.68	2246	1.00
Rho_T[1,1]	1.00	0.00	1.00	1.00	NaN	NaN
Rho_T[1,2]	0.35	0.07	0.24	0.45	1351	1.00
Rho_T[2,1]	0.35	0.07	0.24	0.45	1351	1.00
Rho_T[2,2]	1.00	0.00	1.00	1.00	NaN	NaN
sigma_T	1.44	0.06	1.35	1.55	1249	1.01





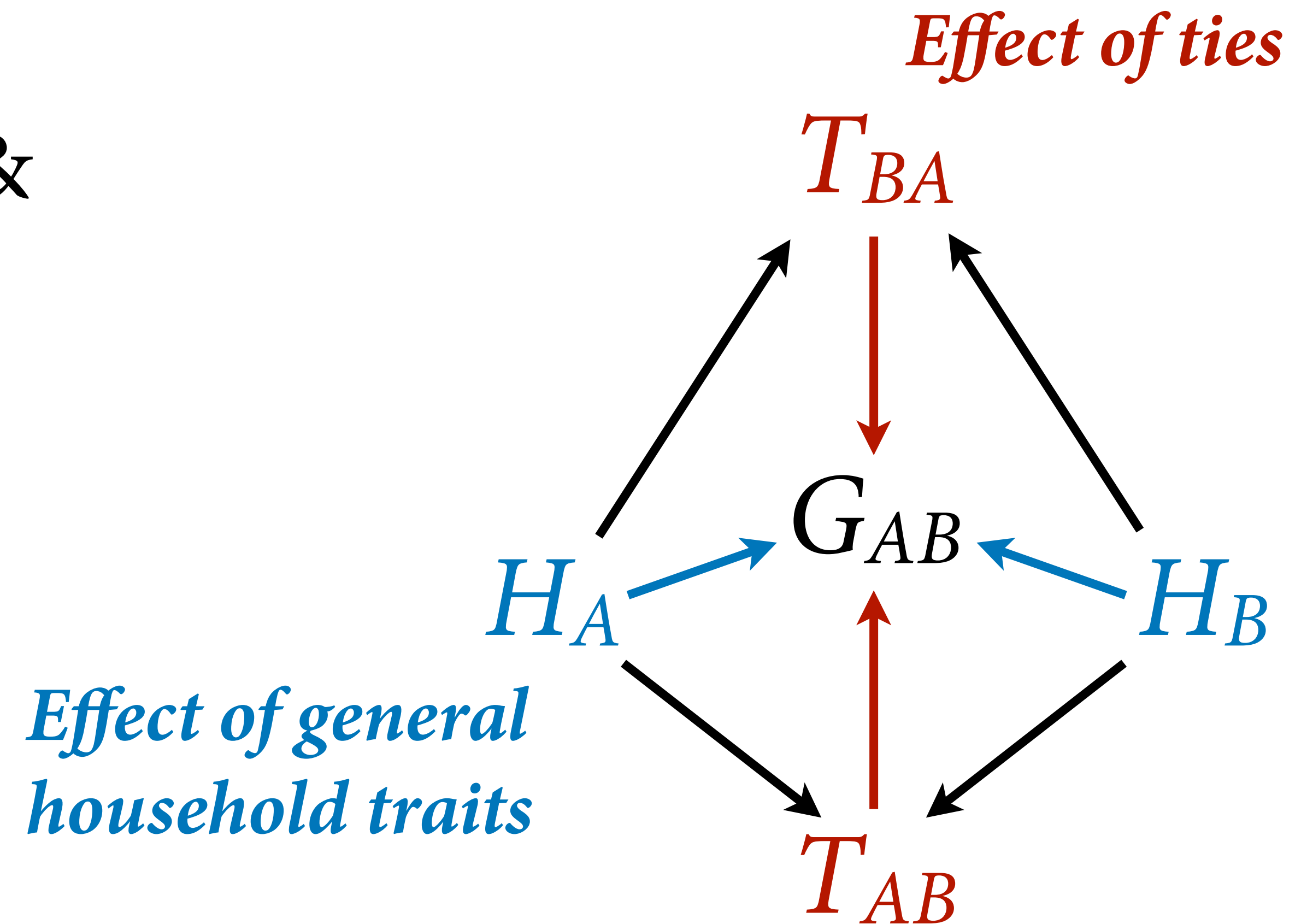
Drawing the Social Owl

(1) Estimand: Reciprocity & what explains it

(2) Generative model

(3) Statistical model

(4) Analyze sample



```

# N households
N <- 25
dyads <- t(combn(N,2))
N_dyads <- nrow(dyads)

# simulate "friendships" in which ties are shared
f <- rbern(N_dyads,0.1) # 10% of dyads are friends

# now simulate directed ties for all individuals
# there can be ties that are not reciprocal
alpha <- (-3) # base rate of ties; -3 ~= 0.05
y <- matrix(NA,N,N) # edge list
for ( i in 1:N ) for ( j in 1:N ) {
  if ( i != j ) {
    # directed tie from i to j
    ids <- sort( c(i,j) )
    the_dyad <- which( dyads[,1]==ids[1] & dyads[,2]==ids[2] )
    p_tie <- f[the_dyad] + (1-f[the_dyad])*inv_logit( alpha )
    y[i,j] <- rbern( 1 , p_tie )
  }
} #ij

```

```

# simulate wealth
W <- rnorm(N) # standardized relative wealth in community
bWG <- 0.5 # effect of wealth on giving - rich give more
bWR <- (-1) # effect of wealth on receiving - rich get less / poor get more

# now simulate gifts
giftsAB <- rep(NA, N_dyads)
giftsBA <- rep(NA, N_dyads)
lambda <- log(c(0.5, 2)) # rates of giving for y=0, y=1
for ( i in 1:N_dyads ) {
  A <- dyads[i, 1]
  B <- dyads[i, 2]
  giftsAB[i] <- rpois( 1 , exp( lambda[1+y[A,B]] + bWG*W[A] + bWR*W[B] ) )
  giftsBA[i] <- rpois( 1 , exp( lambda[1+y[B,A]] + bWG*W[B] + bWR*W[A] ) )
}

```

```
# simulate wealth
```

```
W <- rnorm(N) # standardized relative wealth in community
```

```
bWG <- 0.5 # effect of wealth on giving - rich give more
```

```
bWR <- (-1) # effect of wealth on receiving - rich get less / poor get more
```

```
# now simulate gifts
```

```
giftsAB <- rep(NA, N_dyads)
```

```
giftsBA <- rep(NA, N_dyads)
```

```
lambda <- log(c(0.5, 2)) # rates of giving for y=0, y=1
```

```
for ( i in 1:N_dyads ) {
```

```
  A <- dyads[i, 1]
```

```
  B <- dyads[i, 2]
```

```
  giftsAB[i] <- rpois( 1 , exp( lambda[1+y[A,B]] + bWG*W[A] + bWR*W[B] ) )
```

```
  giftsBA[i] <- rpois( 1 , exp( lambda[1+y[B,A]] + bWG*W[B] + bWR*W[A] ) )
```

```
}
```

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB}$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA}$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

*A's generalized
giving*

*B's generalized
receiving*

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\text{log}(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\text{log}(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} G_A \\ R_A \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_G^2 & r\sigma_G\sigma_R \\ r\sigma_G\sigma_R & \sigma_R^2 \end{bmatrix} \right)$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$\begin{pmatrix} G_A \\ R_A \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_G^2 & r\sigma_G\sigma_R \\ r\sigma_G\sigma_R & \sigma_R^2 \end{bmatrix} \right)$$

A's giving & receiving

Covariance matrix of household giving & receiving

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$\begin{pmatrix} G_A \\ R_A \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{R}_{GR}, \mathbf{S}_{GR} \right)$$

A's giving & receiving

Correlation matrix

Standard deviations

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

$$\alpha \sim \text{Normal}(0,1)$$

$$\begin{pmatrix} G_A \\ R_A \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{R}_{GR}, \mathbf{S}_{GR} \right)$$

$$\mathbf{R}_{GR} \sim \text{LKJCorr}(2)$$

$$\mathbf{S}_{GR} \sim \text{Exponential}(1)$$

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

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$$\mathbf{R}_{GR} \sim \text{LKJCorr}(2)$$

$$\mathbf{S}_{GR} \sim \text{Exponential}(1)$$

25 households

300 dyads

600 counts / 2871 transfers

602 social network parameters

53 household parameters

```

# general model
f_general <- alist(
  GAB ~ poisson( lambdaAB ),
  GBA ~ poisson( lambdaBA ),
  log(lambdaAB) <- a + T[D,1] + gr[HA,1] + gr[HB,2],
  log(lambdaBA) <- a + T[D,2] + gr[HB,1] + gr[HA,2],
  a ~ normal(0,1),

  ## dyad effects - non-centered
  transpars> matrix[N_dyads,2]:T <-
    compose_noncentered(rep_vector(sigma_T,2),L_Rho_T,Z),
  matrix[2,N_dyads]:Z ~ normal( 0 , 1 ),
  cholesky_factor_corr[2]:L_Rho_T ~ lkj_corr_cholesky( 2 ),
  sigma_T ~ exponential(1),

  ## gr matrix of varying effects
  transpars> matrix[N_households,2]:gr <-
    compose_noncentered( sigma_gr , L_Rho_gr , Zgr ),
  matrix[2,N_households]:Zgr ~ normal( 0 , 1 ),
  cholesky_factor_corr[2]:L_Rho_gr ~ lkj_corr_cholesky( 2 ),
  vector[2]:sigma_gr ~ exponential(1),

  ## compute correlation matrixes
  gq> matrix[2,2]:Rho_T <- Chol_to_Corr( L_Rho_T ),
  gq> matrix[2,2]:Rho_gr <- Chol_to_Corr( L_Rho_gr )
)
mGDGR <- ulam(f_general,data=sim_data,chains=4,cores=4,iter=2000)

```

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

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$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

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```

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  transpars> matrix[N_households,2]:gr <-
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  cholesky_factor_corr[2]:L_Rho_gr ~ lkj_corr_cholesky( 2 ),
  vector[2]:sigma_gr ~ exponential(1),

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```

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```

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  cholesky_factor_corr[2]:L_Rho_gr ~ lkj_corr_cholesky( 2 ),
  vector[2]:sigma_gr ~ exponential(1),

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```

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$$\sigma \sim \text{Exponential}(1)$$

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```

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  matrix[2,N_households]:Zgr ~ normal( 0 , 1 ),
  cholesky_factor_corr[2]:L_Rho_gr ~ lkj_corr_cholesky( 2 ),
  vector[2]:sigma_gr ~ exponential(1),

  ## compute correlation matrixes
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  gq> matrix[2,2]:Rho_gr <-<- Chol_to_Corr( L_Rho_gr )
)
mGDGR <- ulam(f_general,data=sim_data,chains=4,cores=4,iter=2000)

```

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$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

$$\sigma \sim \text{Exponential}(1)$$

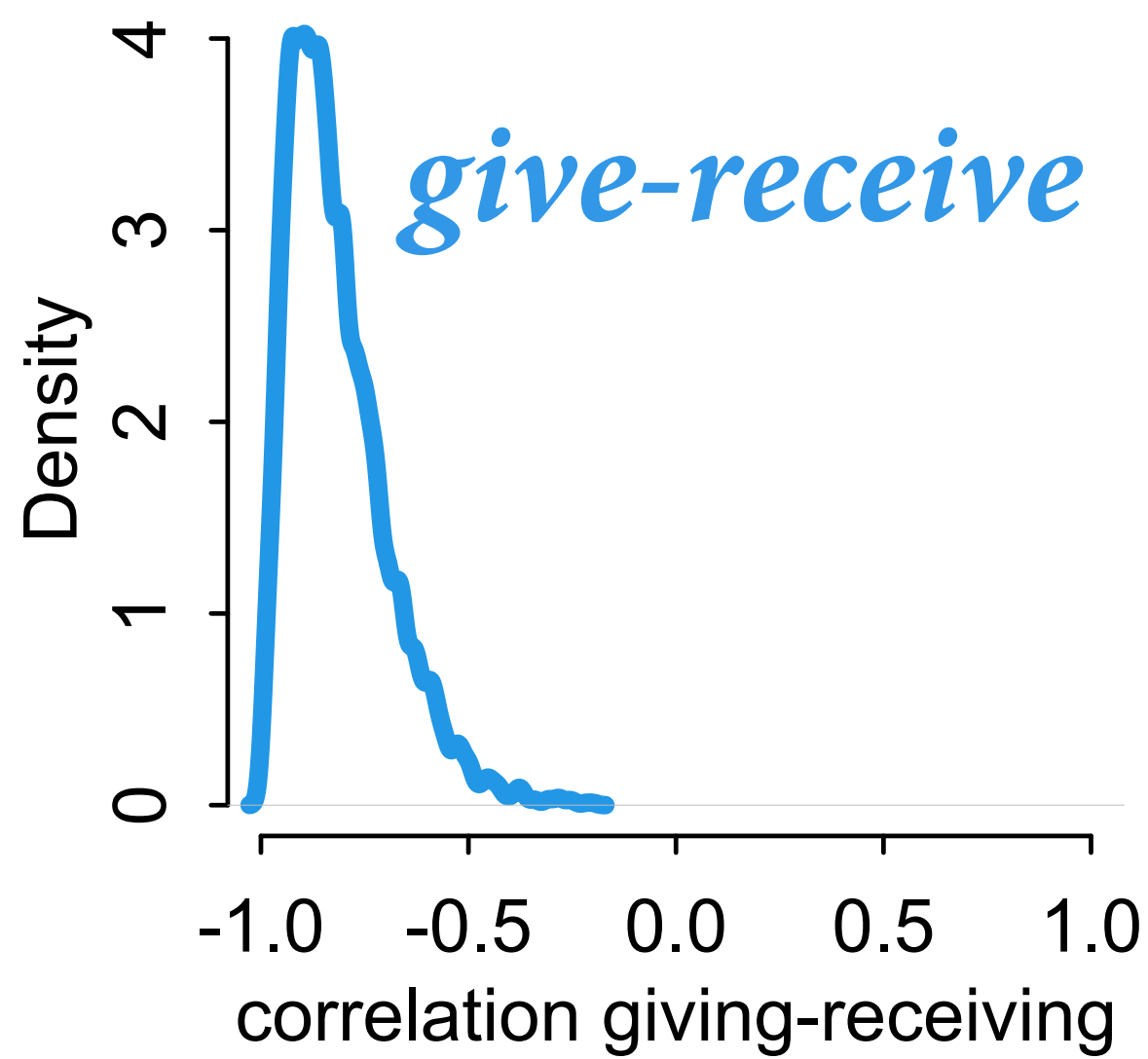
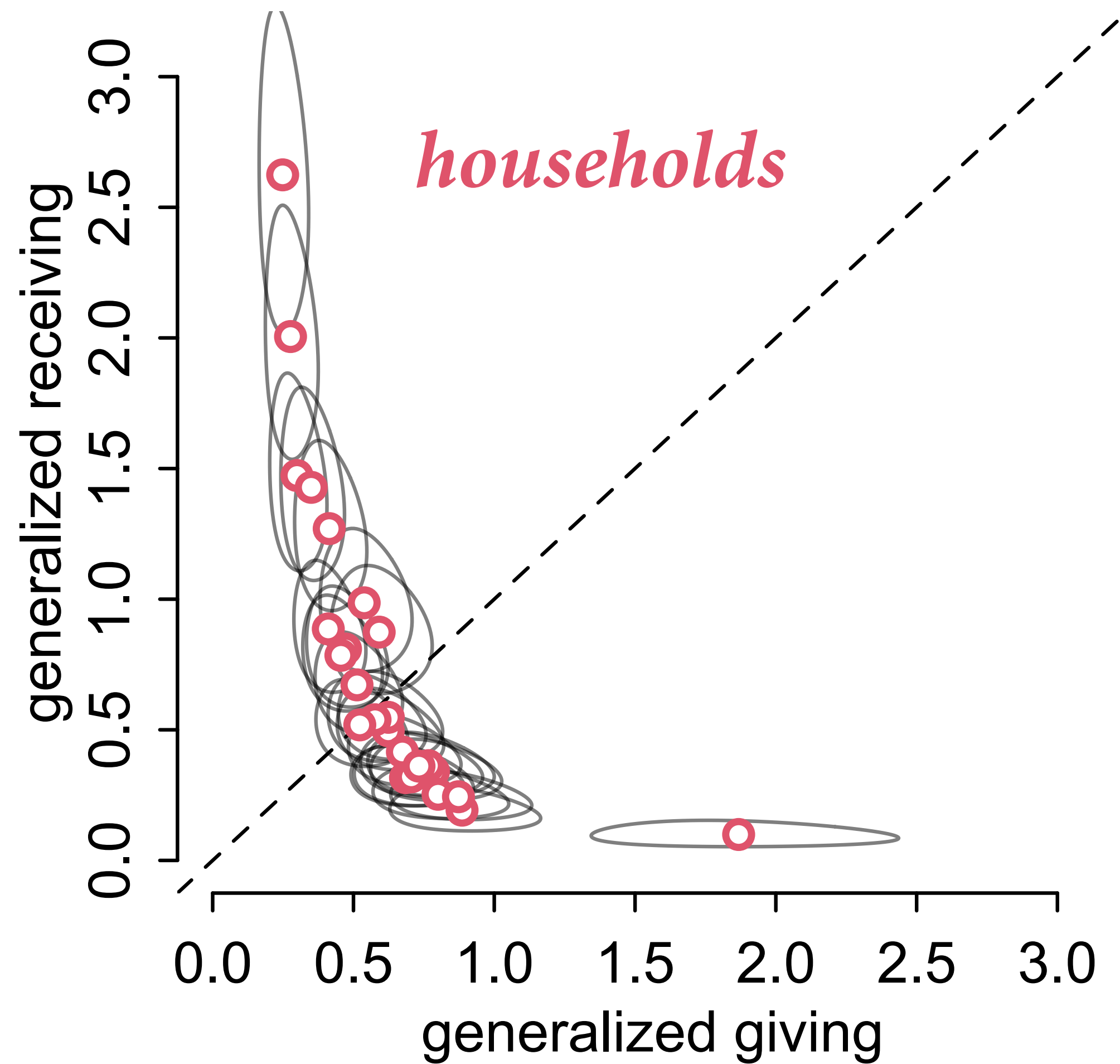
$$\alpha \sim \text{Normal}(0,1)$$

$$\begin{pmatrix} G_A \\ R_A \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{R}_{GR}, \mathbf{S}_{GR} \right)$$

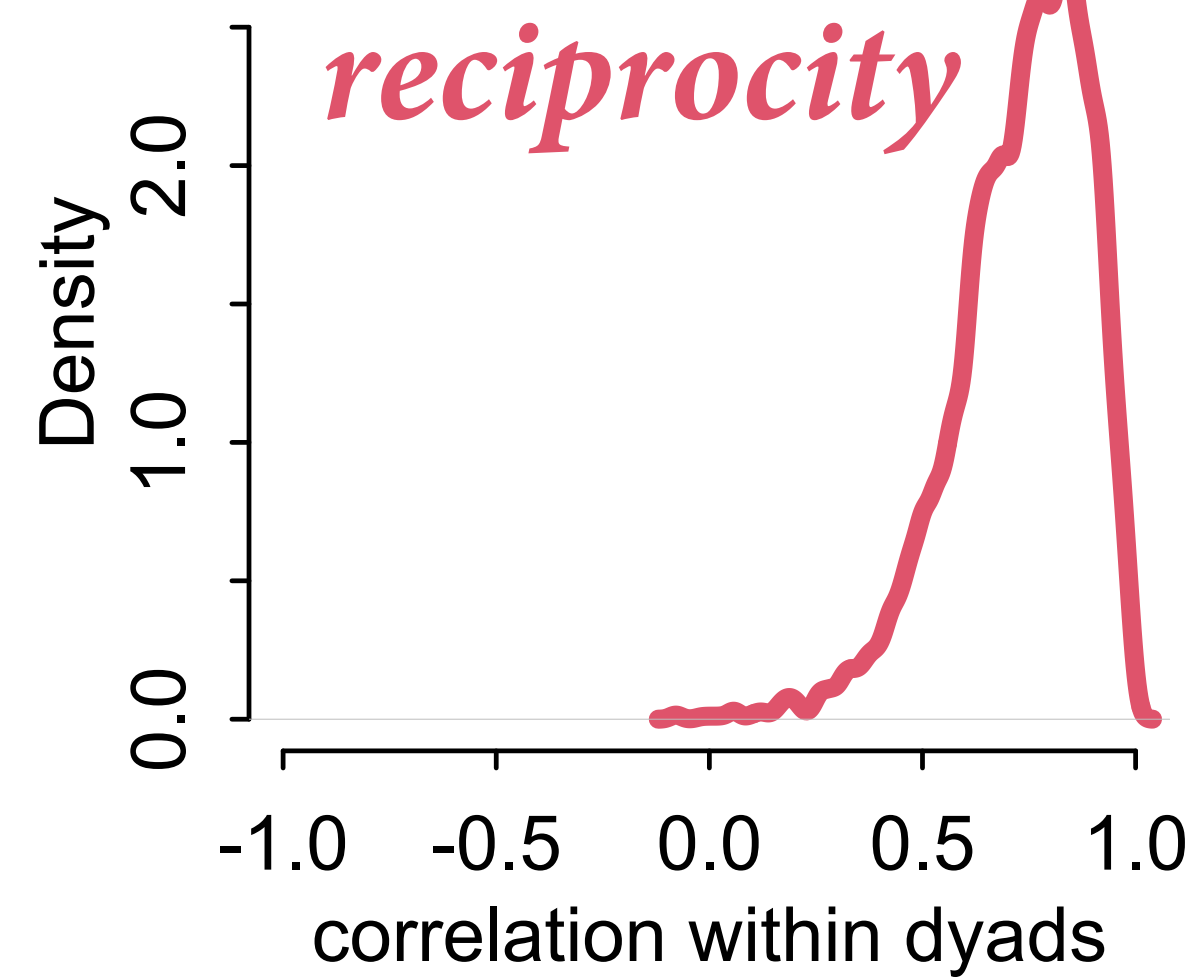
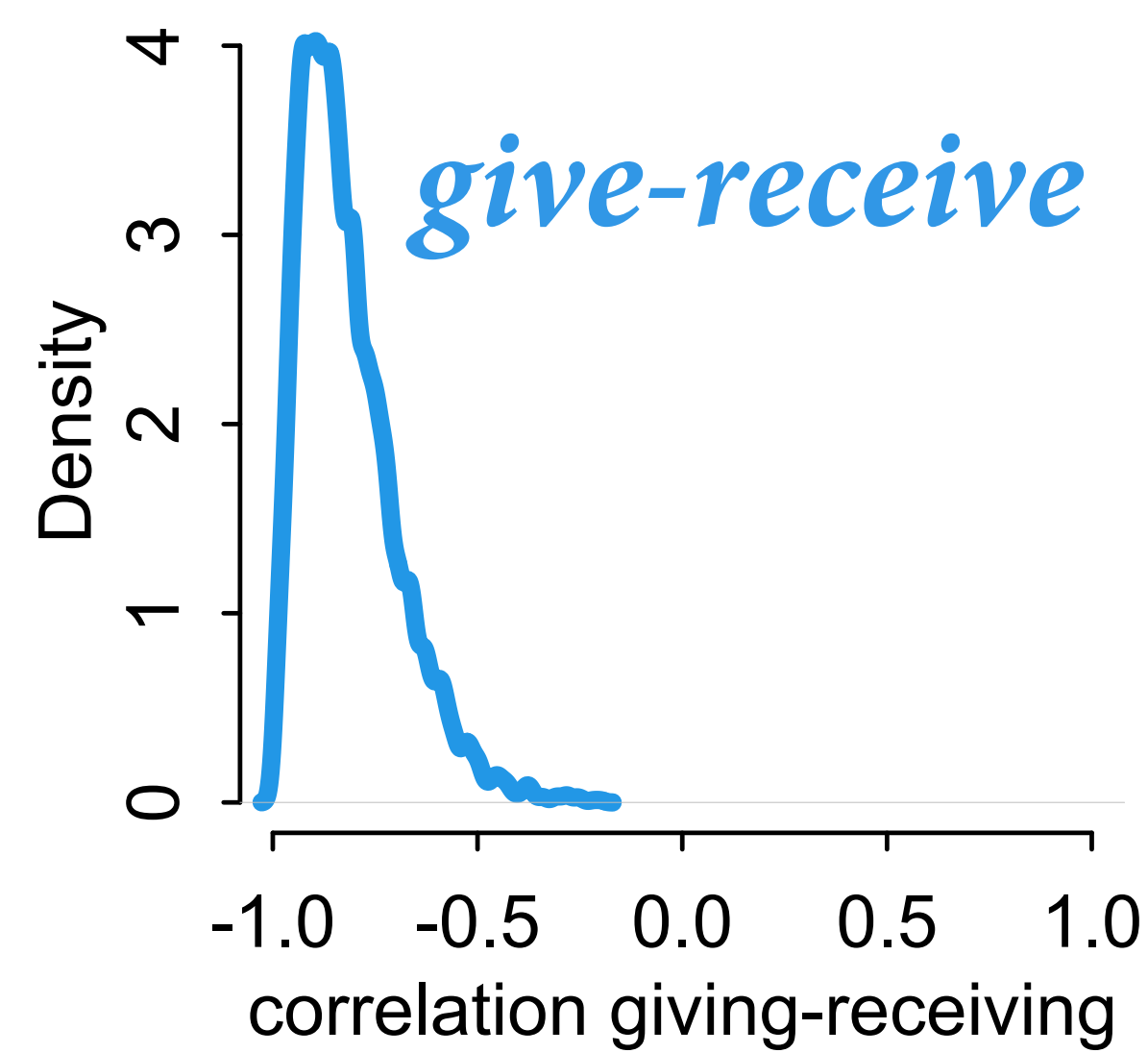
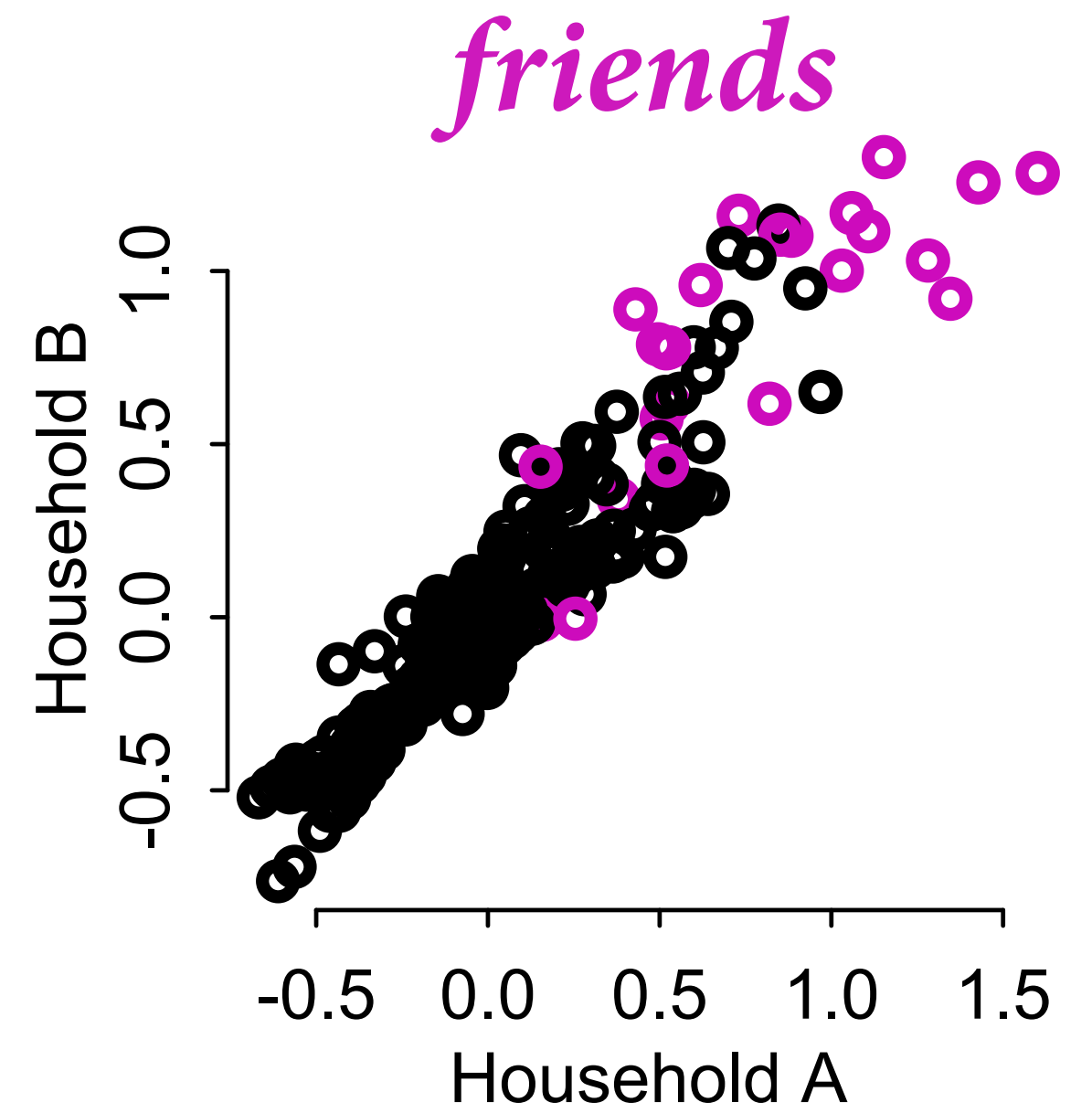
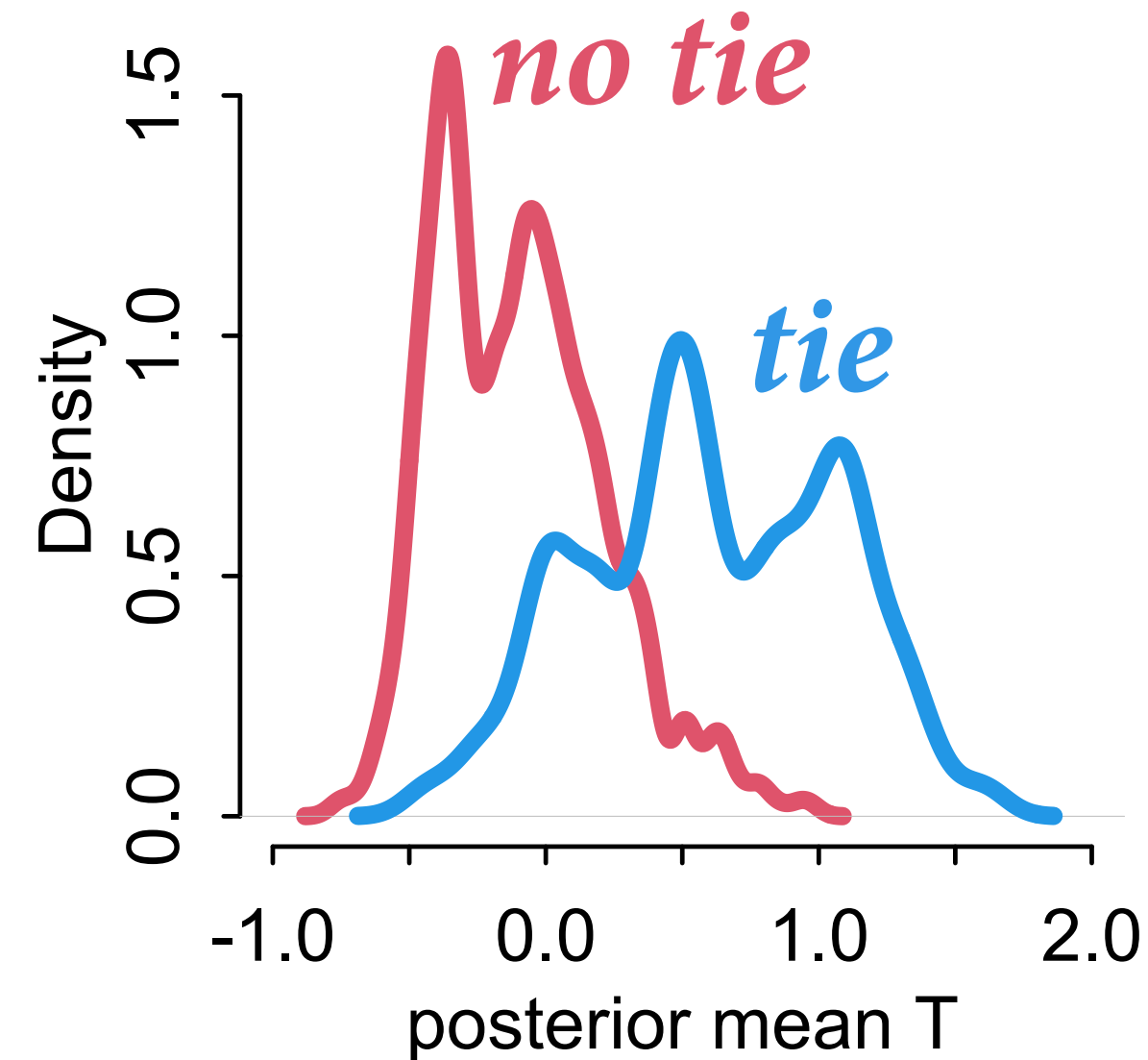
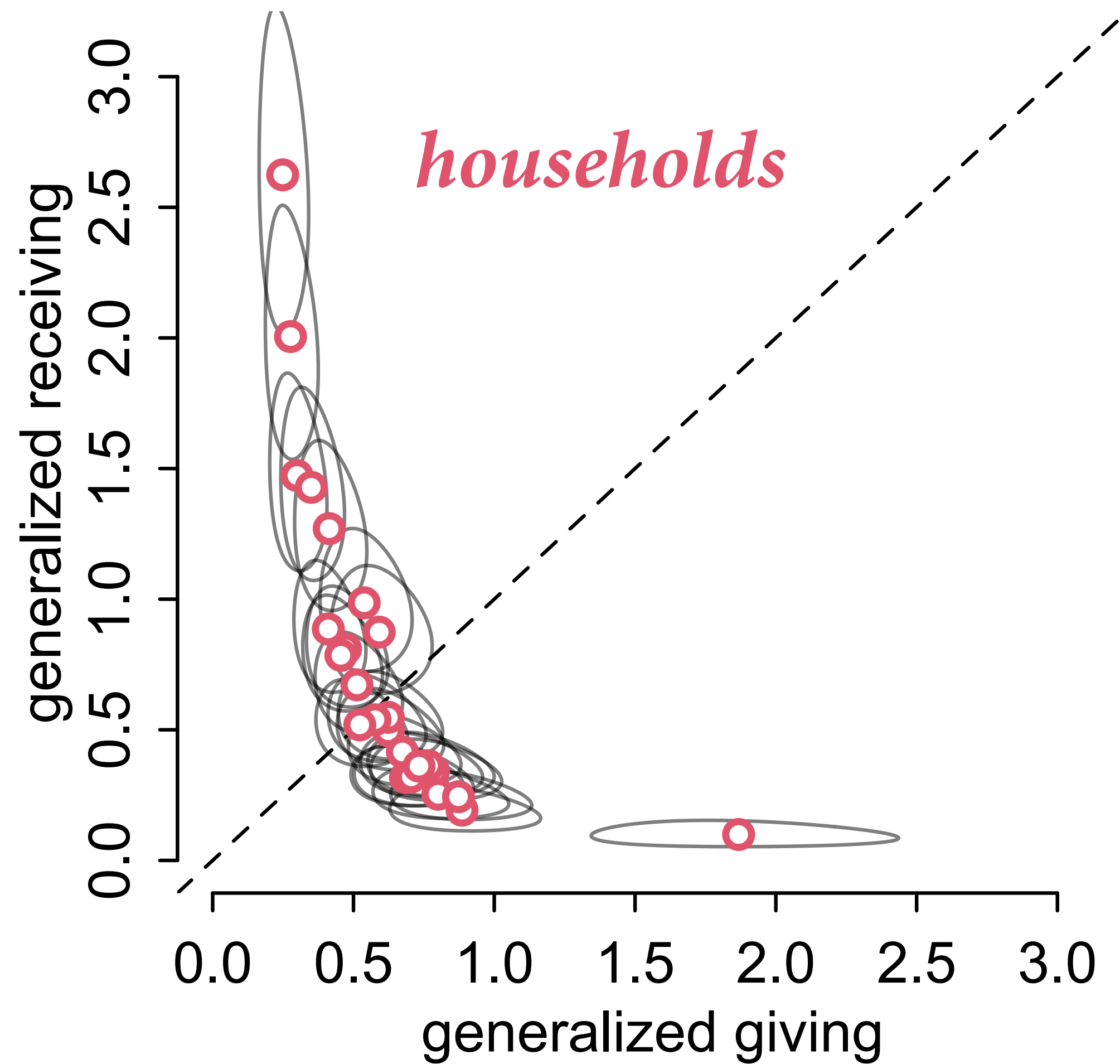
$$\mathbf{R}_{GR} \sim \text{LKJCorr}(2)$$

$$\mathbf{S}_{GR} \sim \text{Exponential}(1)$$

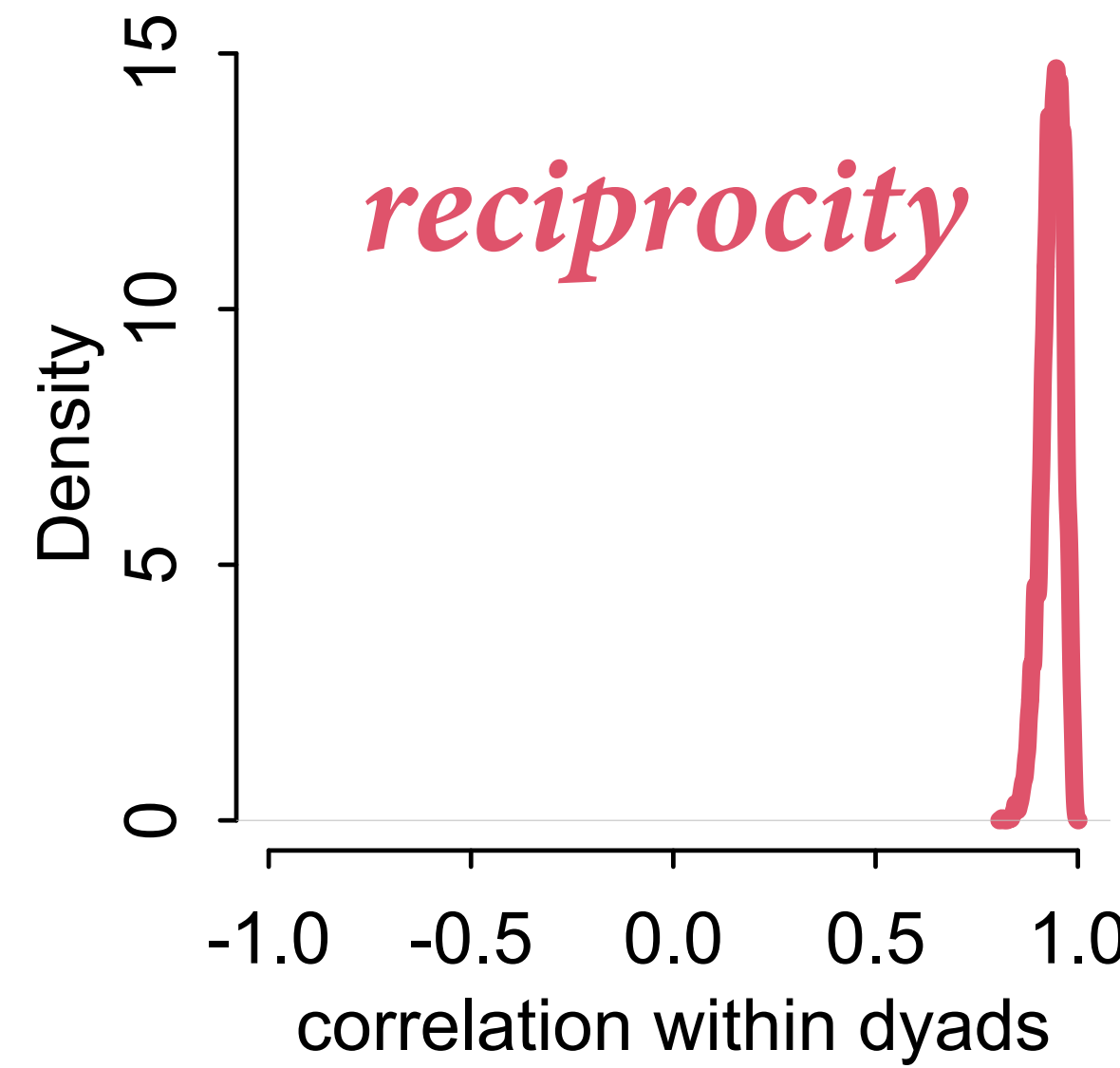
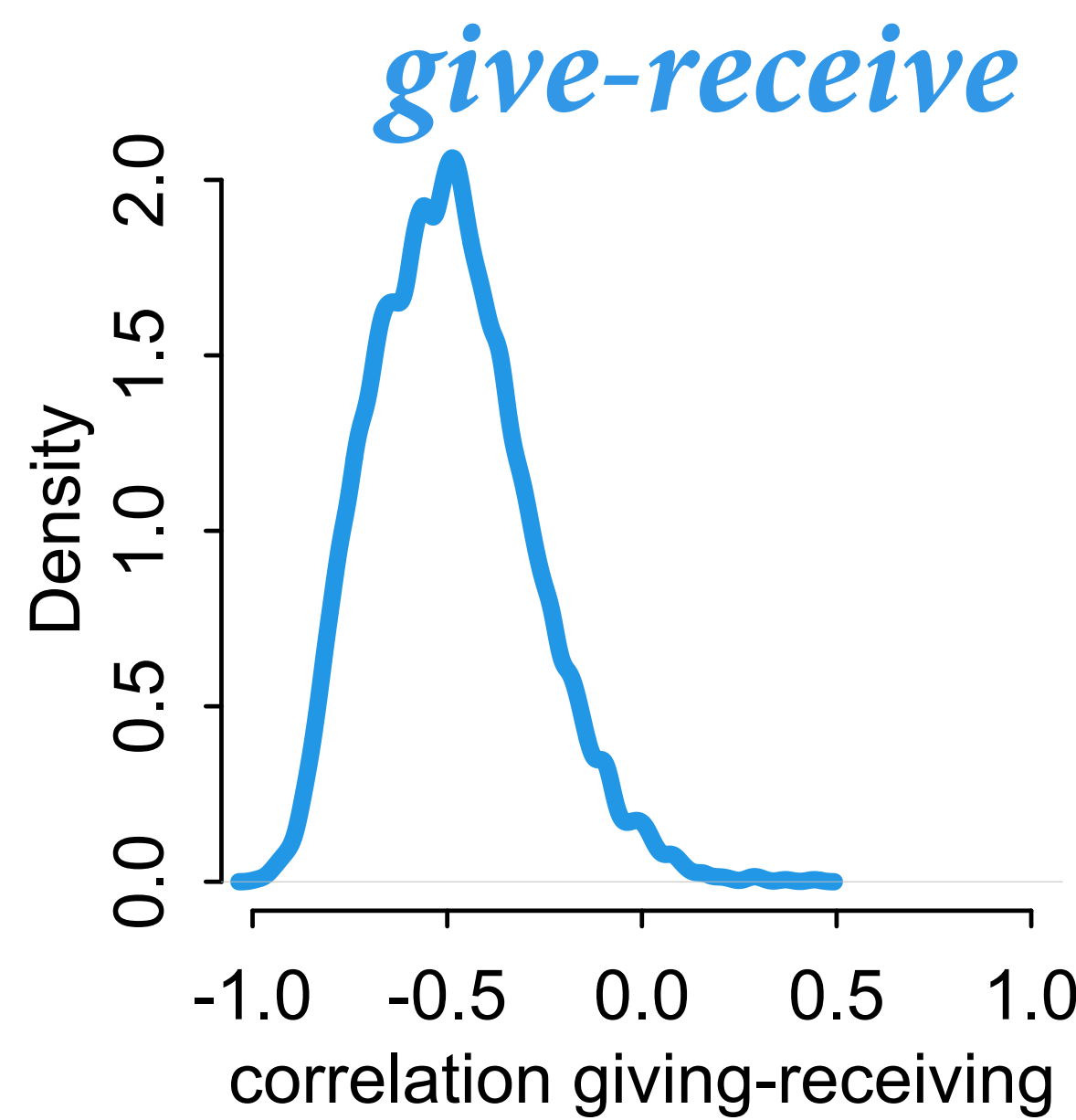
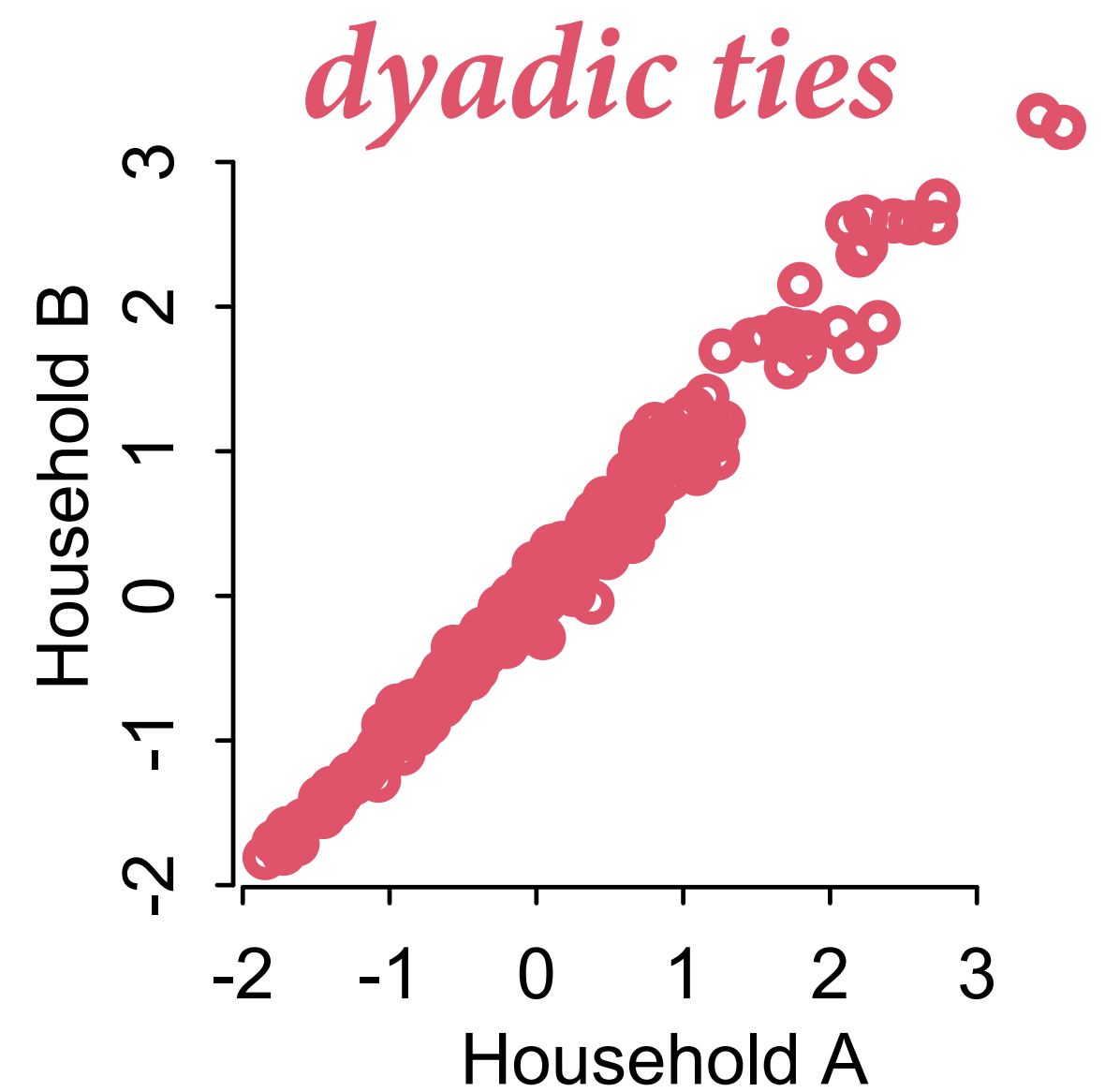
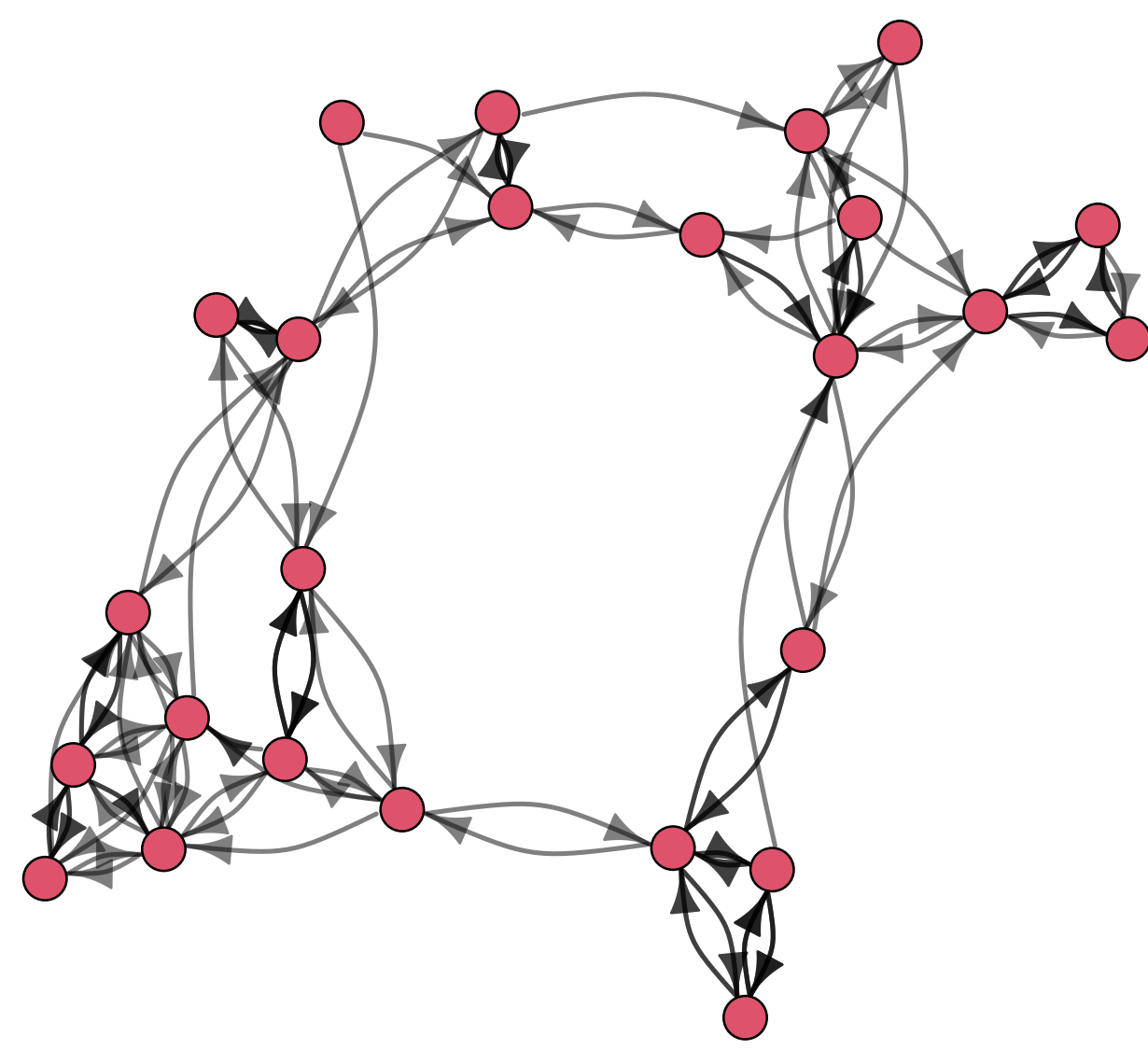
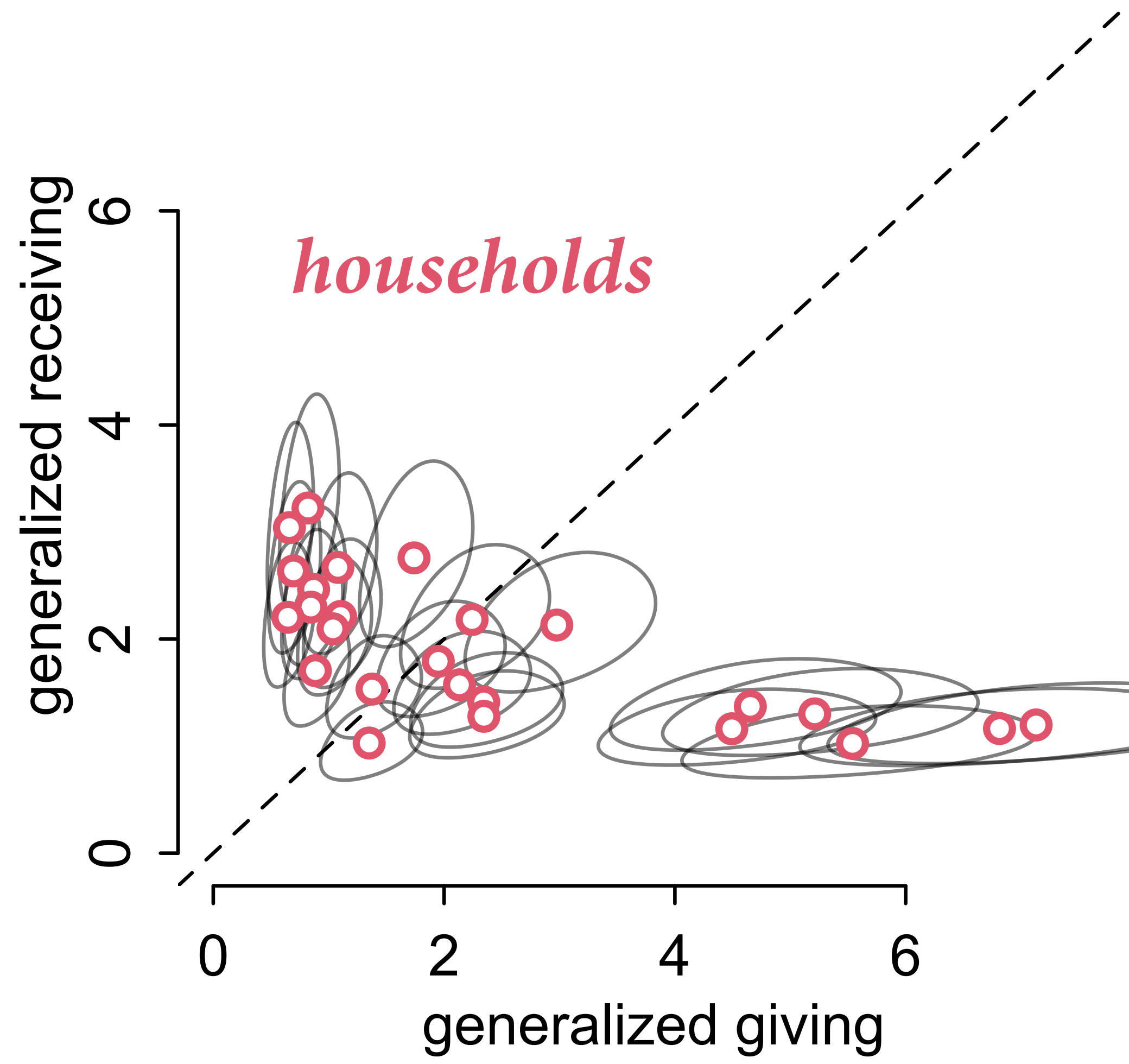
Synthetic data (validation)



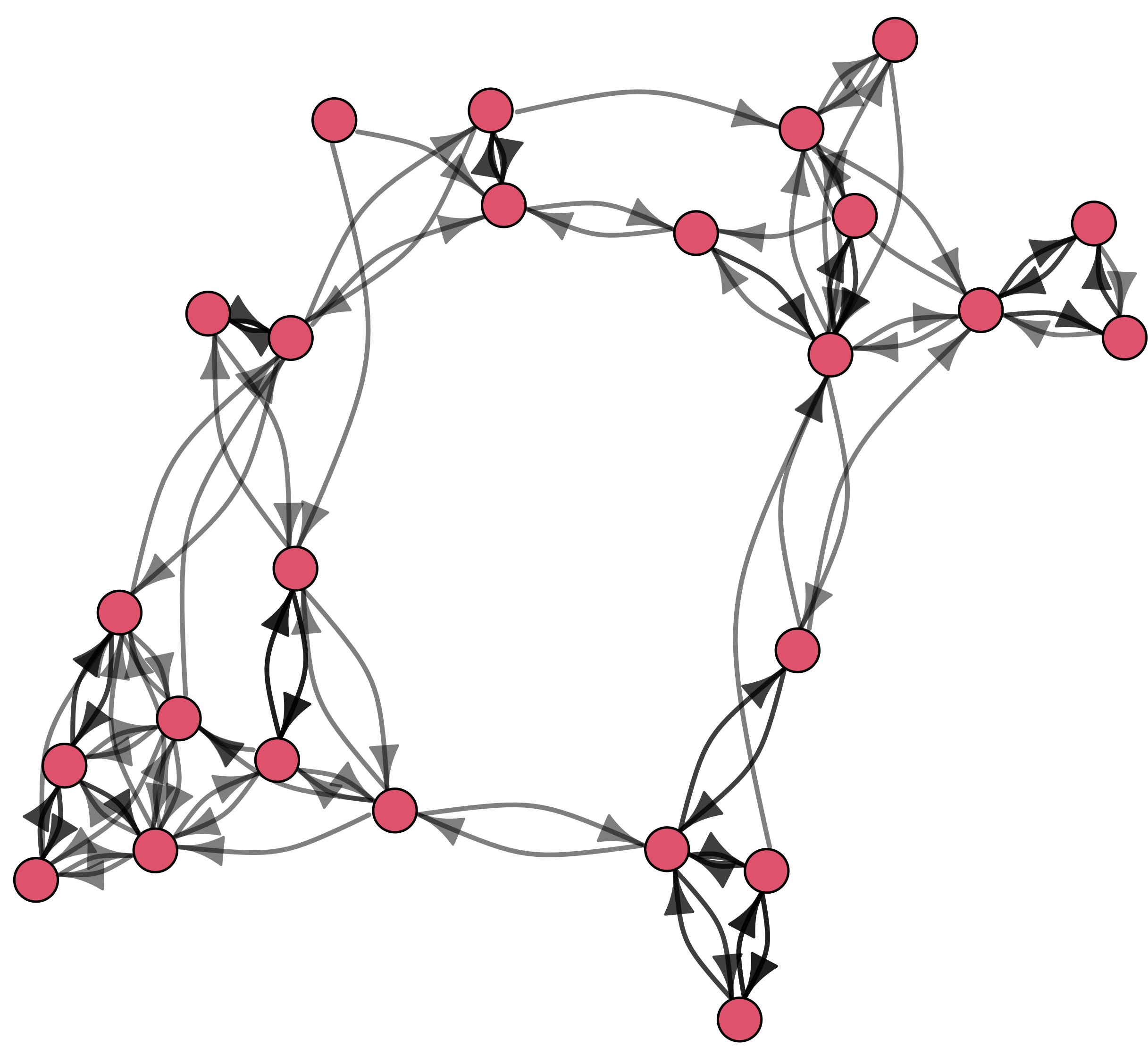
Synthetic data (validation)



Real data (analysis)

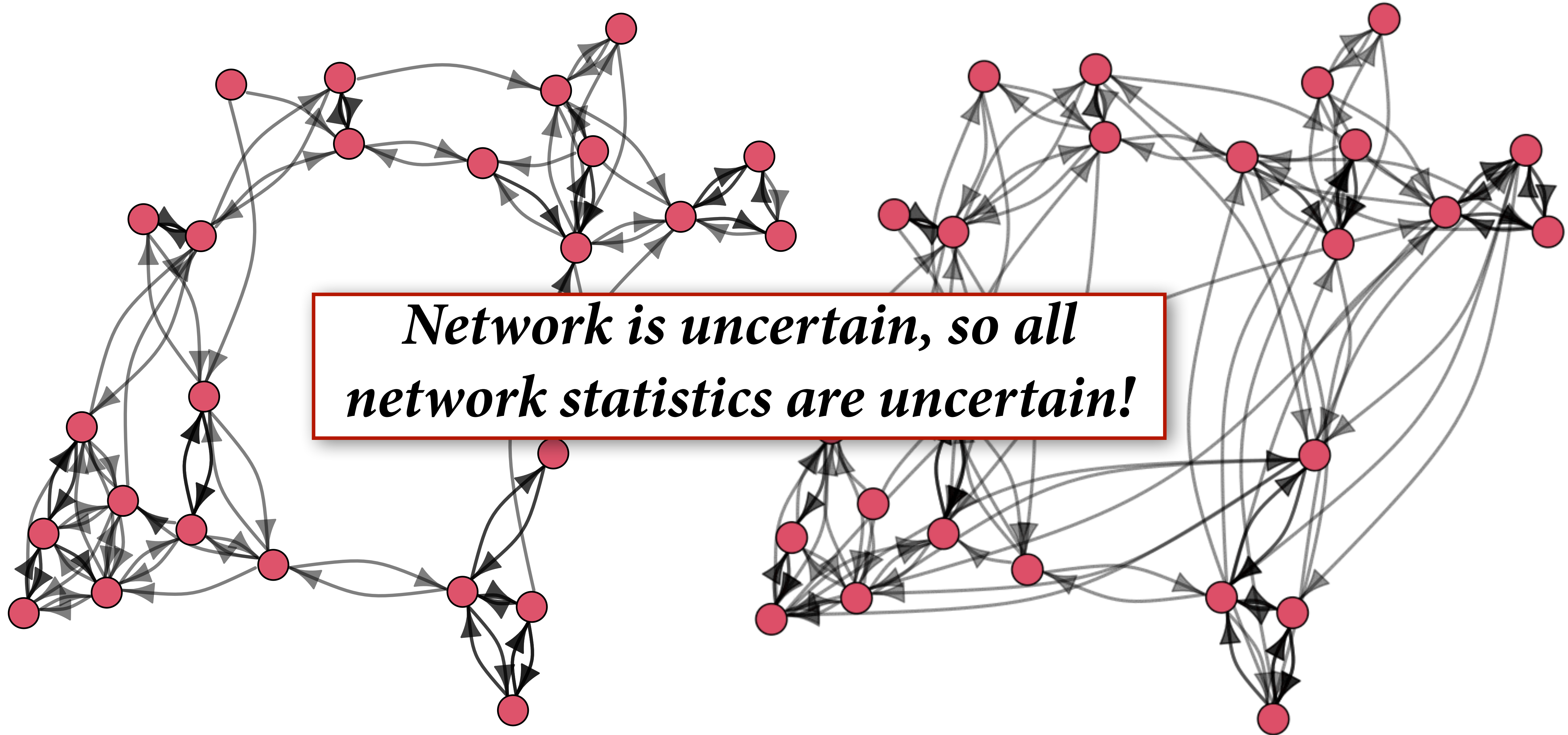


Posterior mean network



Posterior mean network

Samples from posterior



Social Networks Don't Exist

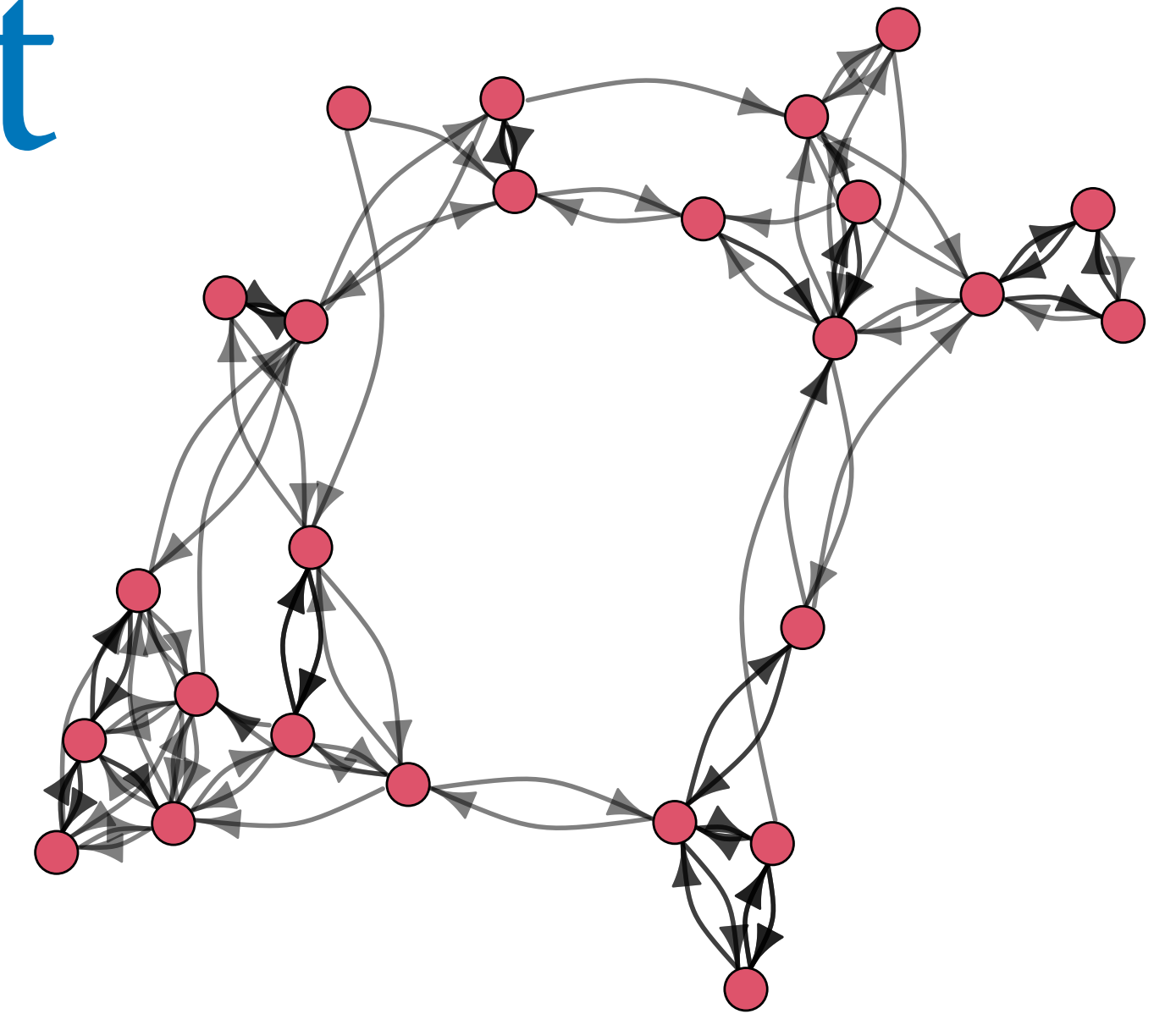
> They are concepts we use to explain and describe

Varying effects are placeholders

Can model the network ties
(using dyad features)

Can model the giving/receiving
(using household features)

Relationships can cause other
relationships



$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + T_{AB} + G_A + R_B$$

$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{BA}) = \alpha + T_{BA} + G_B + R_A$$

$$\begin{pmatrix} T_{AB} \\ T_{BA} \end{pmatrix} \sim \text{MVNormal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

$$\rho \sim \text{LKJCorr}(2)$$

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$$\mathcal{T}_{AB} = T_{AB} + \beta_A A_{AB}$$

*linear model
for tie strength*

*varying
effect*

*effect of
association
between A & B*

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

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$$\mathcal{G}_A = G_A + \beta_{W,G} W_A$$

*linear model
for giving*

*varying
effect*

*effect of A's
wealth on giving*

$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

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$$\mathcal{R}_B = R_B + \beta_{W,R} W_B$$

*linear model
for receiving*

*varying
effect*

*effect of B's wealth
on receiving*

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$$\log(\lambda_{AB}) = \alpha + \mathcal{T}_{BA} + \mathcal{G}_B + \mathcal{R}_A$$

$$\mathcal{T}_{BA} = T_{BA} + \beta_A A_{AB}$$

$$\mathcal{G}_B = G_B + \beta_{W,G} W_B$$

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$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

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$$\mathcal{G}_B = G_B + \beta_{W,G} W_B$$

$$\mathcal{R}_A = R_A + \beta_{W,R} W_A$$

```
# general model with features
```

```
f_houses <- alist(
```

```
  GAB ~ poisson( lambdaAB ),
```

```
  GBA ~ poisson( lambdaBA ),
```

```
# A to B
```

```
log(lambdaAB) <- a + TAB + GA + RB,
```

```
  TAB <- T[D,1] + bA*A,
```

```
  GA <- gr[HA,1] + bW[1]*W[HA] ,
```

```
  RB <- gr[HB,2] + bW[2]*W[HB] ,
```

```
# B to A
```

```
log(lambdaBA) <- a + TBA + GB + RA,
```

```
  TBA <- T[D,2] + bA*A,
```

```
  GB <- gr[HB,1] + bW[1]*W[HB] ,
```

```
  RA <- gr[HA,2] + bW[2]*W[HA] ,
```

```
# priors
```

```
a ~ normal(0,1),
```

```
vector[2]:bW ~ normal(0,1),
```

```
bA ~ normal(0,1),
```

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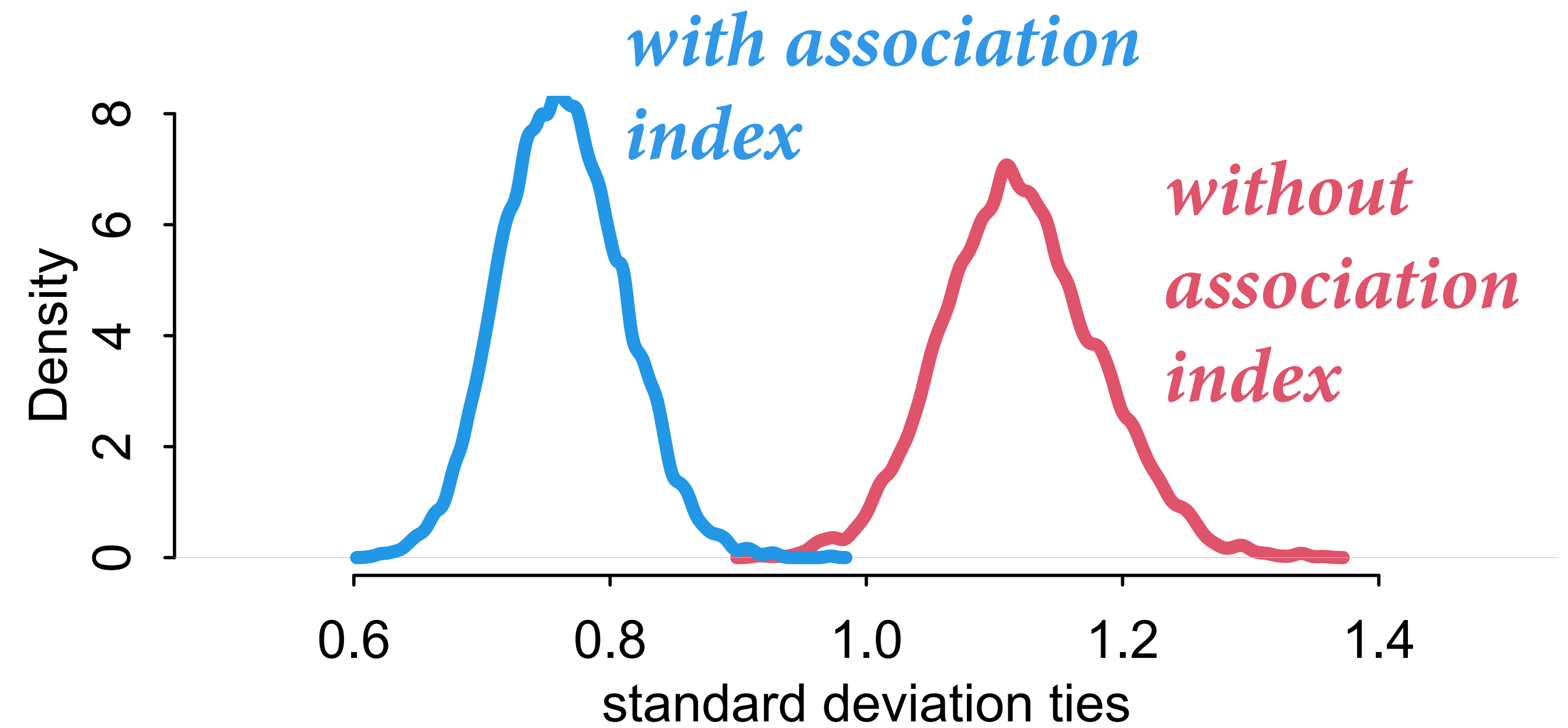
$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{AB}) = \alpha + \mathcal{T}_{BA} + \mathcal{G}_B + \mathcal{R}_A$$

$$\mathcal{T}_{BA} = T_{BA} + \beta_A A_{AB}$$

$$\mathcal{G}_B = G_B + \beta_{W,G} W_B$$

$$\mathcal{R}_A = R_A + \beta_{W,R} W_A$$



$$G_{AB} \sim \text{Poisson}(\lambda_{AB})$$

$$\log(\lambda_{AB}) = \alpha + \mathcal{T}_{AB} + \mathcal{G}_A + \mathcal{R}_B$$

$$\mathcal{T}_{AB} = T_{AB} + \beta_A A_{AB}$$

$$\mathcal{G}_A = G_A + \beta_{W,G} W_A$$

$$\mathcal{R}_B = R_B + \beta_{W,R} W_B$$

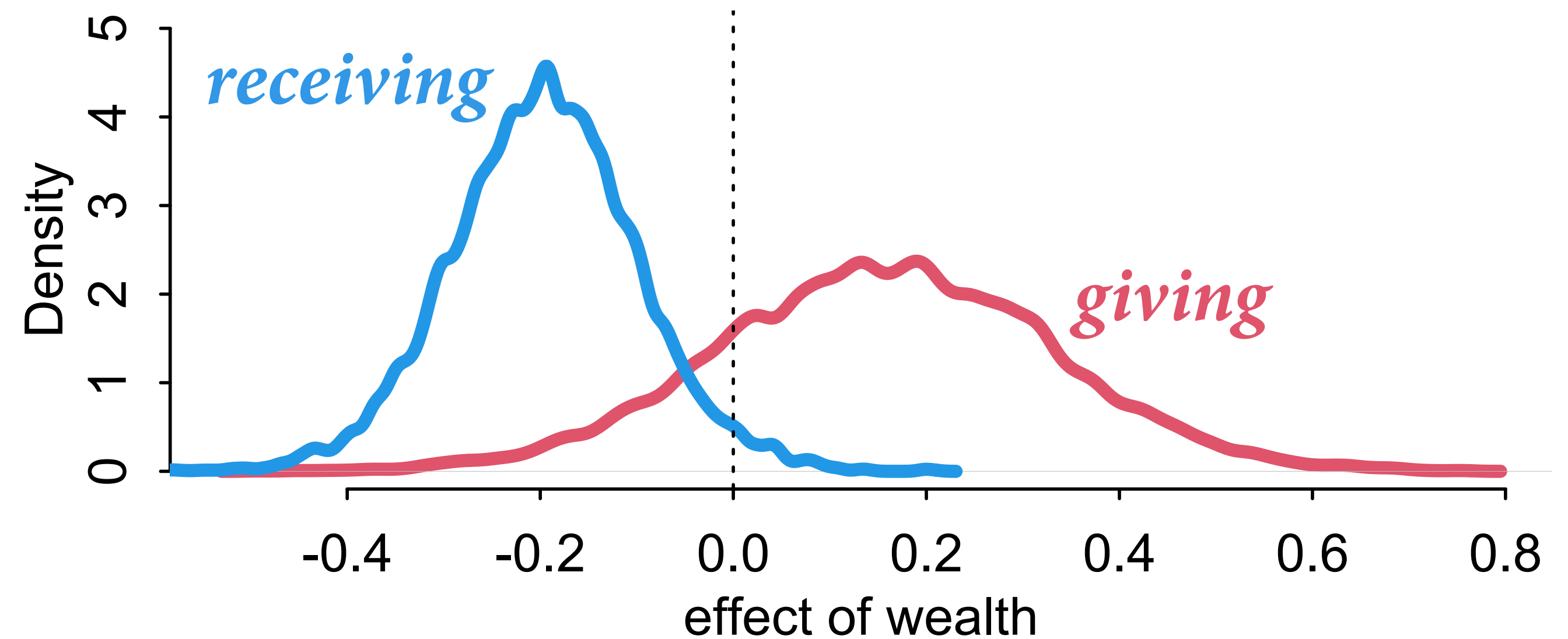
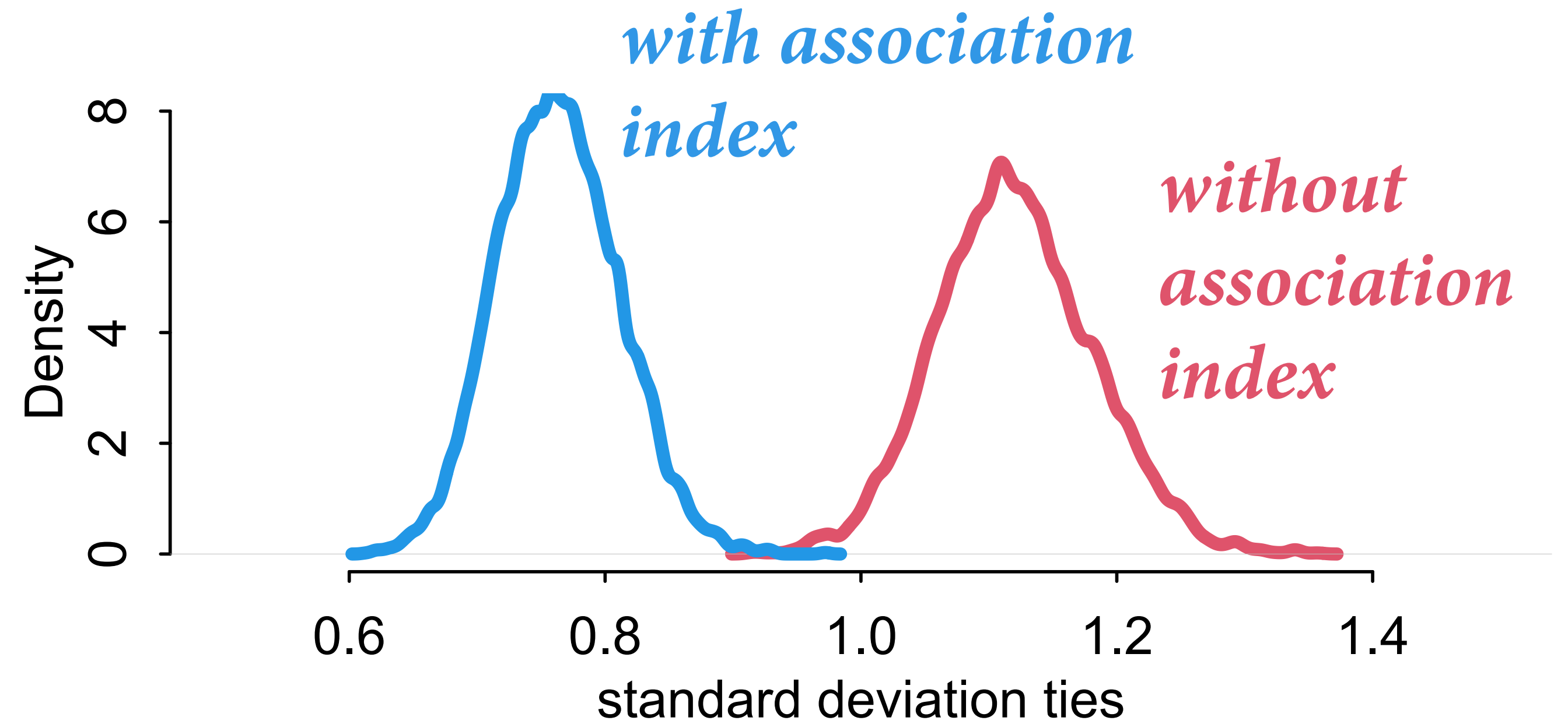
$$G_{BA} \sim \text{Poisson}(\lambda_{BA})$$

$$\log(\lambda_{AB}) = \alpha + \mathcal{T}_{BA} + \mathcal{G}_B + \mathcal{R}_A$$

$$\mathcal{T}_{BA} = T_{BA} + \beta_A A_{AB}$$

$$\mathcal{G}_B = G_B + \beta_{W,G} W_B$$

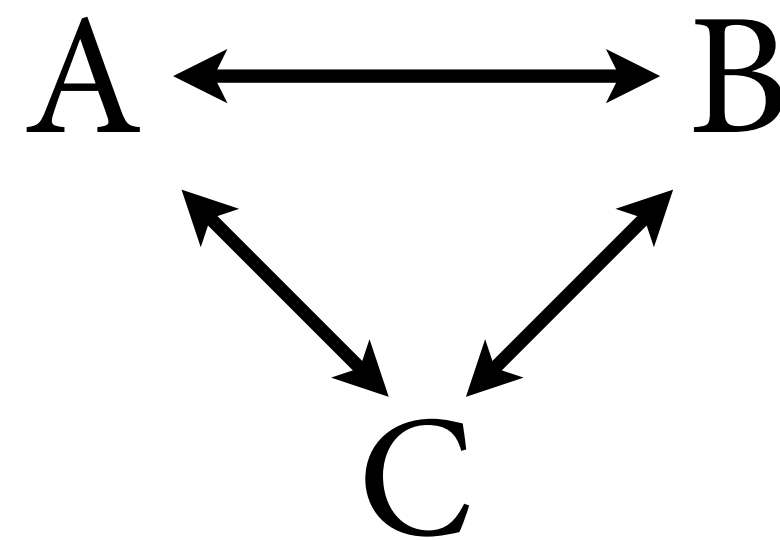
$$\mathcal{R}_A = R_A + \beta_{W,R} W_A$$



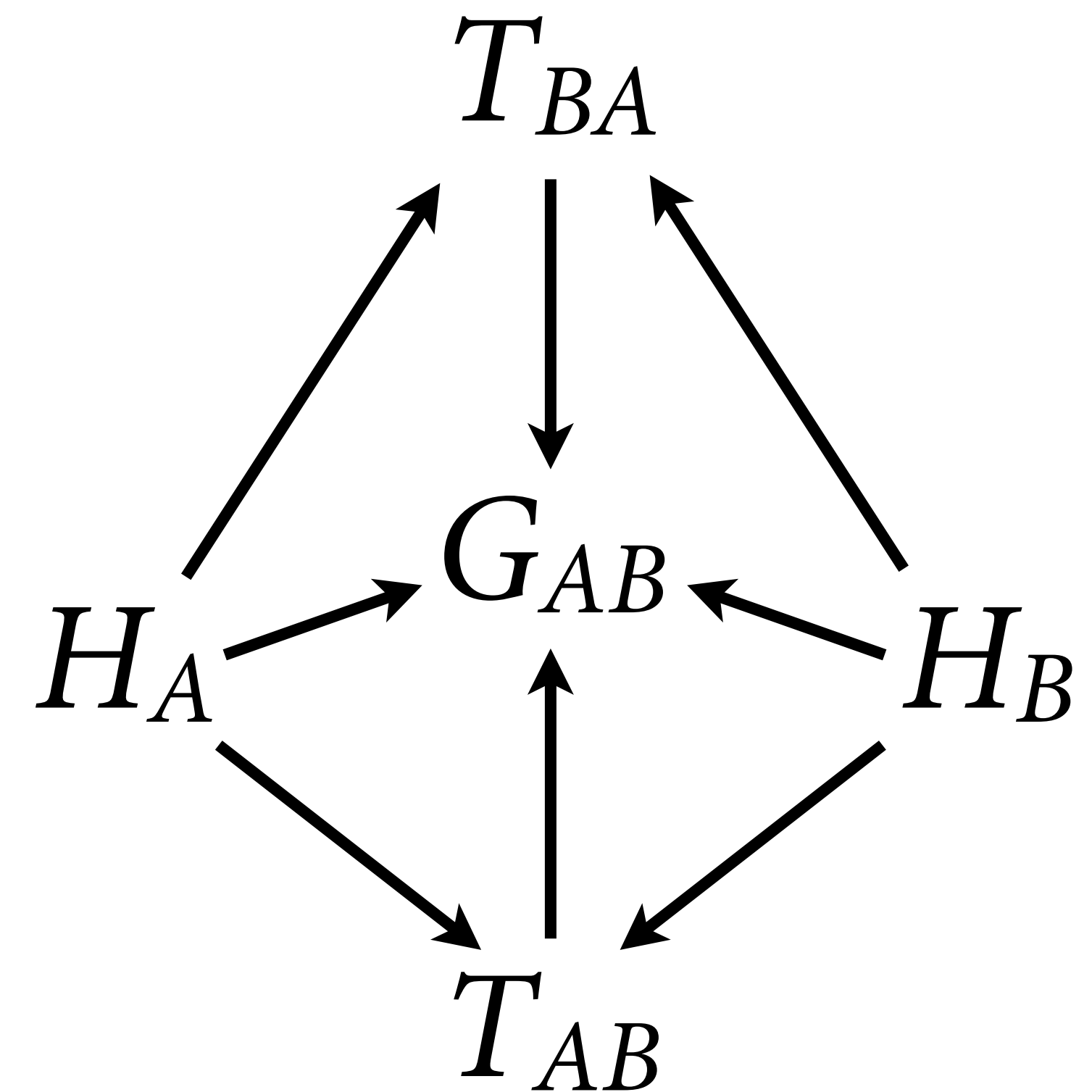
Additional Structure: Triangles

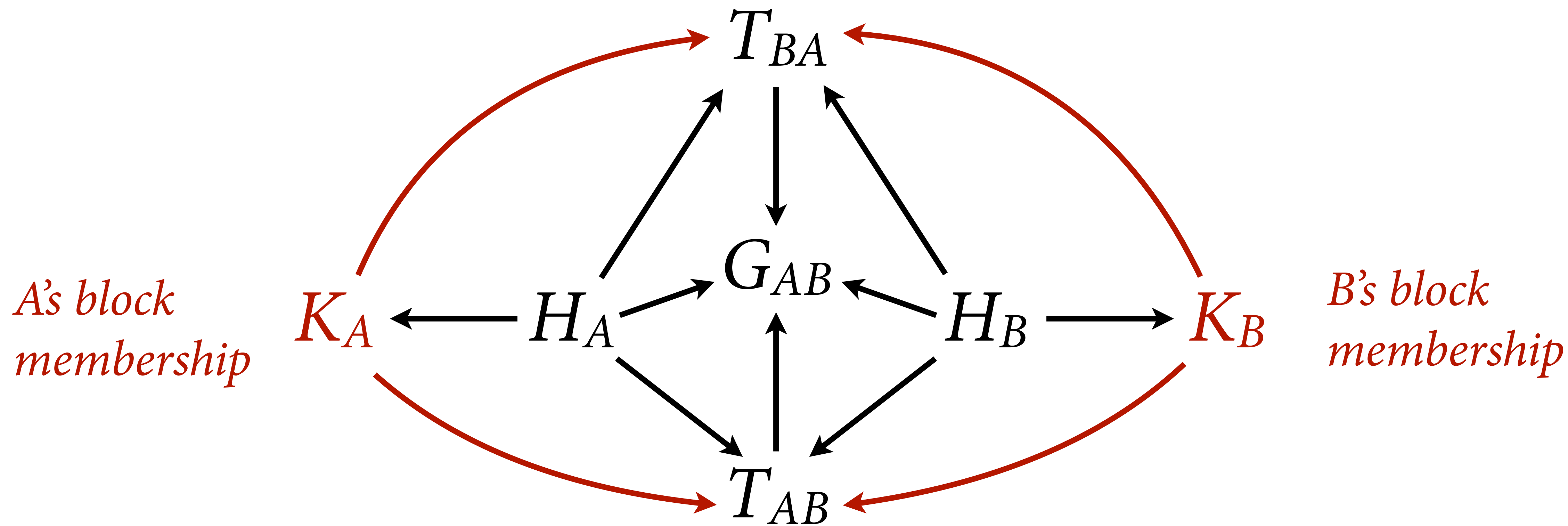
Relationships tend to come in triangles

Triangle closure:

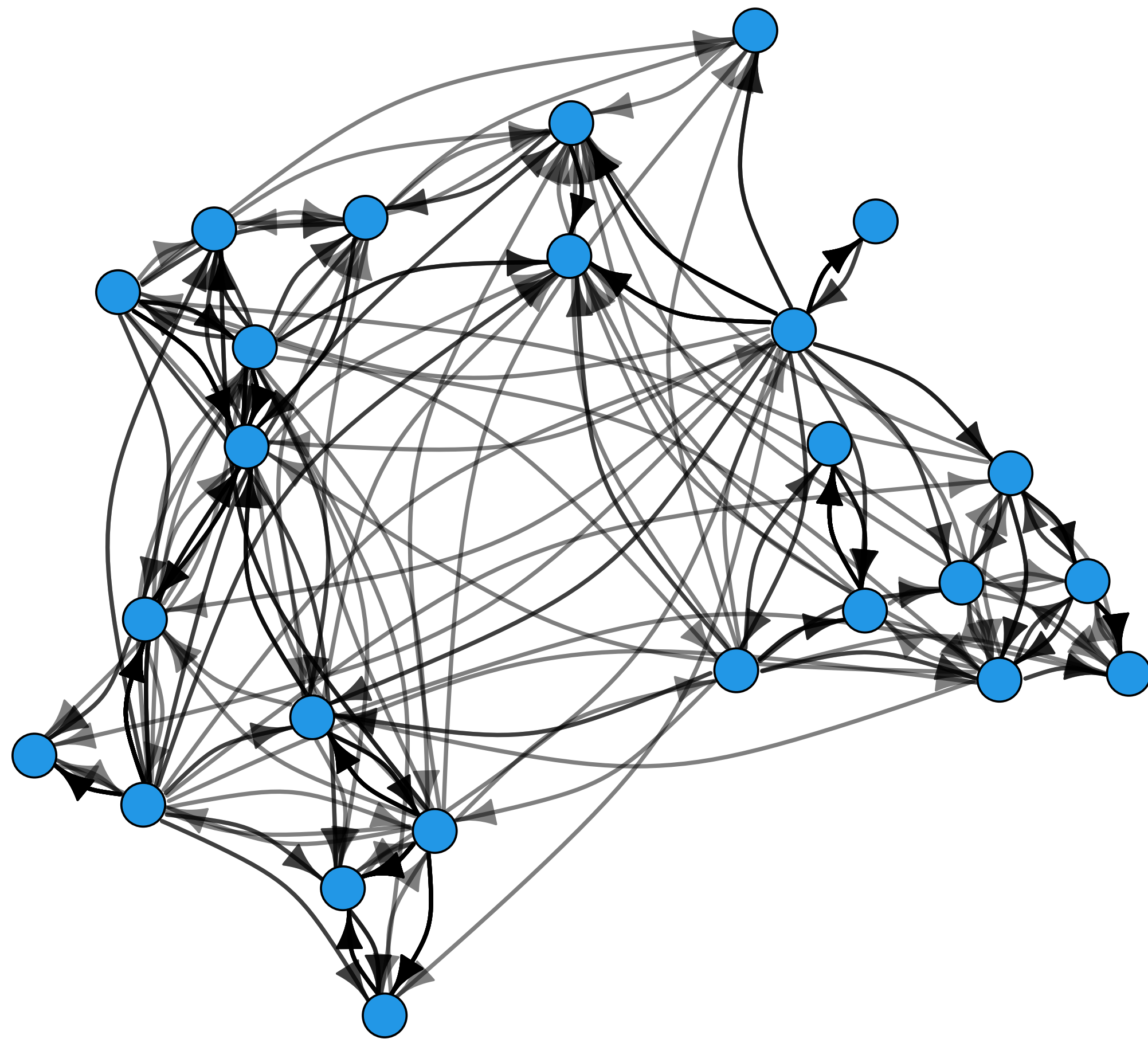


Block models: Ties more common within certain groups (family, office, *Stammtisch*)

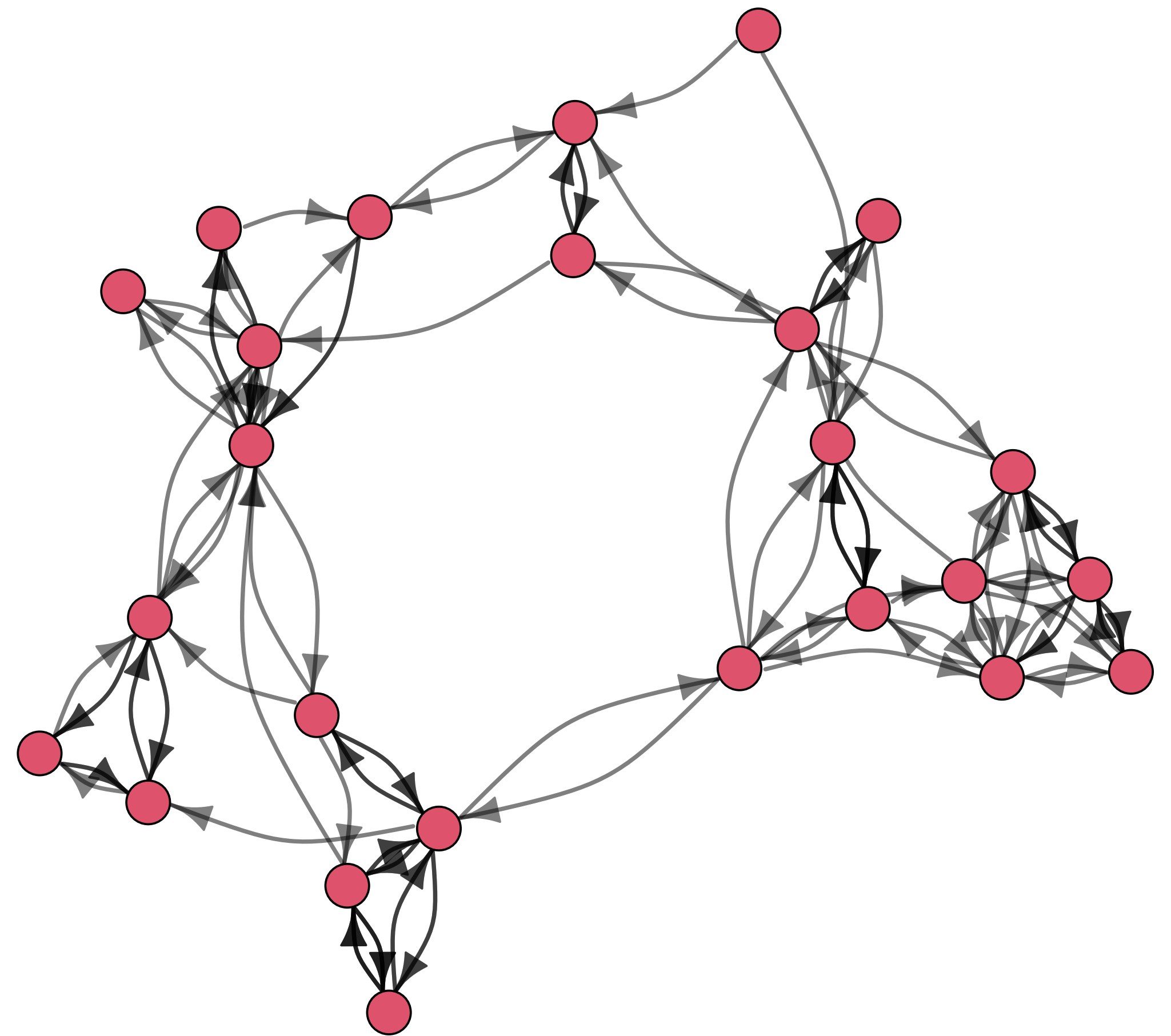




Raw data



Reciprocity
Posterior mean network



Varying effects as technology

Social networks try to express *regularities* of observations

Inferred social network is *regularized*, a structured varying effect

Analogous problems: phylogeny, space, heritability, knowledge, personality

What happens when the clusters are not *discrete* but *continuous*? Age, distance, time, similarity

